PRACTICAL 5
ANALYZING DATA USING TABLE CALCULATIONS IN TABLEAU

Aim:- To analyze data using Tableau table calculations and understand how to perform running totals, percentage calculations, moving averages, cumulative metrics, year-over-year growth, ranking, window functions, addressing, partitioning, and dynamic titles and labels.

Objectives:

After completing this practical, students will be able to:

1. Apply Quick Table Calculations such as Running Total and Percent of Total.
2. Create Moving Averages and cumulative metrics.
3. Perform Year-over-Year Growth Analysis.
4. Use ranking functions to identify Top N and Bottom N records.
5. Implement window functions such as WINDOW_SUM and WINDOW_AVG.
6. Customize addressing and partitioning for table calculations.
7. Create dynamic titles and labels using Tableau calculations.

(a) APPLY QUICK TABLE CALCULATIONS

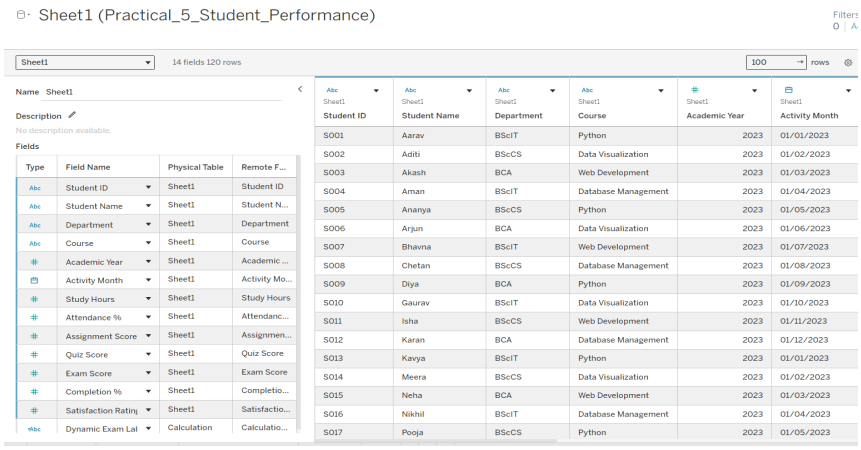
There are two parts:

- **(a1) Running Total**
- **(a2) Percent of Total**

(a1) Running Total of Study Hours

Aim: To calculate the cumulative Study Hours month by month using Tableau's Quick Table Calculation.

Excel Sheet:



Step 1:
Create a Worksheet
1. Open the Tableau workbook.
2. Click New Worksheet.
3. Rename the worksheet as:
Running Total

Step 2:
Add Activity Month
1. From the **Data Pane,** locate **Activity Month.**
2. Drag Activity **Month** to the **Columns** shelf.
3. If Tableau displays individual dates, click the drop-down arrow beside **Activity Month.**
4. Select **Month.**
The months will appear on the horizontal axis.

Step 3:
Add Study Hours
1. From the **Data Pane**, locate **Study Hours.**
2. Drag **Study Hours** to the **Rows** shelf.
3. Tableau will display:
SUM (Study Hours)

Step 4:
Apply Running Total
1. Right-click SUM(Study Hours) on the **Rows** shelf.
2. Select:
Quick Table Calculation → Running Total
Tableau will now calculate the cumulative Study Hours for each month.

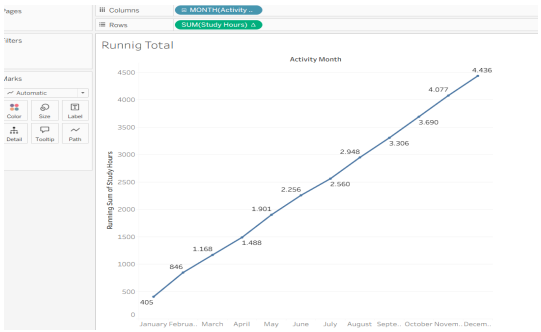
Example:
Suppose the monthly Study Hours are:

Month	Monthly Study Hours	Running Total
January	405	405
February	441	846
March	322	1,168
April	320	1,488

For February:
Running Total = 405 + 441 = 846

Important Note About Labels
Sometimes the graph may display the original monthly value, such as 441, instead of the running total **846.**
To correct this:
1. Go to the **Marks** card.
2. Click **Label.**
3. Remove the existing label field if required.
4. Drag SUM(Study Hours) from the Data Pane to **Label**.
5. Click the drop-down arrow on the SUM(Study Hours) pill on the Label shelf.
6. Select:
Quick Table Calculation → Running Total
Now both the graph and the labels will display the running-total values.

Output:



(a2) Percent of Total Exam Score

Aim: To calculate the percentage contribution of each month's Exam Score to the total Exam Score using Tableau's Quick Table Calculation.

Step 1:
Create a New Worksheet
1. Click **New Worksheet (+)**.
2. Rename the worksheet:
Percent of Total

Step 2:
Add Activity Month to Columns
1. Drag **Activity Month** to the **Columns** shelf.
2. Select **Month.**
3. If required, click the drop-down arrow on MONTH(Activity Month).
4. Select **Discrete.**
The columns should appear as:
Jan | Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct | Nov | Dec

Step 3:

Add Exam Score to Rows

1. Drag **Exam Score** to the **Rows** shelf.
2. Tableau will initially display:

SUM (Exam Score)

At this stage, Tableau is displaying the total Exam Score for each month.

Step 4:

Apply Percent of Total

1. Right-click the green SUM (Exam Score) pill on the **Rows** shelf.
2. Select:

Quick Table Calculation → Percent of Total

Tableau will convert the monthly Exam Scores into percentages of the overall total.

Step 5:

Display Percentage Values

1. Go to the **Marks** card.
2. Click **Label.**
3. Select **Show mark labels.**

The percentage values will now appear directly on the graph.

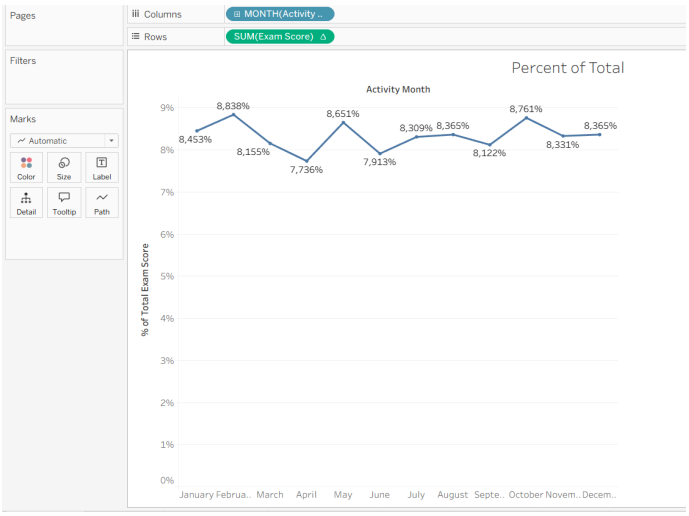
<u>Example Output:</u>

The graph may display values such as:

- January – 8.5%
- February – 8.8%
- March – 8.2%

The exact values depend on the dataset.

Output:



(b) MOVING AVERAGE AND CUMULATIVE METRICS
(b1) Three-Month Moving Average

Aim: To calculate a three-month moving average of Exam Score using Tableau.
A moving average helps identify the overall trend by averaging the current month with previous months.

Step 1:
Create a New Worksheet
1. Click **New Worksheet (+).**
2. Rename the worksheet:
Moving Average

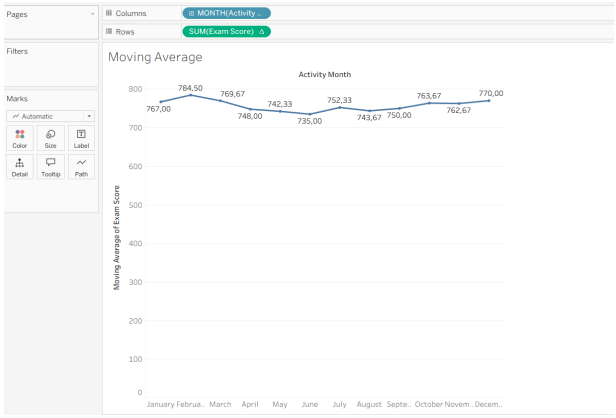
Step 2:
Add Activity Month
1. Drag **Activity Month** to **Columns.**
2. Select **Month.**
3. Make it **Discrete**, if necessary.
The columns should appear as:
Jan | Feb | Mar | Apr | May | Jun |...

Step 3:
Add Exam Score
1. Drag **Exam Score** to **Rows**.
2. Tableau will initially display the monthly Exam Score.

Step 4:
Apply Moving Average
1. Right-click SUM(Exam Score).
2. Select:
Quick Table Calculation → Moving Average
Tableau will now calculate a moving average.

Step 5:
Customize the Moving Average to Three Months
1. Right-click SUM (Exam Score) again.
2. Select **Edit Table Calculation**.
3. Choose **Moving Average**.
4. Set:
Previous values = 2
Next values = 0
This creates a three-month moving average because the calculation includes:
Previous 2 months + Current month

Output:



(b2) Cumulative Metric – Study Hours

Aim: To calculate cumulative Study Hours month by month.

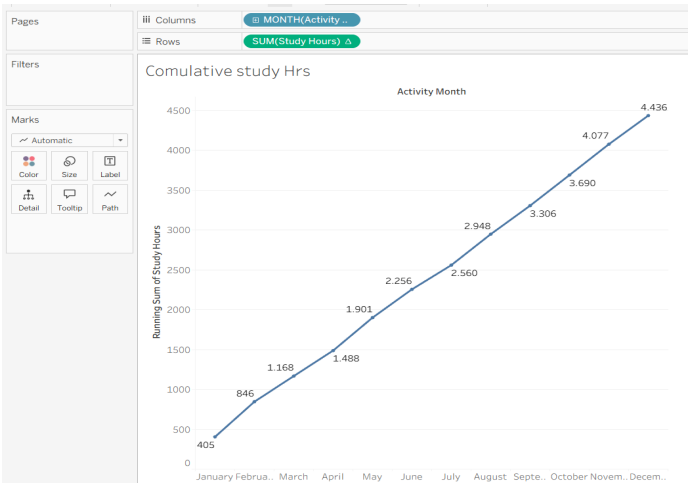
Step 1:
Create a New Worksheet
1. Create another worksheet.
2. Rename it:
Cumulative Study Hours

Step 2:
Add Activity Month
Drag **Activity Month** to **Columns** and select **Month.**

Step 3:
Add Study Hours
Drag **Study Hours** to **Rows.**

Step 4:
Apply Running Total
1. Right-click SUM(Study Hours).
2. Select:
Quick Table Calculation → Running Total
The resulting values represent cumulative Study Hours over time.
This calculation is similar to the Running Total calculation in part (a1), but here it is interpreted specifically as a cumulative metric.

Output:



(C) YEAR-OVER-YEAR GROWTH ANALYSIS

Aim: To compare the Exam Score of one academic year with the previous academic year and calculate the percentage growth.

The dataset contains the academic years:
2023, 2024, and 2025
Therefore, year-over-year growth can be calculated.

Step 1:
Create a New Worksheet
1. Click **New Worksheet (+).**
2. Rename the worksheet:
Year-over-Year Growth

Step 2:
Add Academic Year
1. From the Data Pane, locate **Academic Year**.
2. Drag it to **Columns**.
The columns should show:
2023 | 2024 | 2025

Step 3:
Add Exam Score
1. Drag **Exam Score** to **Rows.**
2. Tableau will display:
SUM (Exam Score)
The output will conceptually look like:

Academic Year	Exam Score
2023	Total for 2023
2024	Total for 2024
2025	Total for 2025

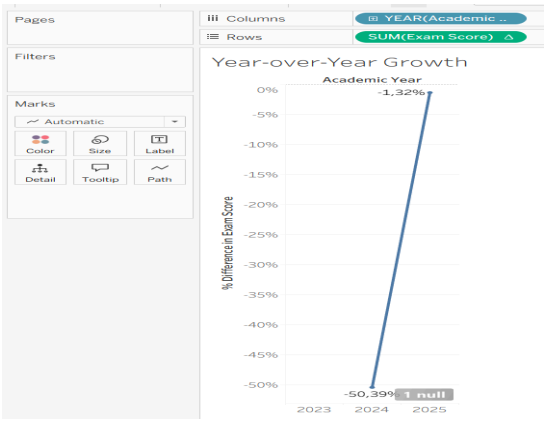
Step 4:
Apply Year-over-Year Growth
1. Right-click SUM (Exam Score).
2. Select:
Quick Table Calculation → Year-over-Year Growth
Tableau will calculate the percentage change compared with the previous year.

Example Output:

Academic Year	Exam Score	YoY Growth
2023	2,800	—
2024	2,950	5.36%
2025	3,100	5.08%

The first year is blank because there is no previous year for comparison.
Calculation for 2024
YoY Growth = ((2,950 − 2,800) / 2,800) × 100
= 5.36%
Calculation for 2025
YoY Growth = ((3,100 − 2,950) / 2,950) × 100
= 5.08%
Interpretation
A positive value such as **+5.36%** means that the Exam Score increased by 5.36% compared with the previous year.
A negative value such as **−5.36%** would mean that the Exam Score decreased by 5.36%.
Result
Year-over-Year growth in Exam Score was successfully calculated using Tableau table Calculations.

Output:



(d) RANKING FUNCTIONS – TOP N AND BOTTOM N

Two worksheets will be created:
1. Top 5 Students by Exam Score
2. Bottom 5 Students by Attendance
Because each student may appear in multiple records, AVG is used instead of SUM.
This provides a more meaningful student-level ranking.

(d1) Top 5 Students by Average Exam Score
Aim: To identify the five students with the highest average Exam Score.

Step 1:
Create a New Worksheet
Create a new worksheet and rename it:
Top 5 Students

Step 2:
Add Student Name
Drag **Student Name** to **Rows**.

Step 3:
Add Exam Score
Drag **Exam Score** to **Columns**.
Initially, Tableau will display:
SUM (Exam Score)

Step 4:
Change SUM to AVG
1. Click the drop-down arrow on SUM (Exam Score).
2. Select:

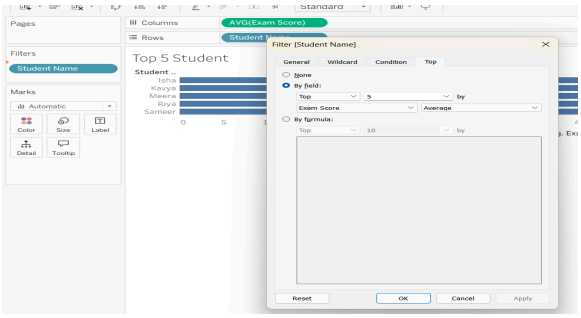
Measure → Average
The field will now become:
AVG (Exam Score)

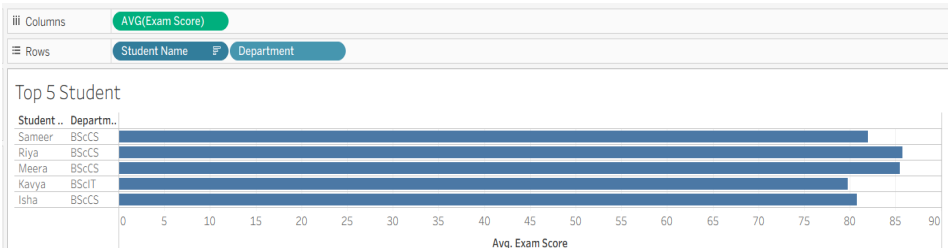
Step 5:
Create the Top 5 Filter
1. Drag **Student Name** to **Filters.**
2. In the Filter dialog box, select the **Top** tab.
3. Select **By field.**
4. Choose: **Top 5**
5. Select: **AVG (Exam Score)**
6. Click **OK**.

Step 6:
Sort the Results
Click the **Sort Descending** button to display the students from the highest average Exam Score to the lowest.

<u>Expected Result</u>
The worksheet will display the five students having the highest average Exam Score.

Output:





(d2) Bottom 5 Students by Average Attendance

Aim: To identify the five students with the lowest average Attendance Percentage.

Step 1:
Create a New Worksheet
Create a new worksheet and rename it:
Bottom 5 Students

Step 2:
Add Student Name
Drag **Student Name** to **Rows.**

Step 3:
Add Attendance %
Drag **Attendance %** to **Columns**.

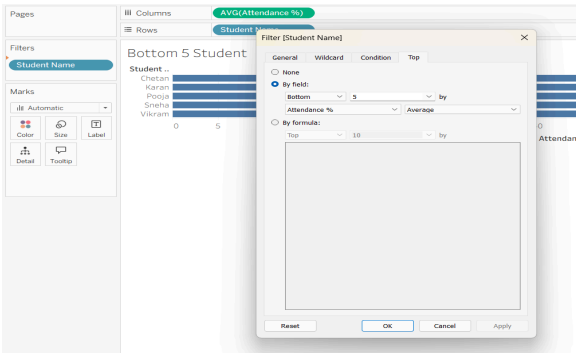
Step 4:
Change SUM to AVG
1. Click the drop-down arrow on SUM (Attendance %).
2. Select:
Measure → Average
The field will now become:
AVG (Attendance %)

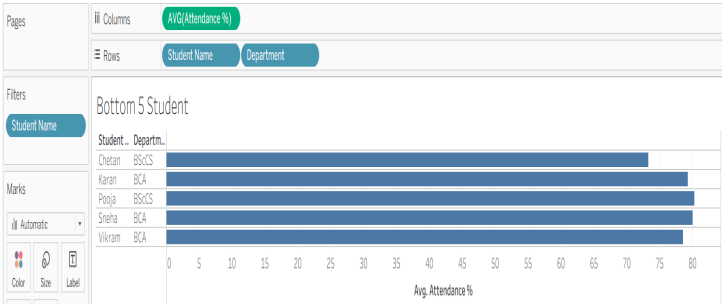
Step 5:
Apply Bottom 5 Filter
1. Drag **Student Name** to **Filters.**
2. Select the **Top** tab.
3. Choose **By field.**
4. Select:
Bottom 5
5. Select:
AVG(Attendance %)
6. Click **OK.**

Step 6:
Sort the Results
Click **Sort Ascending** to display the students with the lowest average Attendance first.

Expected Result
The worksheet will display the five students having the lowest average Attendance Percentage.

Output:





(e) IMPLEMENT WINDOW FUNCTIONS

Two window functions will be created:

1. WINDOW_SUM

2. WINDOW_AVG

(e1) WINDOW_SUM – Three-Month Study Hours

Aim: To calculate the total Study Hours within a three-month window.

Step 1:

Create a New Worksheet

Create a new worksheet and rename it:

WINDOW_SUM

Step 2:

Create a Calculated Field

1. Go to:

Analysis → Create Calculated Field

2. Name the calculated field:

- **3 Month Study Hours**

3. Enter the following formula:

WINDOW_SUM (SUM ([Study Hours]), -2, 0)

4. Click **OK**.
Explanation of the Formula

The formula:
WINDOW_SUM (SUM ([Study Hours]), -2, 0)

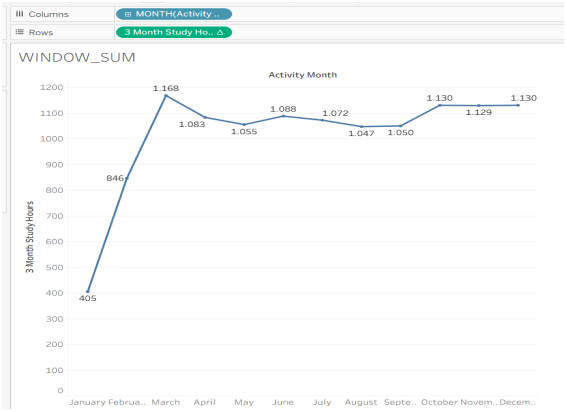
Means:

- -2 = include two previous values.
- 0 = include the current value.

Therefore, the calculation includes:
Previous 2 months + Current month
Step 3:
Add Activity Month
1. Drag Activity **Month** to **Columns**.
2. Select **Month.**

Step 4:
Add the Calculated Field
Drag **3 Month Study Hours** to **Rows.**

Output:



(e2) WINDOW_AVG – Three-Month Average Exam Score

Aim: To calculate the average Exam Score within a three-month window.

Step 1:
Create a Calculated Field
1. Go to:
Analysis → Create Calculated Field
2. Name the calculated field:

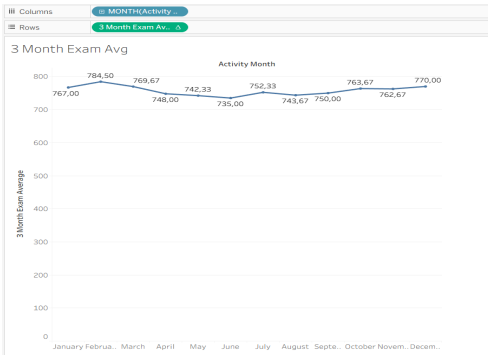
- **3 Month Exam Average**

3. Enter:
WINDOW_AVG (SUM([Exam Score]), -2, 0)
4. Click **OK.**

Step 2:
Add Activity Month
1. Drag **Activity Month** to **Columns.**
2. Select **Month.**

Step 3:
Add the Calculated Field
Drag **3 Month Exam Average** to **Rows.**
Tableau will calculate the average Exam Score for the current month and the previous two months.

Output:



(f) CUSTOMIZE ADDRESSING AND PARTITIONING

Aim: To understand how Tableau determines the direction of a table calculation and where the calculation should restart.
A three-month moving average will be calculated separately for each Course.

Addressing
Addressing tells Tableau the direction in which the table calculation should be performed.
In this practical:
Activity Month is the addressing dimension.
Therefore, Tableau calculates the values in the sequence:
January → February → March → April →…

Partitioning

Partitioning tells Tableau where the table calculation should restart.
In this practical:
Course is the partitioning dimension.
The calculation will be performed separately for each course, such as:

- Python
- Data Visualization
- Web Development
- Database Management

Steps to Customize Addressing and Partitioning

Step 1:
Create a New Worksheet
Create a new worksheet and rename it:
Addressing and Partitioning

Step 2:
Add Course
Drag **Course** to **Rows.**

Step 3:
Add Activity Month
Drag **Activity Month** to **Columns** and select **Month.**

Step 4:
Add Exam Score
Drag **Exam Score** to **Rows.**
Change the aggregation from:
SUM (Exam Score)
To:
AVG (Exam Score)

Step 5:
Apply Moving Average
Right-click the Exam Score field and select:
Quick Table Calculation → Moving Average

Step 6:
Edit the Table Calculation
1. Right-click the table-calculated field.
2. Select:

Edit Table Calculation

Step 7:
Select Specific Dimensions
Select:
Specific Dimensions

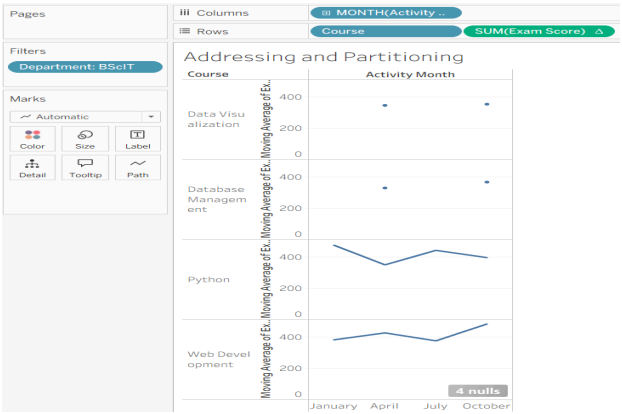
You will see:

- Course
- Activity Month

Make sure **Activity Month** is selected as the dimension through which Tableau addresses the calculation.
The **Course** dimension acts as the partition.

Result
Tableau will not calculate one moving average across all courses.
Instead, it will calculate a separate moving average for each course:
Python:
January → February → March → April →...
Data Visualization:
January → February → March → April →...
Web Development:
January → February → March → April →...
Database Management:
January → February → March → April →...
Expected Output
The worksheet will display separate lines or values for each course, with the moving average calculated independently for each course.

Output:



(g) CREATE DYNAMIC TITLES AND LABELS

This section demonstrates how Tableau can automatically change titles and labels based on the selected Course or other data values.

(g1) Dynamic Title

Aim: To create a worksheet title that changes dynamically according to the selected Course.

Step 1:
Create a New Worksheet
Create a new worksheet and rename it:
Dynamic Title

Step 2:
Add Course to Filters
1. Drag **Course** to **Filters.**
2. Select All.
3. Click **OK.**

Step 3:
Show the Course Filter
1. Right-click **Course** in the Filters area.
2. Select:
Show Filter
A Course filter will appear on the worksheet.

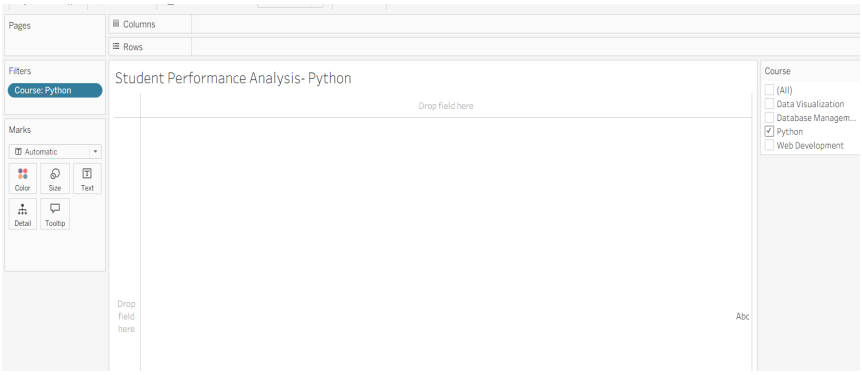
Step 4:
Show the Worksheet Title
Go to:
Worksheet → Show Title
The worksheet title will now be visible.

Step 5:
Edit the Title
1. Right-click the worksheet title.
2. Select **Edit Title**.
3. Enter:
Student Performance Analysis –
4. Select:
Insert → Course
The title will now contain the selected Course.

For example:
Student Performance Analysis – <Course>

Step 6:
Test the Dynamic Title
Selec**t Python** from the Course filter.
The title should display:
Student Performance Analysis – Python
Now select **Web Development.**
The title should automatically change to:
Student Performance Analysis – Web Development
This demonstrates that the worksheet title changes according to the selected filter Value.

Output:



(g2) Dynamic Label

Aim: To create a dynamic label that displays the Exam Score according to the selected Course or filter.

Step 1:
Create a Calculated Field
Go to:
Analysis → Create Calculated Field

Step 2:
Name the Calculated Field
Enter:
Dynamic Exam Label

Step 3:
Enter the Formula

Use the following formula:
"Exam Score:" + STR (ROUND (SUM ([Exam Score]), 0))
Click **OK**.

Explanation:
The formula performs the following operations:

- SUM ([Exam Score]) calculates the total Exam Score.
- ROUND (..., 0) rounds the result to a whole number.
- STR (...) converts the number into text.
- "Exam Score:" + adds descriptive text before the value.

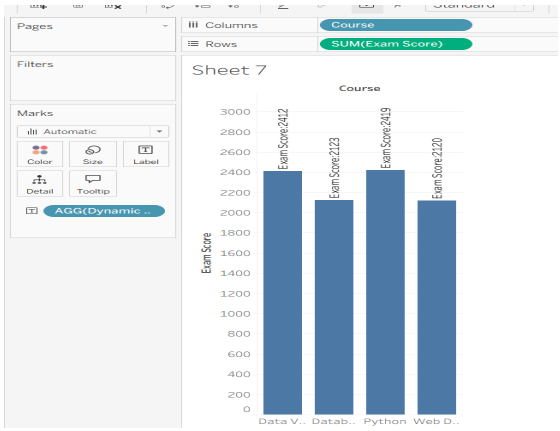
Step 4:
Create a Simple Chart
For example:

Drag **Course** to **Columns.**
Drag **Exam Score** to **Rows**.

Step 5:
Add the Dynamic Label
1. From the Data Pane, locate **Dynamic Exam Label**.
2. Drag it to **Label** on the **Marks** card.

Expected Output
The labels may display values such as:
Exam Score: 2,845
Or
Exam Score: 3,120
The exact values depend on the selected Course and the underlying data.
If a different Course is selected, the label automatically updates.

Output:



Conclusion:

Thus, the practical was successfully completed using Tableau table calculations. We learned to perform running totals, percentage of total, moving averages, cumulative metrics, year-over-year growth, ranking, window functions, addressing and partitioning. We also created dynamic titles and labels to make the data analysis more interactive and informative. These calculations helped us analyze student performance data effectively and understand trends and comparisons over time.

PRACTICAL 7. CREATING INTERACTIVE DASHBOARDS AND DATA STORIES IN TABLEAU

Aim:- To create an interactive dashboard and data story for analyzing retail store performance using Tableau.

Objectives:

After completing this practical, students will be able to:

1. Design a basic dashboard layout using multiple worksheets.
2. Add filters and interactive controls.
3. Implement Highlight and Filter Actions.
4. Use URL Actions and Set Actions for interactivity.
5. Create dynamic dashboards using parameters.
6. Design dashboards for multiple devices.
7. Build a complete data story using the Tableau Story feature.

STEP 1: CONNECT THE EXCEL FILE

1. Open **Tableau Public/Desktop**.
2. Click **Microsoft Excel**.
3. Select the Excel file:

   Retail_Store_Performance_500 (1).xlsx

4. Drag the required Excel sheet to the Data Source area, if required.
5. Click **Sheet 1** to begin creating visualizations.

STEP 2: CHECK DATA TYPES

Before creating the worksheets, verify that the fields are correctly identified.

Dimensions:

- Region
- State
- City
- Customer Segment
- Gender
- Age Group
- Product Category
- Product Name
- Shipping Mode

- Payment Mode
- Returned

Measures:

- Quantity
- Unit Price
- Discount (%)
- Sales
- Profit
- Customer Rating
- Sales Target

Date Field:

- Order Date

If **Order Date** is not recognized as a date, click its data type icon and select **Date**.

PART A: CREATE WORKSHEETS

The following worksheets will be created:

1. Sales by Region
2. Sales by Category
3. Monthly Sales Trend
4. Profit by Region
5. Customer Rating Analysis

SHEET 1: SALES BY REGION

Step 1:-
Rename **Sheet 1** as: **Sales by Region**

Step 2:-
Drag **Region** to the **Rows** shelf.

Step 3:-
Drag **Sales** to the **Columns** shelf.

Step 4:-
Click **Show Me**.

Step 5:-
Select **Bar Chart**.

Step 6:-
Drag **Sales** to **Label** on the Marks card.

Expected Output:-
The worksheet will display sales for each region.

Region	Sales
North	Sales Value
South	Sales Value
East	Sales Value
West	Sales Value

SHEET 2: SALES BY PRODUCT CATEGORY

Step 1:-
Click **New Worksheet**.

Step 2:-
Rename the worksheet: **Sales by Category**

Step 3:-
Drag **Product Category** to **Rows**.

Step 4:-
Drag **Sales** to **Columns**.

Step 5:-
Select **Bar Chart** from Show Me.

Step 6:-
Drag **Profit** to **Color** on the Marks card.

Step 7:-
Drag **Sales** to **Label**.

Expected Output:-
A bar chart will display sales for different product categories, with the bars colored according to Profit.

SHEET 3: MONTHLY SALES TREND

Step 1:-
Create a new worksheet.

Step 2:-
Rename it: **Monthly Sales Trend**

Step 3:-
Drag **Order Date** to **Columns**.

Step 4:-
Click the drop-down arrow on **Order Date**.

Step 5:-
Select **Month**.

Step 6:-
Drag **Sales** to **Rows**.

Step 7:-
Select **Line Chart**.

Expected Output:-
The worksheet will display the monthly sales trend over time.

SHEET 4: PROFIT BY REGION

Step 1:-
Create a new worksheet.

Step 2:-
Rename it: **Profit by Region**

Step 3:-
Drag **Region** to **Columns**.

Step 4:-
Drag **Profit** to **Rows**.

Step 5:-
Select **Bar Chart**.

Step 6:-
Drag **Profit** to **Label**.

Expected Output:-
A bar chart will display the profit generated by each region.

SHEET 5: CUSTOMER RATING ANALYSIS

Step 1:-
Create a new worksheet.

Step 2:-
Rename it: **Customer Rating Analysis**

Step 3:-
Drag **Product Category** to **Columns**.

Step 4:-
Drag **Customer Rating** to **Rows**.

Step 5:-
Change the aggregation from **SUM** to **Average**.
The field should now display:
AVG (Customer Rating)

Step 6:-
Select **Bar Chart**.

Expected Output:-
A bar chart will display the average customer rating for each product category.

PART A: DESIGN A BASIC DASHBOARD

STEP 3: CREATE THE DASHBOARD

Step 1:-
Click the **New Dashboard** button at the bottom of Tableau.

Step 2:-
Rename the dashboard: **Retail Store Performance Dashboard**

Step 3:-
Set the dashboard size to:
Automatic
Or
Desktop Browser

STEP 4: ARRANGE THE WORKSHEETS

From the left-side dashboard pane, drag the required worksheets into the dashboard.

Suggested Dashboard Layout:-

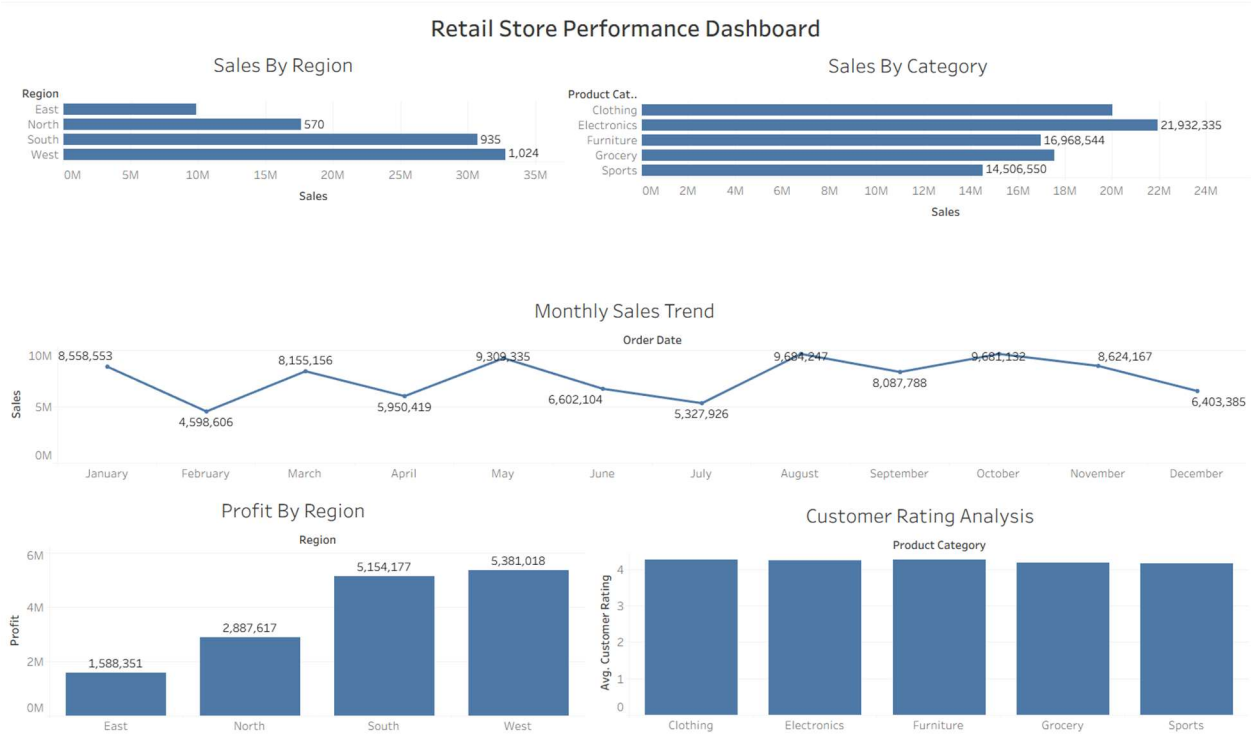
RETAIL STORE PERFORMANCE DASHBOARD

Region Filter Product Category Filter

Sales by Region	**Sales by Category**
Regional Sales	Category Sales
Monthly Sales Trend	
Monthly Sales Analysis	
Profit by Region	**Customer Rating Analysis**
Regional Profit	Customer Ratings

Use **Horizontal Containers** and **Vertical Containers** where required to organize the dashboard properly.

Expected Output:-



PART B: ADD FILTERS AND INTERACTIVE CONTROLS

STEP 5: ADD REGION FILTER

Step 1:-
Open any worksheet, such as **Sales by Region**.

Step 2:-
Drag **Region** to the **Filters** shelf.

Step 3:-
Keep all values selected.

Step 4:-
Right-click **Region** on the Filters shelf.

Step 5:-
Select: **Show Filter**
The Region filter will appear on the worksheet.

STEP 6: APPLY THE FILTER TO ALL WORKSHEETS

On the Region filter card:

1. Click the small drop-down arrow.
2. Select **Apply to Worksheets**.
3. Select **All Using This Data Source**.

Now the Region filter will control all worksheets that use the same data source.

Example

If **North** is selected, all charts in the dashboard will display data related to the North region.

STEP 7: ADD PRODUCT CATEGORY FILTER

Repeat the same process for Product Category.

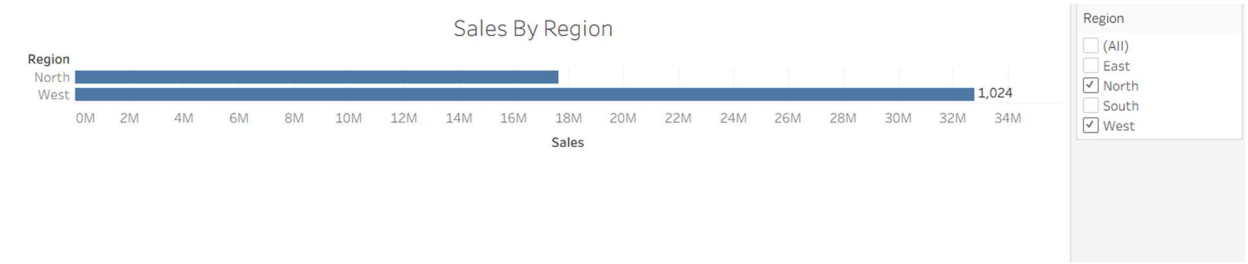
1. Drag **Product Category** to **Filters**.
2. Select **Show Filter**.
3. On the filter card, select:

Apply to Worksheets → All Using This Data Source

The dashboard will now contain two interactive filters:

- Region
- Product Category

Expected Output:-



PART C: HIGHLIGHT ACTION

A Highlight Action highlights related data without completely removing the other data from the visualization.

STEP 8: CREATE HIGHLIGHT ACTION

Step 1:-

Go to the dashboard.

Step 2:-

Select: **Dashboard → Actions**

Step 3:-

Click: **Add Action → Highlight**

Step 4:-

Configure the action as follows:

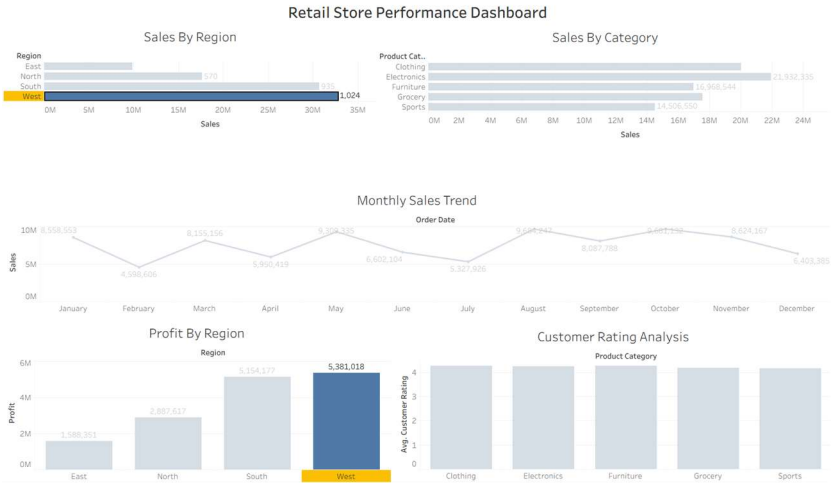
Option	Selection
Source Sheet	Sales by Region
Target Sheets	All other sheets
Run action on	Select
Clearing Selection	Show All Values

Step 5:-

Click **OK**.

Result:-

When a region such as **West** is selected in the Sales by Region chart, related information in the other charts will be highlighted.



PART C: FILTER ACTION

A Filter Action filters the data displayed in other worksheets based on the user's selection.

STEP 9: CREATE FILTER ACTION

Step 1:-

Go to: **Dashboard → Actions**

Step 2:-

Click: **Add Action → Filter**

Step 3:-

Configure the action as follows:

Option	Selection
Source Sheet	Sales by Region
Target Sheets	Sales by Category, Monthly Sales Trend, Profit by Region, Customer Rating Analysis
Run action on	Select
Clearing Selection	Show All Values

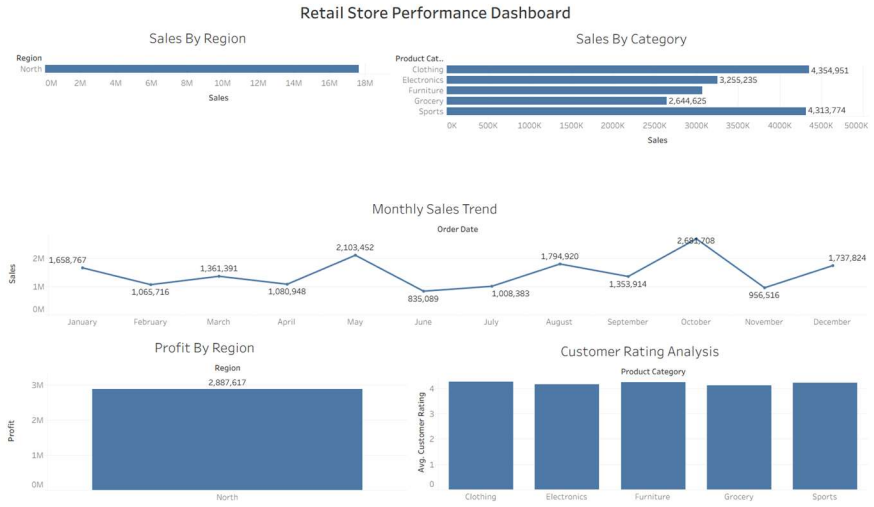
Step 4:-

Click **OK**.

Result:-

If **North** is selected in the Sales by Region chart, the other charts will display only North Region data.

To return to the complete dataset, click in an empty area of the visualization.



PART D: URL ACTION

A URL Action can be used to open an external webpage when a user selects data in a dashboard.

STEP 10: CREATE URL ACTION

Step 1:-

Go to: **Dashboard → Actions**

Step 2:-

Select: **Add Action → Go to URL**

Step 3:-

Enter the action name: **Open Retail Website**

Step 4:-

Select the source sheet: **Sales by Category**

Step 5:-

Enter the website URL: **https://www.tableau.com**

Step 6:-

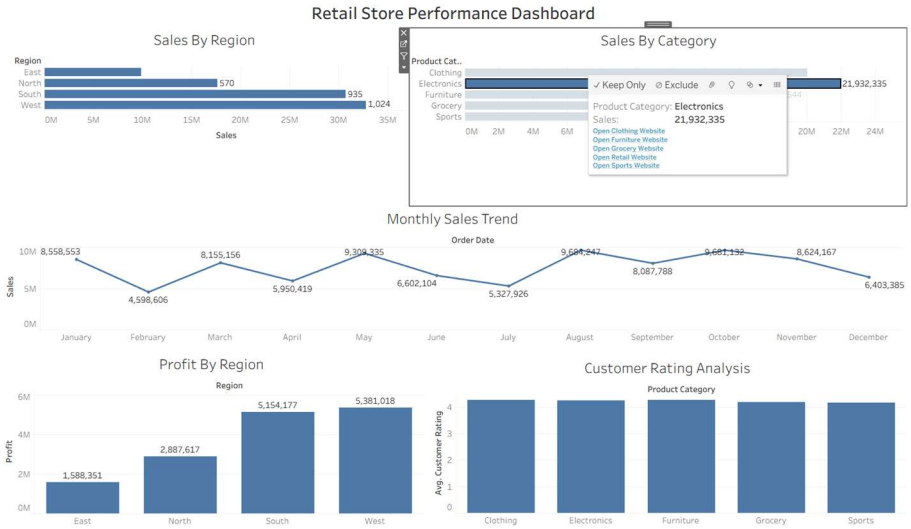
Set: **Run action on → Select**

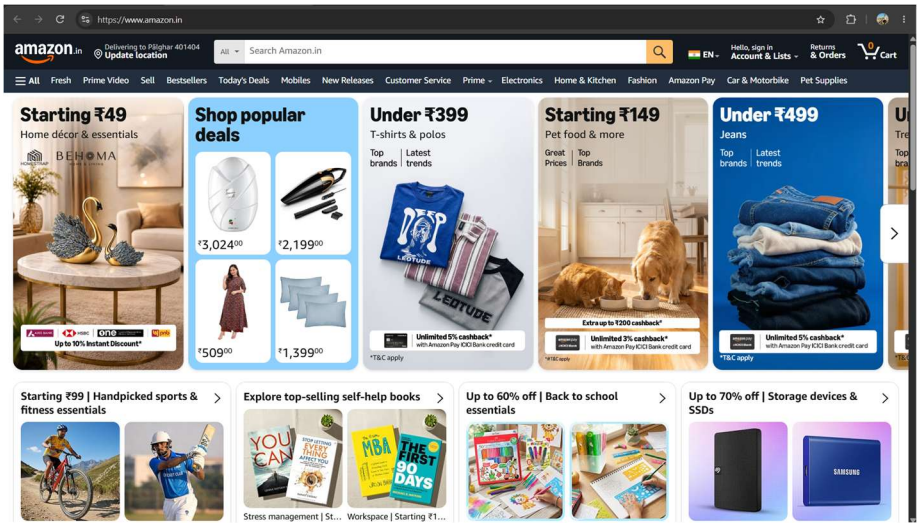
Step 7:-

Click **OK**.

Result:-

When a data item is selected in the selected worksheet, Tableau can open the specified webpage.





PART D: SET ACTION

A Set Action allows users to dynamically change the values contained in a Tableau set. In this practical, a set will be created for **Product Category**.

STEP 11: CREATE PRODUCT CATEGORY SET

Step 1:-

Open the **Sales by Category** worksheet.

Step 2:-

Right-click **Product Category**.

Step 3:-

Select: **Create → Set**

Step 4:-

Enter the set name: **Selected Product Category**

Step 5:-

Select all values.

Step 6:-

Click **OK**.

STEP 12: USE THE SET FOR COLOR

Step 1:-
Drag **Selected Product Category** to **Color** on the Marks card.
Selected and non-selected categories will now have different appearances.

STEP 13: CREATE SET ACTION

Step 1:-
Return to the dashboard.

Step 2:-
Go to:
Dashboard → Actions

Step 3:-
Select: **Add Action → Change Set Values**

Step 4:-
Configure the action:

Option	Value
Source Sheet	Sales by Category
Target Set	Selected Product Category
Run action on	Select
Clearing Selection	Keep Set Values

Step 5:-
Click **OK**.

Result:-
When a product category is selected, the **Selected Product Category** set will change dynamically.

PART E: CREATE A DYNAMIC DASHBOARD USING A PARAMETER

The parameter will allow the user to switch between:

- Sales
- Profit
- Sales Target

STEP 14: CREATE A PARAMETER

Step 1:-

In the Data Pane, right-click in an empty area.

Step 2:-

Select: **Create Parameter**

Step 3:-

Enter the following settings:

Setting	Value
Name	Select Measure
Data Type	String
Allowable Values	List

Step 4:-

Add the following values to the list:

- Sales
- Profit
- Sales Target

Step 5:-

Click **OK**.

Step 6:-

Right-click **Select Measure** and choose:
Show Parameter
The parameter control will now be visible.

STEP 15: CREATE A CALCULATED FIELD

Step 1:-
Right-click in the Data Pane.

Step 2:-
Select: **Create Calculated Field**

Step 3:-
Name the calculated field: **Dynamic Measure**

Step 4:-
Enter the following formula:

```
CASE [Select Measure]

WHEN "Sales" THEN [Sales]

WHEN "Profit" THEN [Profit]

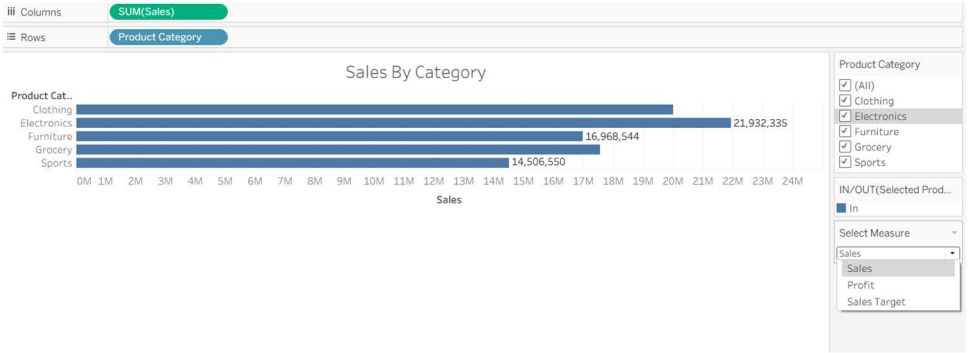
WHEN "Sales Target" THEN [Sales Target]

END
```

Step 5:-
Click **OK**.

TEST THE PARAMETER

Select **Sales**.
The chart will display Sales values.
Select **Profit**.
The chart will automatically change to Profit values.
Select **Sales Target**.
The chart will display Sales Target values.
This demonstrates how a parameter can dynamically change the measure used in a visualization.



STEP 16: CREATE A DYNAMIC CHART

Step 1:-
Create a new worksheet.

Step 2:-
Rename it: **Dynamic Performance Analysis**

Step 3:-
Drag **Region** to **Columns**.

Step 4:-
Drag **Dynamic Measure** to **Rows**.

Step 5:-
Select **Bar Chart**.

Step 6:-
Keep the **Select Measure** parameter visible.
The user can now switch between:
Sales → Profit → Sales Target
The chart will automatically update according to the selected measure.

Step 7:-
Add the **Dynamic Performance Analysis** worksheet to the dashboard.

PART F: DESIGN DASHBOARDS FOR MULTIPLE DEVICES
Tableau allows dashboards to be designed for different screen sizes and devices.
The dashboard can be customized for:

- Desktop
- Tablet
- Phone

STEP 17: CREATE A PHONE LAYOUT

Step 1:-
Open the **Retail Store Performance Dashboard**.

Step 2:-
Open **Device Preview** or **Device Designer**, depending on the Tableau version.

Step 3:-
Select: **Add Phone Layout**

Step 4:-
Arrange the charts vertically.

Suggested Phone Layout
RETAIL STORE PERFORMANCE
Region Filter
Product Category Filter
Sales by Region
Sales by Category
Monthly Sales Trend
Profit by Region
The charts should be arranged vertically so that they are easy to view on a mobile screen.

Other Device Layouts
Similarly, create layouts for:

- Desktop
- Tablet
- Phone

This ensures that the dashboard remains readable and interactive on different devices.



PART G: CREATE A COMPLETE DATA STORY

Tableau's **Story** feature can be used to present the analysis step by step.
The following story points will be created:

1. Overall Retail Performance
2. Regional Sales Analysis
3. Product Category Analysis
4. Monthly Sales Trend
5. Profit Analysis
6. Customer Rating Analysis
7. Dynamic Performance Analysis
8. Final Insights and Conclusion

STEP 18: CREATE A NEW STORY

Step 1:-

At the bottom of Tableau, click the **New Story** icon.

Step 2:-

Rename the story: **Retail Store Performance Story**

Step 3:-

Set the story size to:
Automatic
Or
Desktop Browser

STORY POINT 1: OVERALL RETAIL PERFORMANCE

Step 1:-

Drag the complete **Retail Store Performance Dashboard** into the Story.

Step 2:-

Set the title: **Overall Retail Store Performance**

Caption

This dashboard provides an overall analysis of retail store sales, profit, product categories, customer ratings, and regional performance.

STORY POINT 2: REGIONAL SALES ANALYSIS

Step 1:-
Click **New Blank Point**.

Step 2:-
Drag **Sales by Region** into the Story.
Title: Regional Sales Performance

Caption:-
This analysis compares sales across different regions and helps identify the region with the strongest retail performance.

STORY POINT 3: PRODUCT CATEGORY ANALYSIS

Step 1:-
Click **New Blank Point**.

Step 2:-
Drag **Sales by Category** into the Story.
Title:- Product Category Performance

Caption:-
This analysis compares sales and profit across different product categories to identify high-performing categories.

STORY POINT 4: MONTHLY SALES TREND

Step 1:-
Create a new Story Point.

Step 2:-
Drag **Monthly Sales Trend** into the Story.
Title:- Monthly Sales Trend Analysis

Caption:-
This visualization shows how retail sales change over time and helps identify periods of higher and lower sales performance.

STORY POINT 5: PROFIT ANALYSIS

Step 1:-
Create another Story Point.

Step 2:-
Drag **Profit by Region** into the Story.
Title:- Regional Profit Analysis

Caption:-
This analysis compares profit generated by different regions and identifies the most profitable areas.

STORY POINT 6: CUSTOMER RATING ANALYSIS

Step 1:-
Create a new Story Point.

Step 2:-
Drag **Customer Rating Analysis** into the Story.
Title:- Customer Satisfaction Analysis

Caption:-
This analysis compares average customer ratings across product categories to understand customer satisfaction.

STORY POINT 7: DYNAMIC PERFORMANCE ANALYSIS

Step 1:-
Create another Story Point.

Step 2:-
Add the **Dynamic Performance Analysis** worksheet.
Title:- Dynamic Retail Performance

Step 3:-
Use the parameter to demonstrate switching between:

- Sales
- Profit
- Sales Target

Caption:-

The parameter allows users to dynamically change the measure and analyze different aspects of retail store performance.

FINAL STORY POINT: CONCLUSION

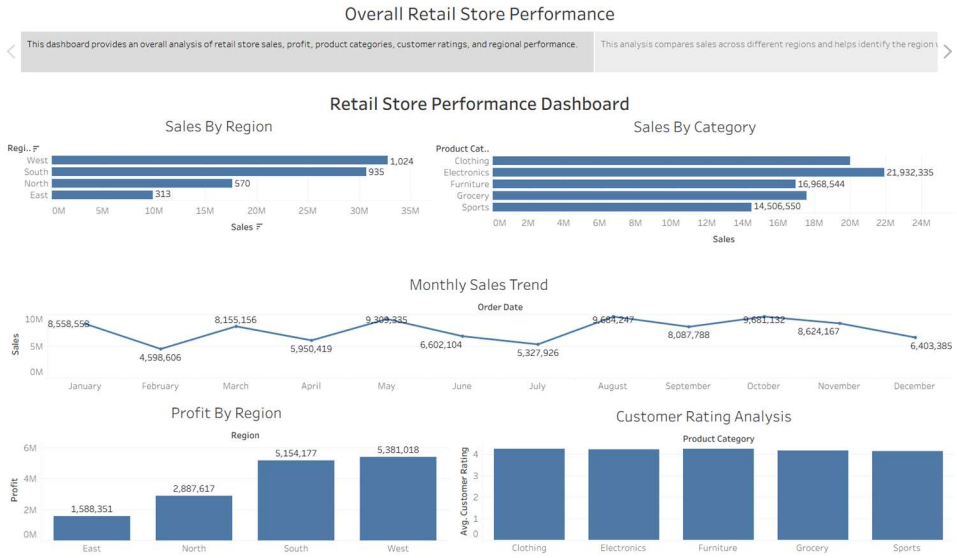
Step 1:-

Add the complete dashboard again or create a summary view.
Title:-Final Insights and Conclusion

Conclusion:-

The Retail Store Performance dashboard provides an interactive analysis of regional sales, product category performance, profit, customer ratings, and sales trends. Interactive filters, Highlight Actions, Filter Actions, Set Actions, URL Actions, and parameters allow users to explore the data dynamically. The Story feature presents the complete analysis step by step and helps users understand important business insights.

Excepted Output:-



OVERALL WORKFLOW:

Retail Store Performance Dataset
↓
Worksheets

1. Sales by Region
2. Sales by Category
3. Monthly Sales Trend
4. Profit by Region
5. Customer Rating Analysis
6. Dynamic Performance Analysis

↓
Interactive Dashboard

- Region Filter
- Product Category Filter
- Highlight Action
- Filter Action
- URL Action
- Set Action
- Parameter

↓
Device Layouts

- Desktop
- Tablet
- Phone

↓
Data Story

1. Overall Retail Performance
2. Regional Sales Analysis
3. Product Category Analysis
4. Monthly Sales Trend
5. Profit Analysis
6. Customer Rating Analysis
7. Dynamic Performance Analysis
8. Final Insights and Conclusion

Conclusion:-

This practical successfully demonstrated how to create an interactive retail store performance dashboard and data story using Tableau. Multiple worksheets were combined into a single dashboard, with filters and interactive controls added to make data analysis easier. Highlight, Filter, URL, and Set Actions were implemented to improve user interaction. Parameters were also used to create dynamic dashboards, while device-specific layouts ensured proper visualization on different screen sizes.

PRACTICAL NO. 1

Aim:- Getting started with Tableau and building basic visualization.

Objectives:-

The Objective of practical is to learn:

- How to connect data to Tableau.
- How Tableau organizes data.
- How to create different types of charts.
- How to build a dashboard using multiple charts.

Before starting, remember the workflow:

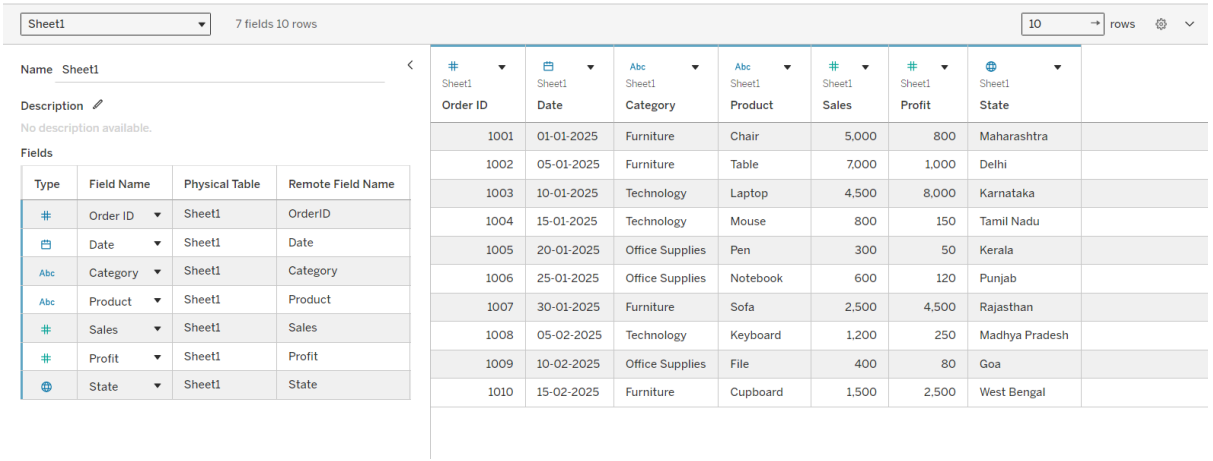
Excel/CSV Data
↓
Connect to Tableau
↓
Analyze Data
↓
Create Charts
↓
Build Dashboard

PRACTICAL NO. 1(A)

Aim:- Connect to a CSV Dataset and explore the Tableau interface.

Steps to Load a Dataset in Tableau Public:-

1. Open Tableau Public on your computer.
2. On the start page, click Connect to Data.
3. Select the appropriate data source (e.g., Microsoft Excel, Text File (CSV), JSON, or another supported format).
4. Browse to the location where your dataset is stored and select the file.
5. Click Open to import the dataset into Tableau Public.
6. The Data Source page will appear, where you can preview the data.



Sheet1 | 7 fields 10 rows | 10 rows

Name: Sheet1

Description: No description available.

Fields

Type	Field Name	Physical Table	Remote Field Name
#	Order ID	Sheet1	OrderID
📅	Date	Sheet1	Date
Abc	Category	Sheet1	Category
Abc	Product	Sheet1	Product
#	Sales	Sheet1	Sales
#	Profit	Sheet1	Profit
🌐	State	Sheet1	State

Order ID	Date	Category	Product	Sales	Profit	State
1001	01-01-2025	Furniture	Chair	5,000	800	Maharashtra
1002	05-01-2025	Furniture	Table	7,000	1,000	Delhi
1003	10-01-2025	Technology	Laptop	4,500	8,000	Karnataka
1004	15-01-2025	Technology	Mouse	800	150	Tamil Nadu
1005	20-01-2025	Office Supplies	Pen	300	50	Kerala
1006	25-01-2025	Office Supplies	Notebook	600	120	Punjab
1007	30-01-2025	Furniture	Sofa	2,500	4,500	Rajasthan
1008	05-02-2025	Technology	Keyboard	1,200	250	Madhya Pradesh
1009	10-02-2025	Office Supplies	File	400	80	Goa
1010	15-02-2025	Furniture	Cupboard	1,500	2,500	West Bengal

PRACTICAL NO. 1(B)

Aim:- Identify and differentiate Measures and Dimensions.

Steps to Check Measures and Dimensions in Tableau Public

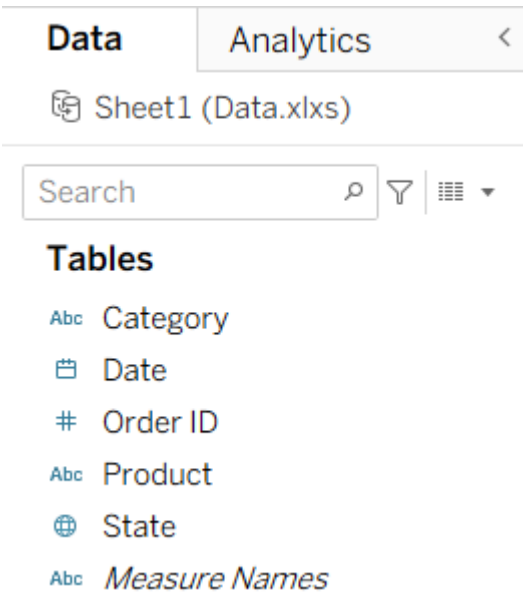
1. Open Tableau Public and load your dataset.
2. Navigate to the Worksheet (Sheet 1).
3. Look at the Data Pane on the left side of the screen.
4. The fields are automatically divided into two sections:

Usually Dimension are in blue color.

Example:

1) Category
2) Date
3) Order ID
4) Product
5) State

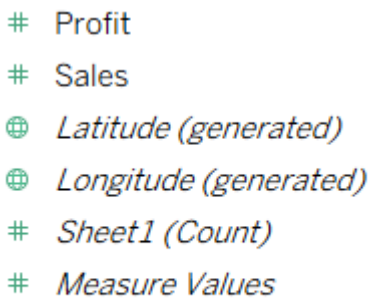
These describes the data.



Measures are generally green in color.

Example:

1) Profit
2) Sales



These measures can be calculated.

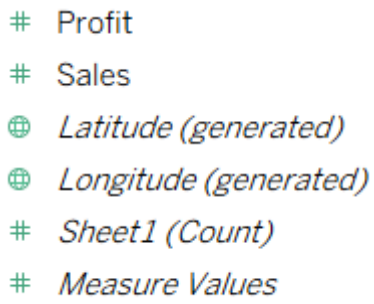
Example:

1) SUM(Sales)
2) AVG(Profit)
3) COUNT(Order ID)

PRACTICAL NO. 1(C)

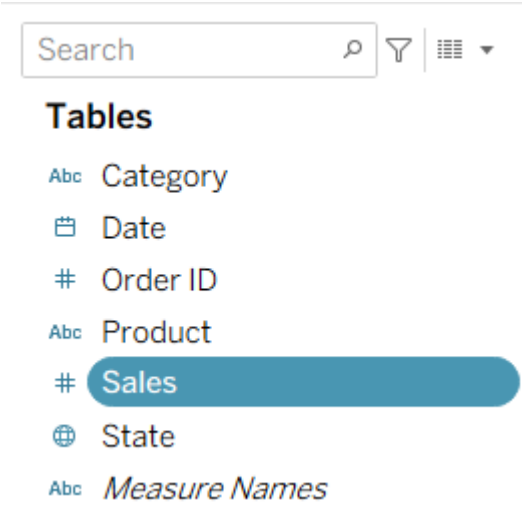
Aim:- Convert fields between discrete and continuous and observe changes.

Suppose “Sales” is under Dimensions



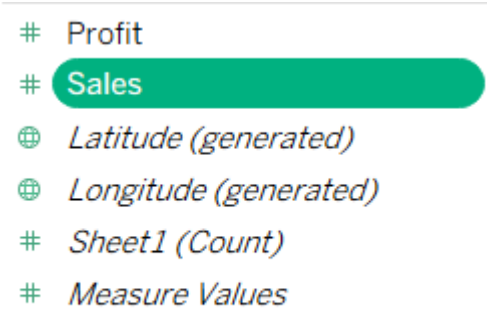
Convert Measures to Dimension:

1) Right - Click on Sales.
2) Click
3) Click on Convert to Dimension.
4) Now it moves under Dimension.



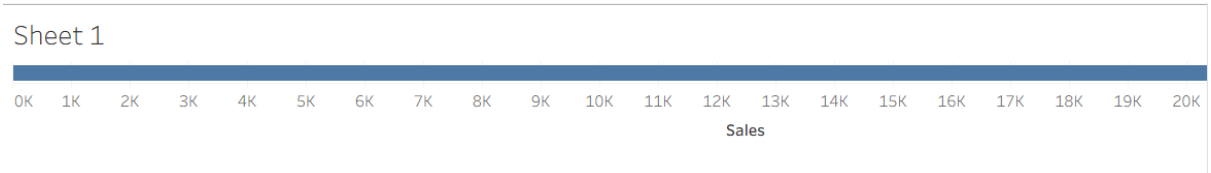
Now, Convert Dimension back to Measures:

1) Right Click on Sales.
2) Click.
3) Click on convert to Measures.
4) Observe.
5) It moves again to Measures.



Continuous & Discrete:

1) Drag Sales into Column.
2) Click the small arrow on Sales.
3) Click on Continuous
4) Observe
5) Green Pill
6) Axis becomes Continue



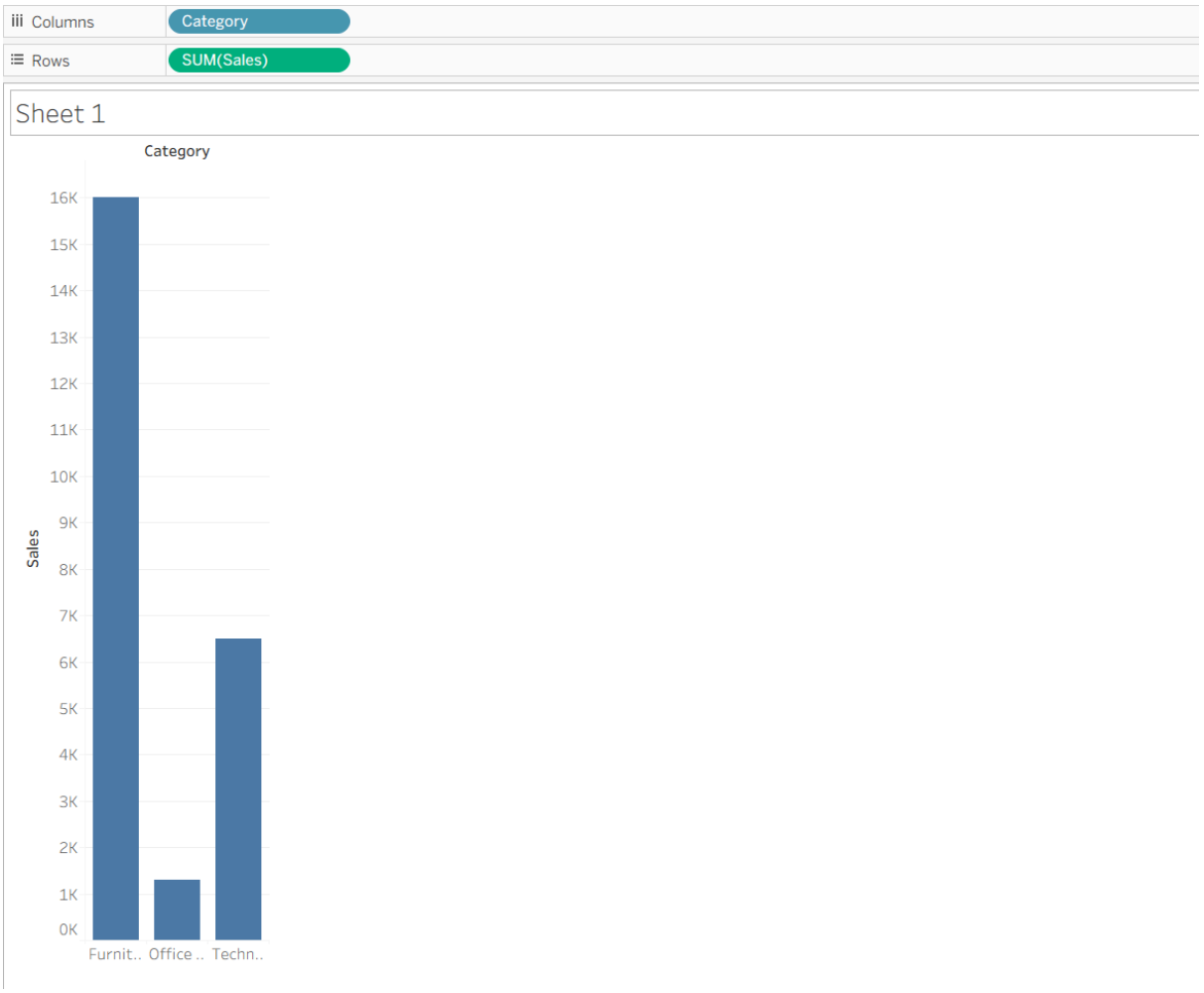
7) Now click on Discrete
8) Observe
9) Blue Pill
10) Separate header appear



PRACTICAL NO. 1(D)

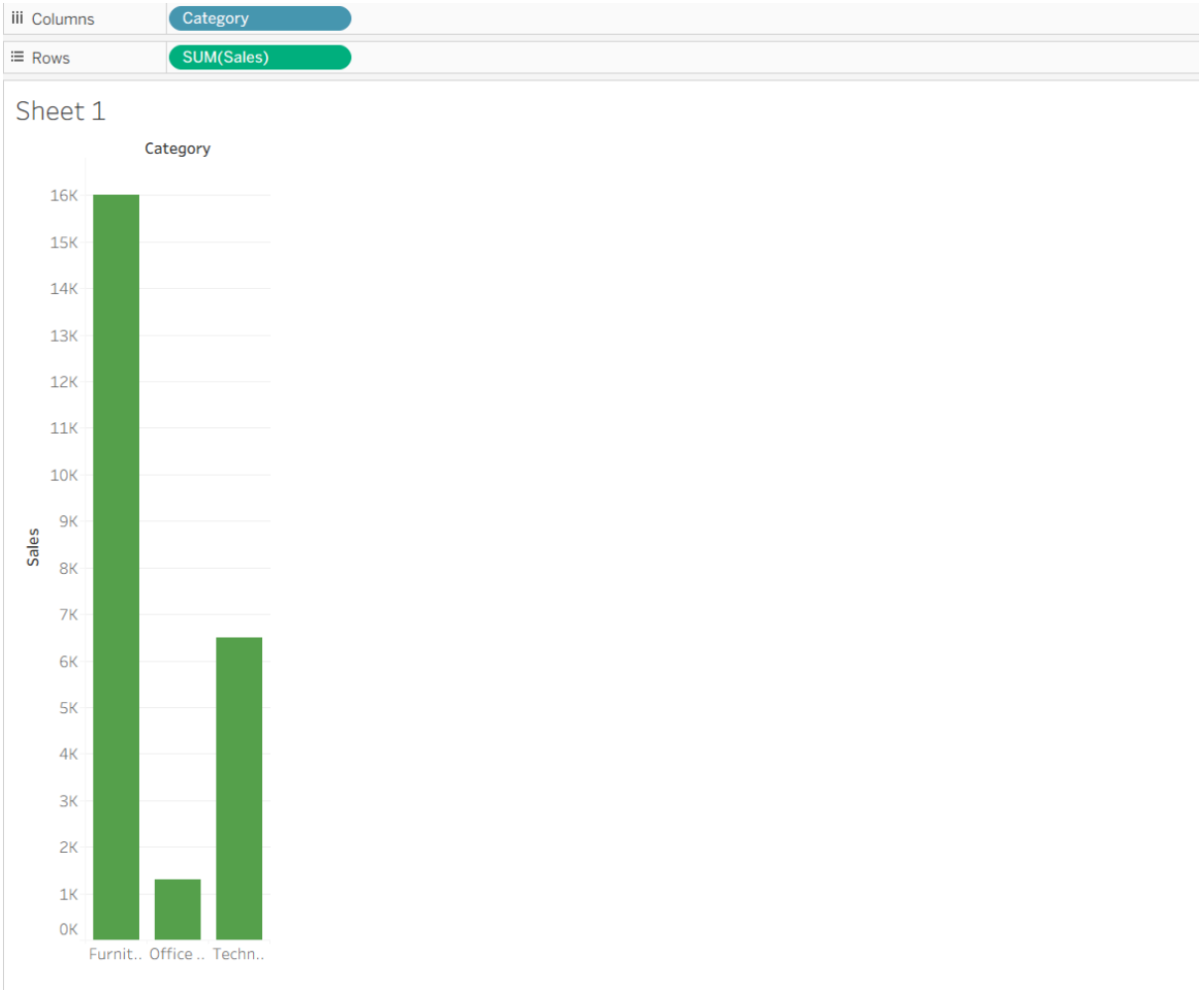
Aim:- Create and customize a basic bar chart.

Drag “Category” to Column and Drag “Sales” to row immediately bars appear.



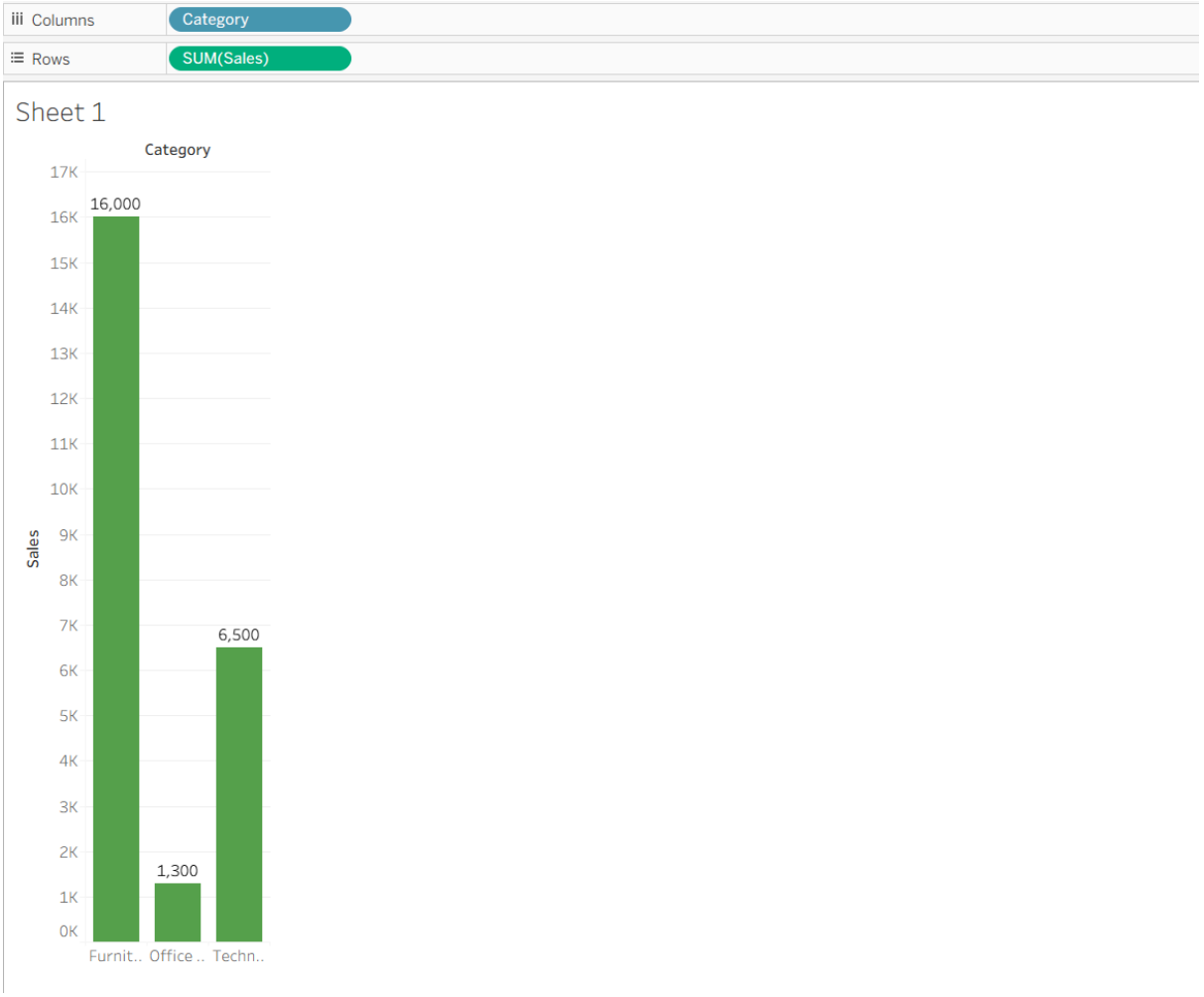
Customize:-

1) Click
2) Color
3) Choose any Color



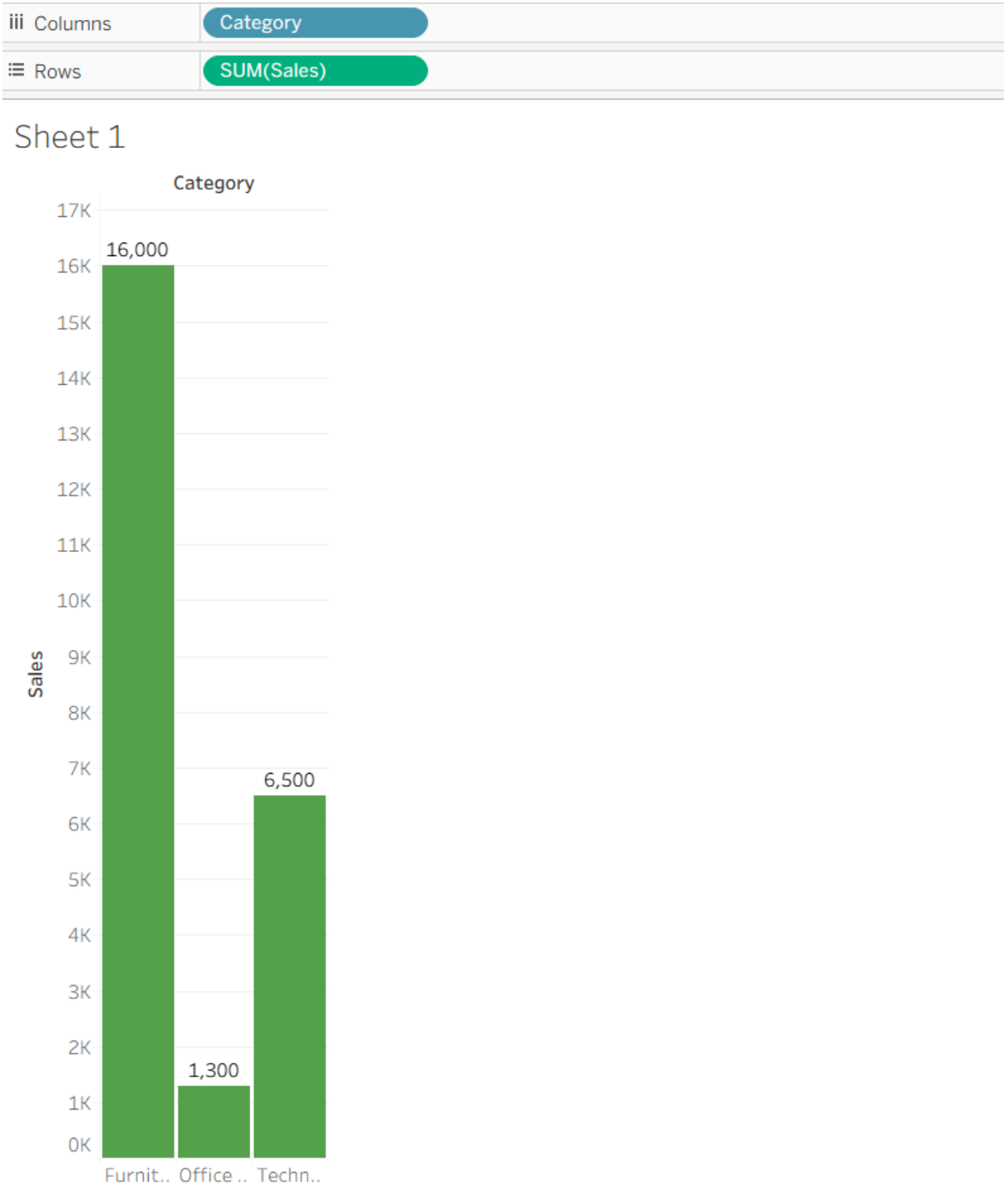
Add MarkUp Labels:-

1) Click
2) Label
3) Check
4) Show Mark Labels
5) Sales value appears



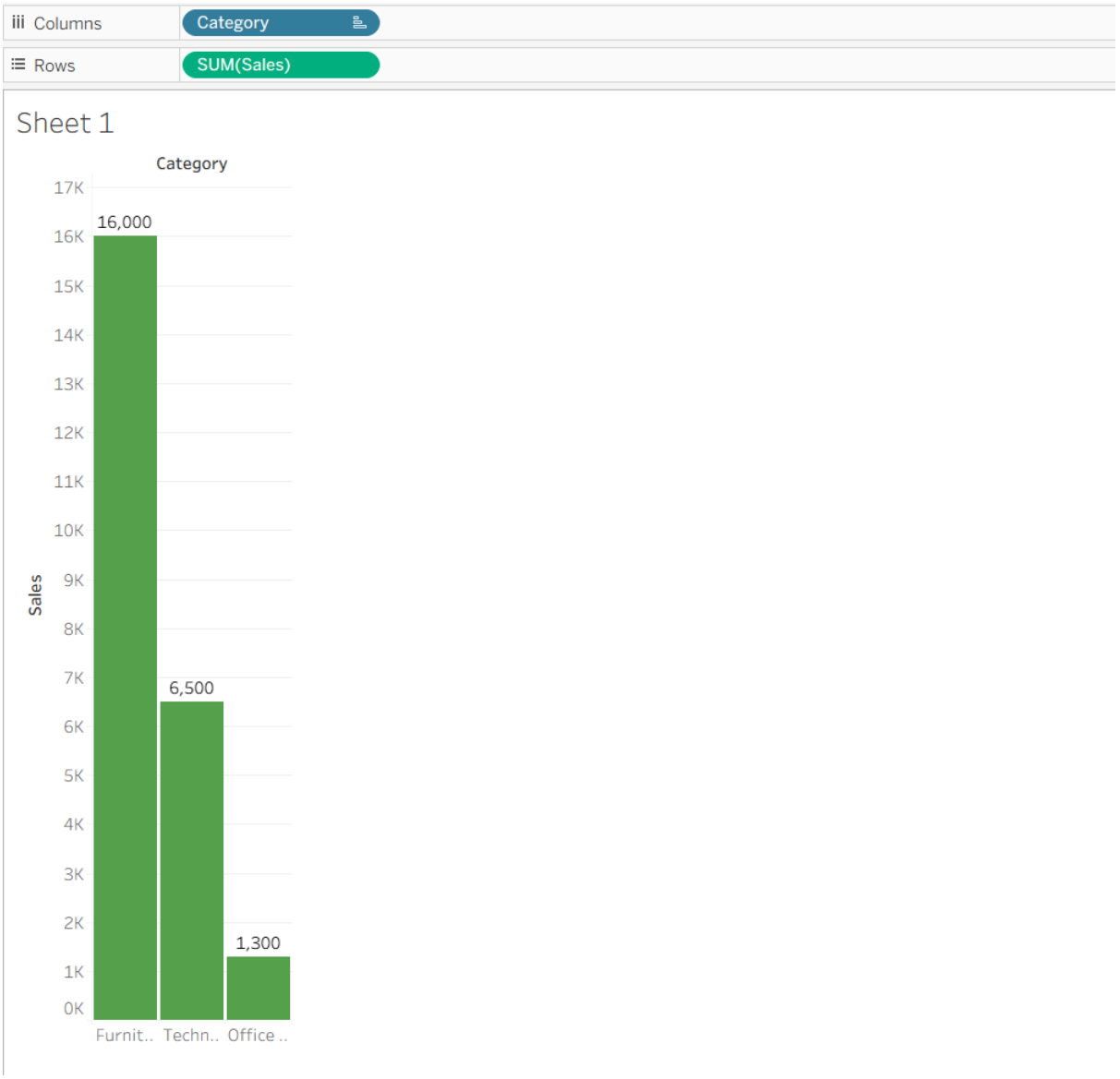
Increase Size:-

1) Click
2) Size
3) Increase Size



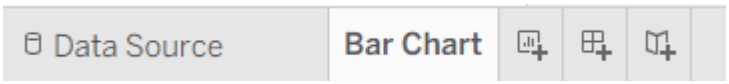
Sorting:-

1) Click
2) Sort Descending
3) Bars arrange from highest to lowest



Rename Sheet:-

1) Double Click on “Sheet1”.
2) Click on rename
3) Enter Name “Bar Chart”.



PRACTICAL NO. 1(E)

Aim:- Build a line chart for time-series analysis.

1) Open new Worksheet.
2) Drag Date to Column.
3) If Tableau asks Year, Month, Day choose Month.
4) Now drag Sales to choose mark line drag Sales to Rows shelf.
5) Now monthly trend appears.



PRACTICAL NO. 1(F)

Aim:- Create a geographic map visualization.

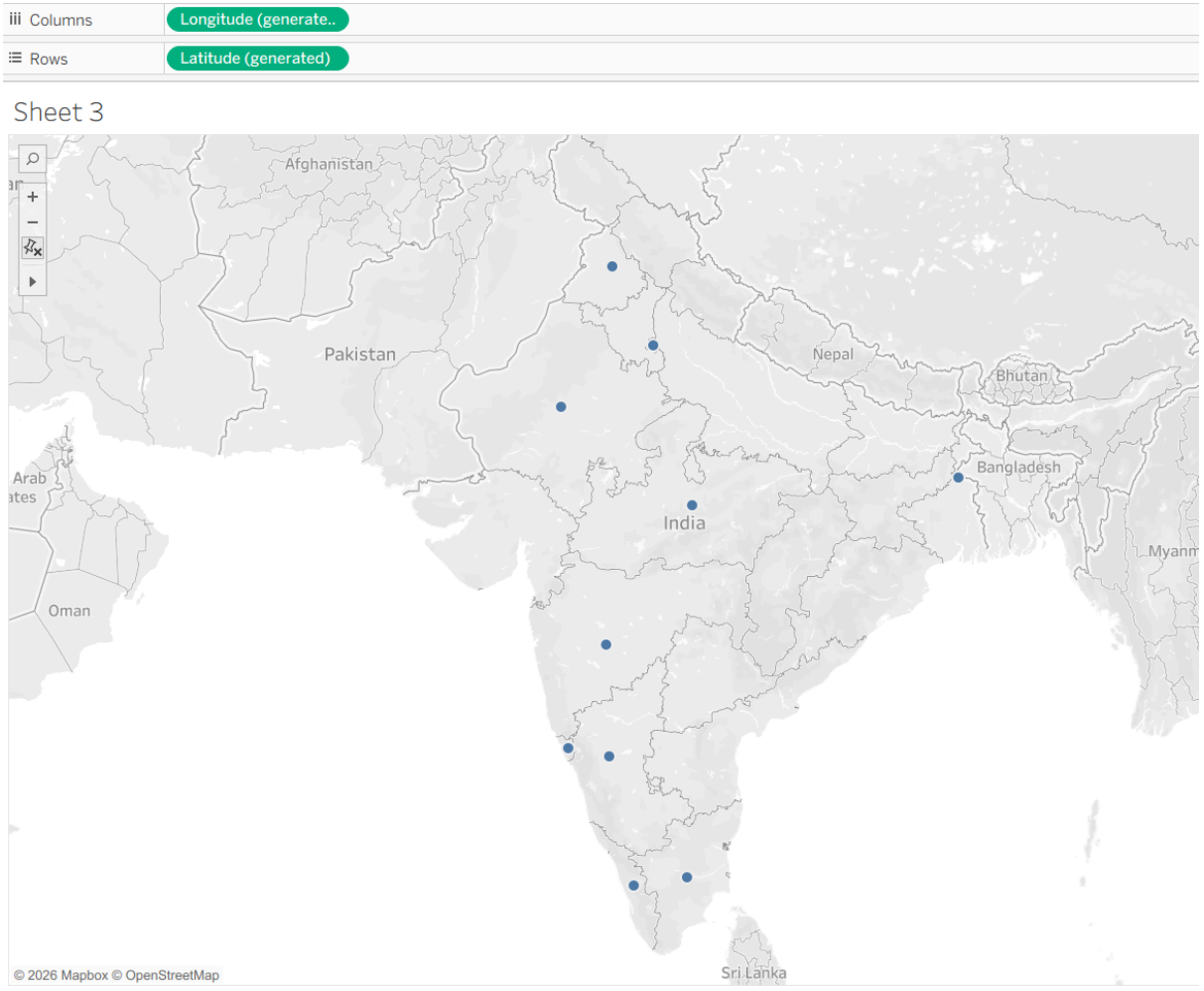
1) Open new worksheet.
2) Drag State to worksheet.
3) If Tableau recognizes state, it creates latitude and longitude automatically.

If not:

1) Right click on State
2) Click on geographic role
3) Then click on State/Province
4) Then double state.

Step 1: Create the Map

1) Drag State to worksheet
2) Tableau recognizes it as geographic field and display a map.
   (It it dosen't right click on State → Geographic Role → State/Province)

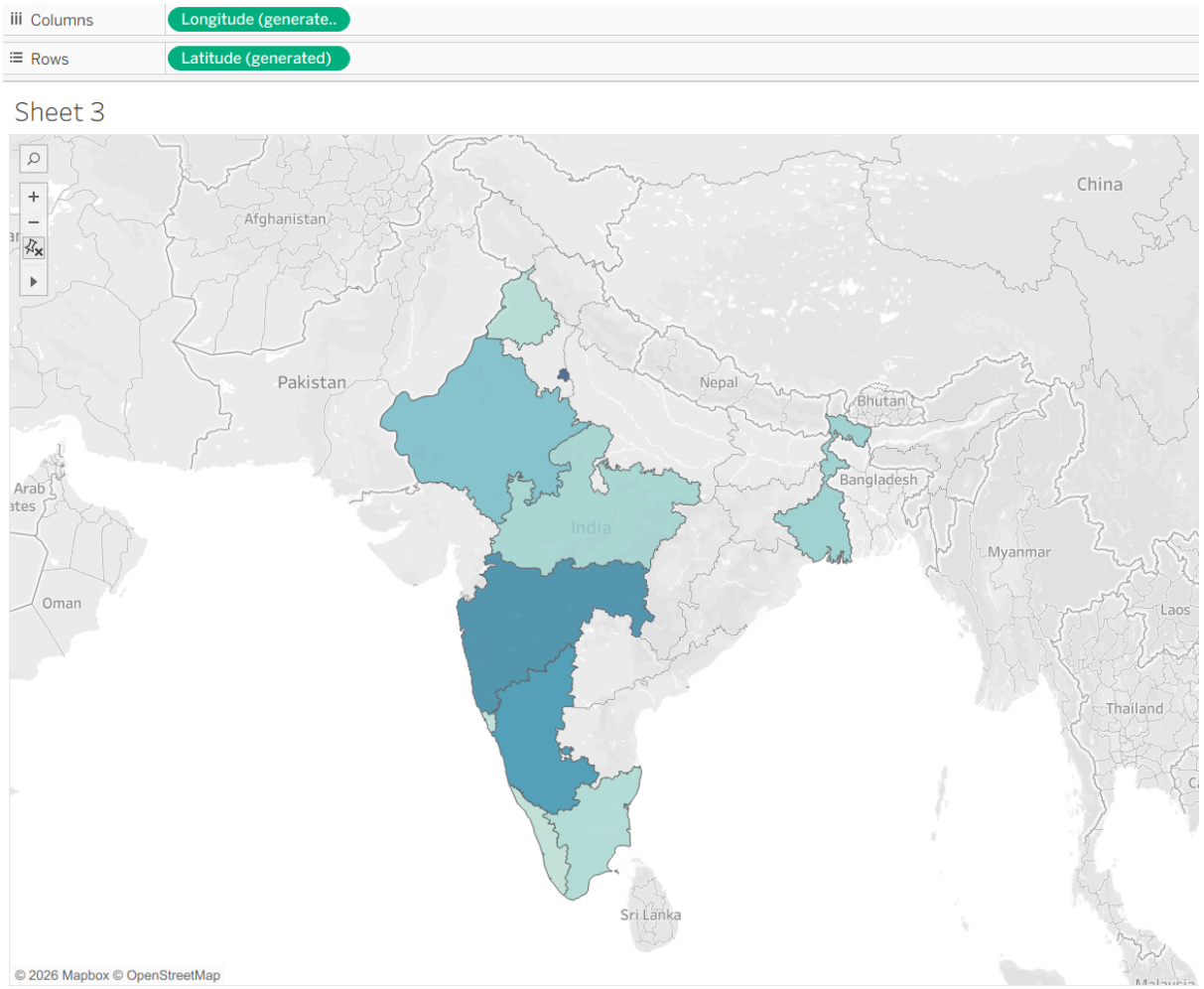


Step 2: Drag Sales to Color

1) In the data pane(left side), find Sales.
2) Click and hold Sales.
3) Drag it to color option on the mark card.
4) Release the mouse.

The Map color will changes automatically:

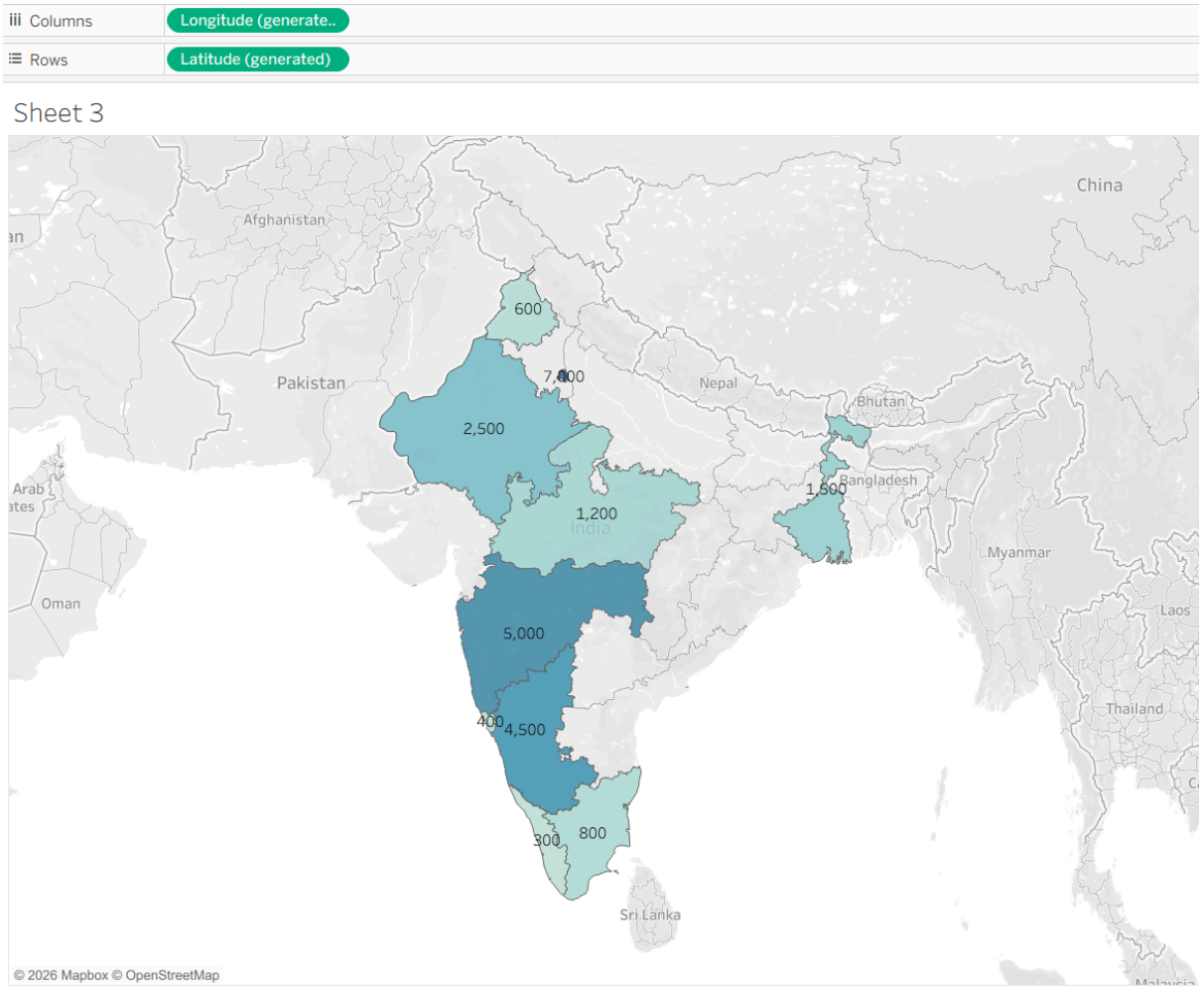
1) Darker color = Higher Sales
2) Lighter color = Lower Sales



Step 3: Drag Sales to Label

1) Again, click and hold Sales from the data pane.
2) Drag it to label on the mark card.
3) Release the mouse.

Each state on the map will display its value.



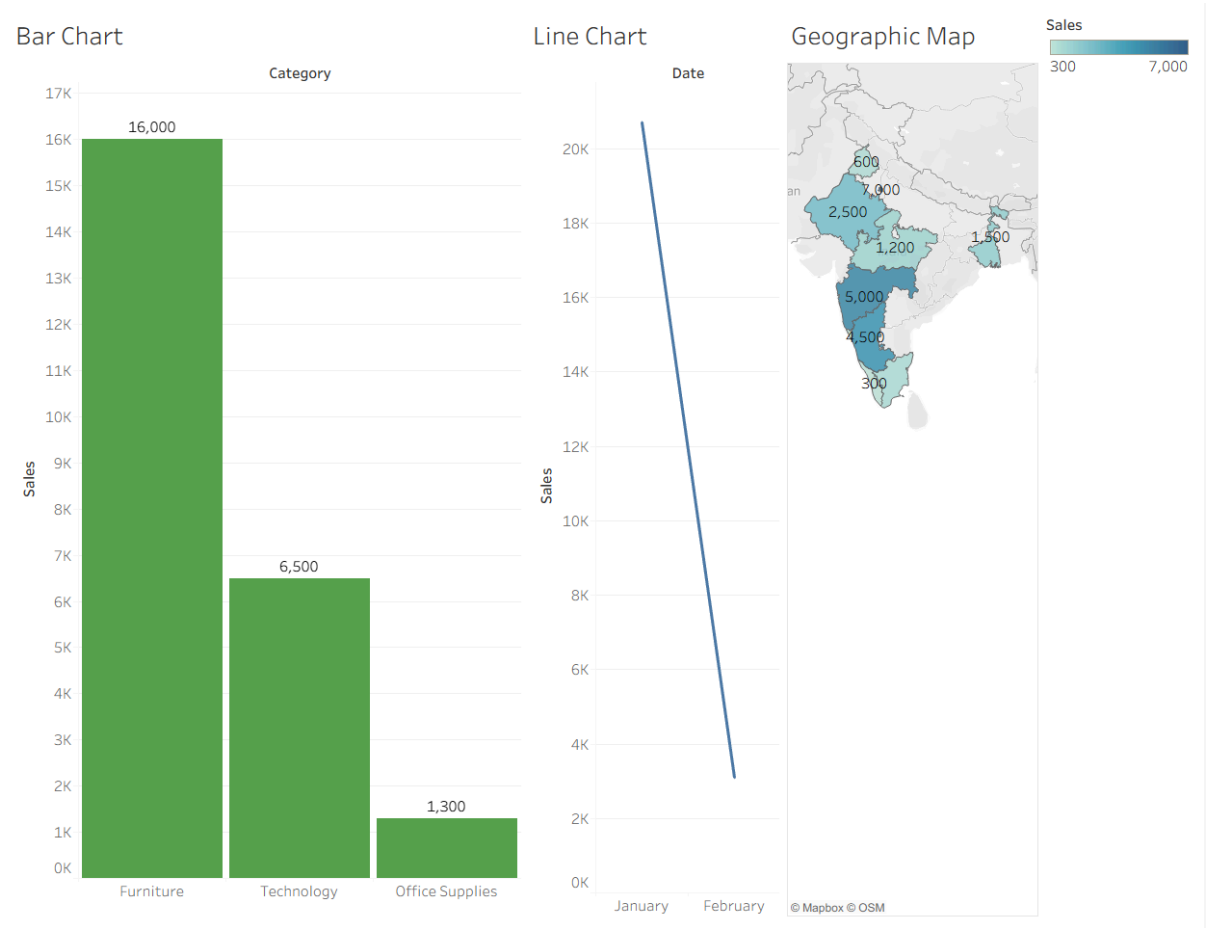
PRACTICAL NO. 1(F)

Aim:- Combine multiple charts into simple interactive dashboard.

Create Dashboard:

1) Click on New Dashboard
2) (Bottom tab with dashboard icon)
3) Empty dashboard opens
4) Left side contains all sheets
5) Drag all those sheets in dashboard
    a) Bar Chart
    b) Line Chart
    c) Geographic Map

Arrange them neatly.



Conclusion:

In this practical, we learned the fundamentals of using Tableau for data visualization and analysis. We successfully connected a CSV dataset, explored the Tableau interface, and understood the difference between Measures and Dimensions. We also observed how converting fields between discrete and continuous affects the visualization. Various charts, including bar charts, line charts, and geographic maps, were created and customized to represent data effectively. Finally, multiple visualizations were combined into an interactive dashboard, enabling users to analyze data from different perspectives in a single view. This practical provided a strong foundation in Tableau and demonstrated how visual analytics can transform raw data into meaningful insights.

PRACTICAL NO.3(A)

Aim:- Create stacked and grouped bar charts for comparision.

Dataset:

Abc Sheet1 Student ID	Abc Sheet1 Department	Abc Sheet1 Activity	# Sheet1 Hours	# Sheet1 Attendance	# Sheet1 Score	Abc Sheet1 Activity Date
S113	BSc IT	Sports	13	94	91	22-08-2024
S114	BSc CS	NSS	15	81	56	28-08-2024
S115	Bcom	Music	17	97	78	5/9/2024
S116	BSc IT	Drama	8	88	86	12/9/2024
S117	BSc CS	Sports	10	83	48	18-09-2024
S118	Bcom	NSS	16	95	57	25-09-2024
S119	BSc IT	Music	12	90	68	2/10/2024
S120	BSc CS	Drama	6	84	95	10/10/2024
S121	Bcom	Sports	9	91	84	18-10-2024
S122	BSc IT	NSS	10	77	76	22-10-2024
S123	BSc CS	Music	15	48	53	20-11-2024
S124	Bcom	Drama	13	98	84	13-11-2024
S125	BSc IT	Sports	14	78	98	25-11-2024
S126	BSc CS	NSS	9	86	86	27-11-2024
S127	Bcom	Music	8	84	84	3/12/2024
S128	BSc IT	Drama	12	75	59	11/12/2024
S129	BSc CS	Sports	13	73	76	13-12-2024
S130	Bcom	NSS	12	87	86	20-12-2024
S131	BSc IT	Music	10	88	88	125-12-2024
S132	BSc CS	Drama	8	90	77	28-12-2024
S133	Bcom	Sports	8	95	84	1/1/2025
S134	BSc IT	NSS	9	96	75	12-01-205
S135	BSc CS	Music	10	80	66	23-01-2025

Objective:-
Compare total activity hours of each department, divided by activity type.

a.) What is stacked bar chart ?
A stacked bar chart is a type of bar chart in which each bar is divided into different colored segments. Each segment represents a category, and together they show both the total value and contribution of each category.

Steps:-

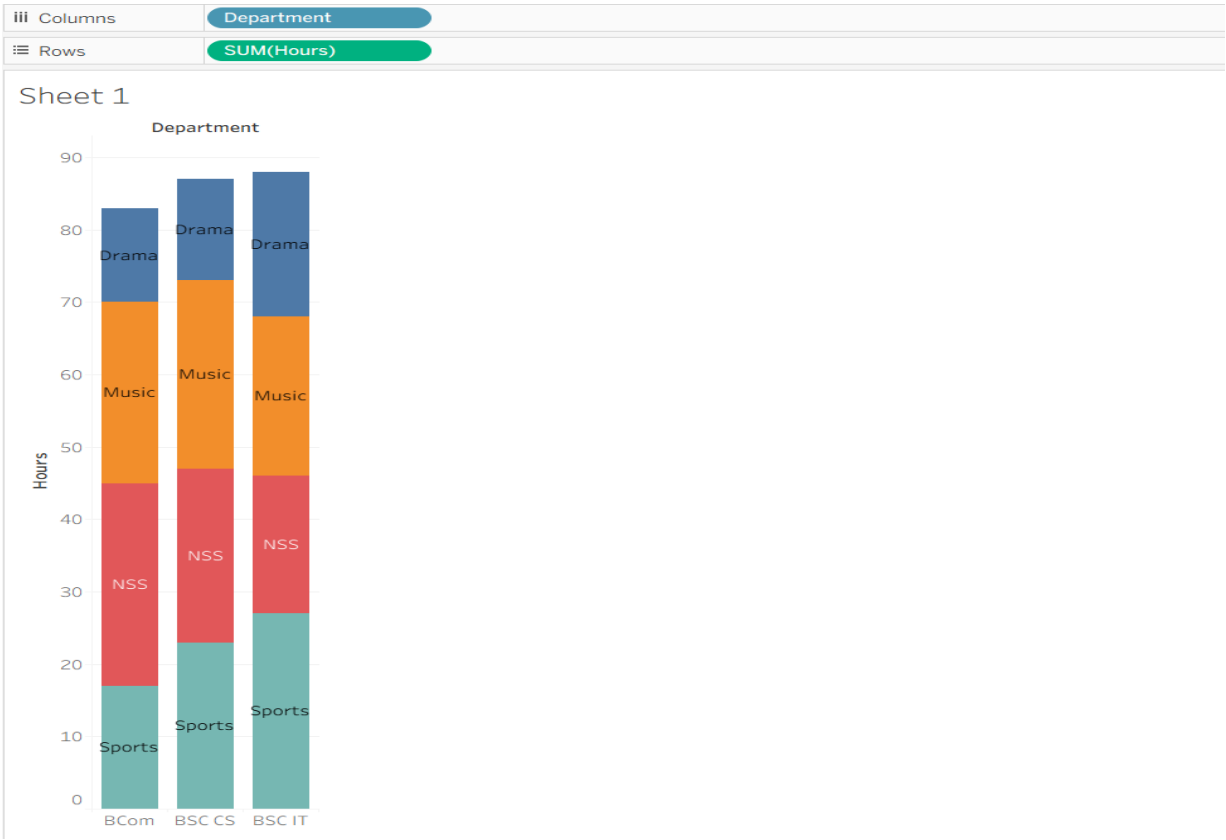
Step 1:
Drag **Department → Column**

Step 2:
Drag **Hours → Rows**
A bar chart appears

Step 3:
Drag **Activity → Color(Marks Card)**
Each department bar is divided into different colors.

Step 4:
If bars are not stacked
Click:
Analysis → Stack Marks → On

Output:-
Each department(BSC IT, BSC CS, BCom) shows total hours divided into sports, music, drama and NSS.



b.) What is grouped bar chart ?

A grouped bar chart is also called side-by-side bar chart, is a chart that displays multiple bar chart next to each other for each category. It is used to compare the values of different groups within the same category.

Objective:-
Compare Activity Hours department wise.

Steps:-

Step 1:
Open **new worksheet**

Step 2:
Drag **Department → Columns**

Step 3:
Drag **Activity → Column**
(Place after Department)

Step 4:
Drag **Hours → Rows**

Step 5:
Drag **Activity → Color**
Sepearte color appears for each activity.

Step 6:
Drag **Activity → Label**
Name of each activity appears

Output:-

Each department displays separate bars for Sports, Music, Drama & NSS.



PRACTICAL NO. 3(B)

Aim:- Build Treemaps and pie chart for part to whole analysis.

a.) What is a treemap ?

A treemap is a visualization that displays hierarchical or categorical data as nested rectangles. The size of each rectangle represents the value of measure (such as Hours or Score), while the color can represent another measure or category.

Objective:-
Show which activity takes more student hours.

Steps:-

Step 1:-
Open new worksheet

Step 2:-
Click → **Show Me**

Step 3:-
Click → **TreeMap**

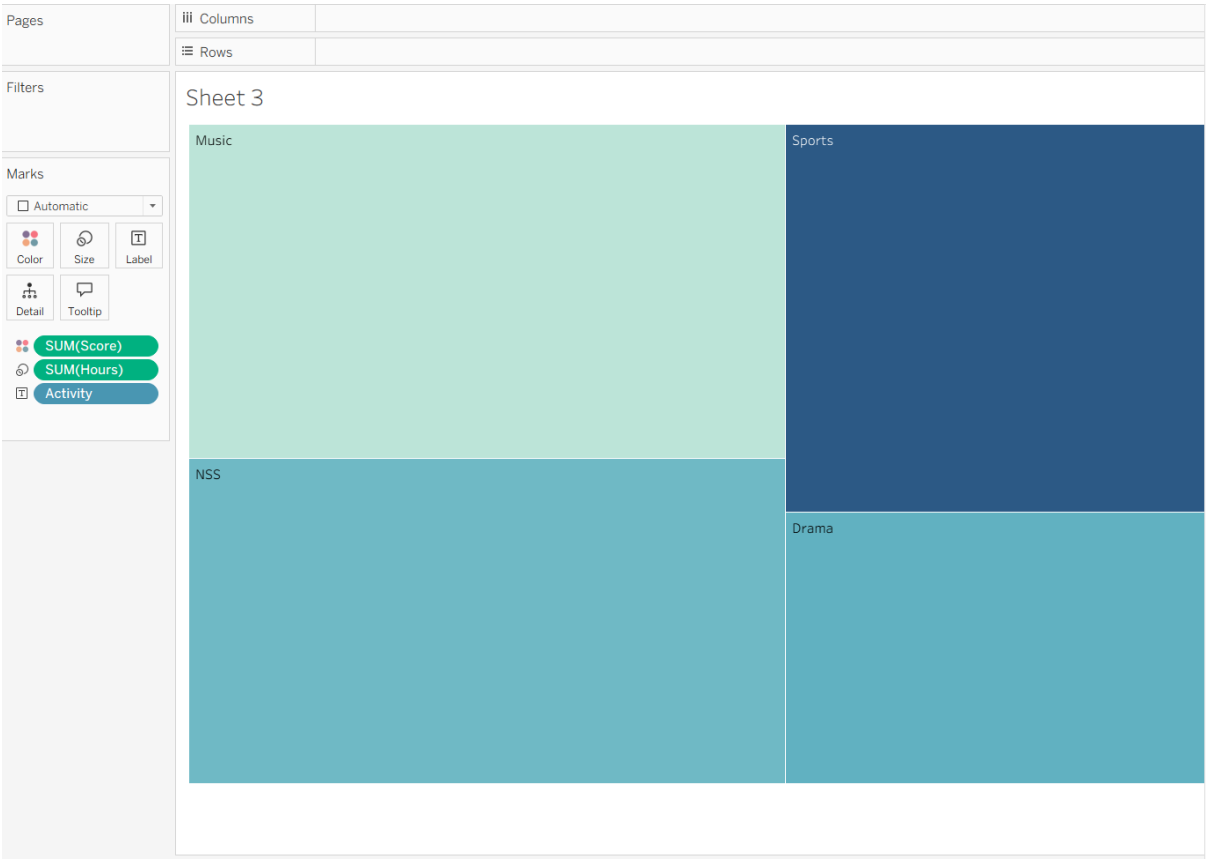
Step 4:-
Drag **Activity → Label**

Step 5:
Drag **Hours → Size**

Step 6:
Drag **Score → Color**

Output:-

Large rectangle represents activities with higher total hours.



b.) What is pie chart ?

A pie chart shows the relationship of parts to the whole for a categorical variable by depicting a circle, or pie, divided into segments. The size of each segment, or slice of the pie, represents the proportional contribution of a specific category to the whole.

Objective:-
Show percentage by activity.

Steps:-

Step 1:-
Open **new worksheet**

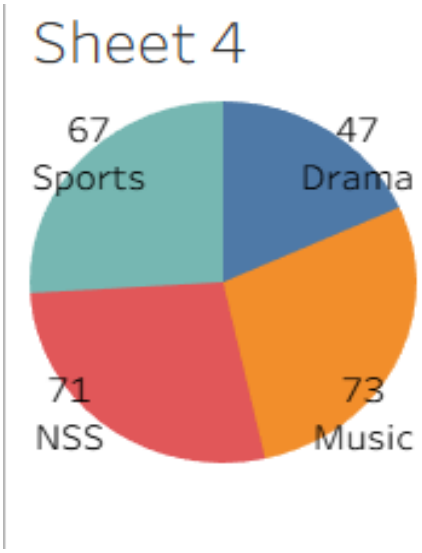
Step 2:
Choose → **PieChart**

Step 3:
Drag **Activity** → **Color**

Step 4:
Drag **Hours** → **Angle**

Step 5:
Drag **Hours** → **Label**

Output:-
Pie Chart showing each activity's share of total hours.



PRACTICAL NO. 3(C)

Aim:- Design Scatter plot to identify relationships.

a.) What is scatterplot ?

A Scatterplot is a graph that displays the relationship between two numerical variables using points. Each point represents one record in the dataset.

Objective:-
Find relationship between attendance and score.

Steps:-

Step 1:-
Open **new worksheet**

Step 2:-
Drag **Attendance → Column**

Step 3:-
Drag **Score → Rows**
An empty scatterplot appears on screen.

Step 4:-
Drag **Department → Color**
4 colors appears on screen each indicating specific department

Step 5:-
Drag **StudentID → Detail**
Multiple small circles appear on screen with specific color of their activity each representing a student

Output:-

Each student appears as a point

Student with higher attendance generally have higher score



PRACTICAL NO. 3(D)

Aim:- Create dual axis and combination chart.

b.) What is dual- axis chart ?

A Dual axis chart is a chart that displays two different measures on the same graph, each with its own axis. It allows you to compare two related measures over the same dimension(such as time).

Objective:-
Compare Hours and Score together.

Steps:-

Step 1:-
Open **new worksheet**

Step 2:-
Drag **Activity Date → Column**
(Choose Month)

Step 3:-
Drag **Hours → Rows**

Step 4:-
Drag **Score → Rows**
(Place besides hours)
Two graphs will appear

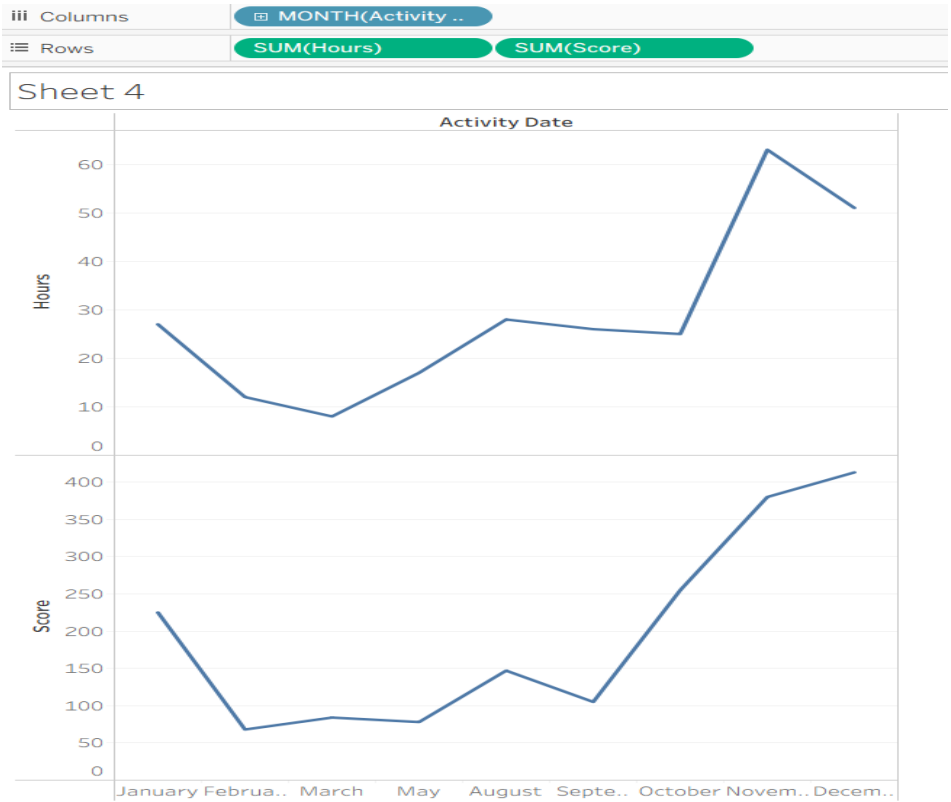
Step 5:-
Right Click
Score **Axis → Dual Axis**

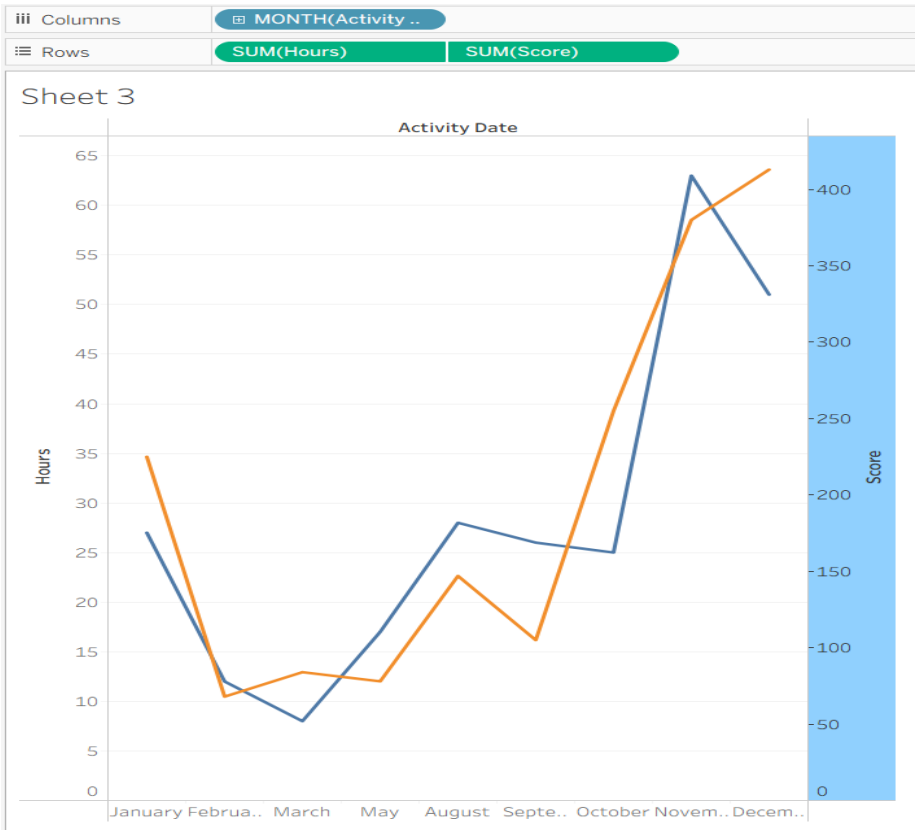
Step 6:-
Right **Axis → Synchronize Axis**

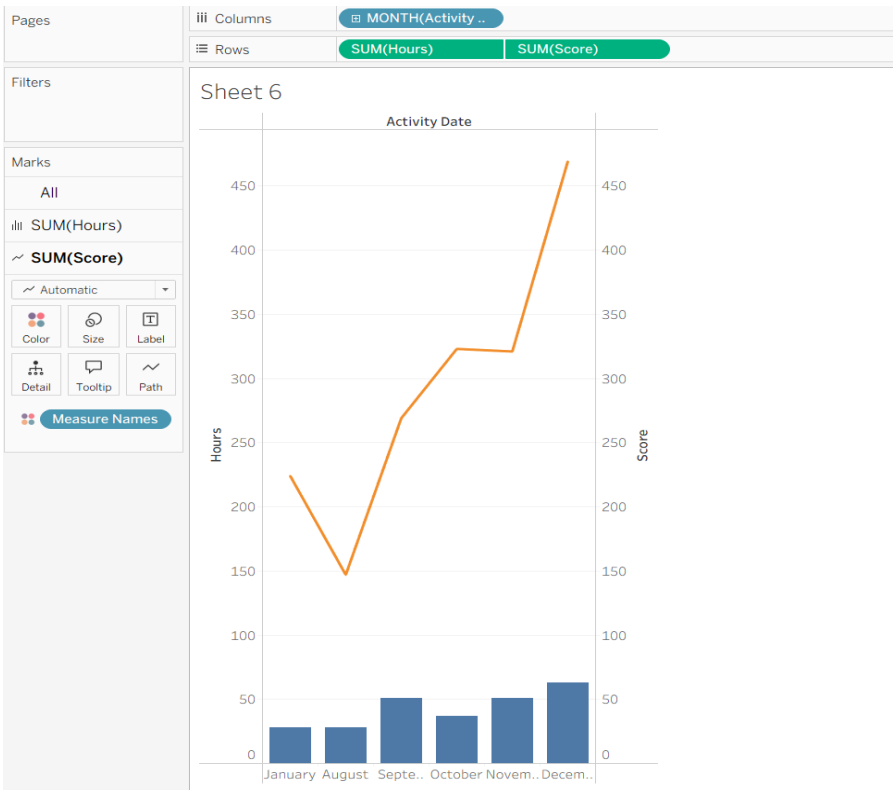
Step 7:-
Marks Card
Hours → Bar

Marks Card
Score → Line

Output:-
Bars shows Hours
Line shows Score




Columns
MONTH(Activity ..
Rows
SUM(Hours)
SUM(Score)
Sheet 3
Activity Date
Hours
Score
65
60
55
50
45
40
35
30
25
20
15
10
5
0
400
350
300
250
200
150
100
50
0
January
Februa..
March
May
August
Septe..
October
Novem..
Decem..


Pages
Columns
MONTH(Activity ..
Rows
SUM(Hours)
SUM(Score)
Filters
Marks
All
SUM(Hours)
SUM(Score)
Automatic
Color
Size
Label
Detail
Tooltip
Path
Measure Names
Sheet 6
Activity Date
Hours
Score
450
400
350
300
250
200
150
100
50
0
January
August
Septe..
October
Novem..
Decem..

PRACTICAL NO. 3(E)

Aim:- Develop histogram for distribution analysis.

a.) What is histogram ?

A histogram is a chart that shows the distribution(frequency) of a numerical variable by grouping values into continuous ranges called bins.

Objective:-
See the distribution of student scores.

Steps:-

Step 1:-
Open new worksheet

Step 2:-
Right Click
Score → Create → Bins

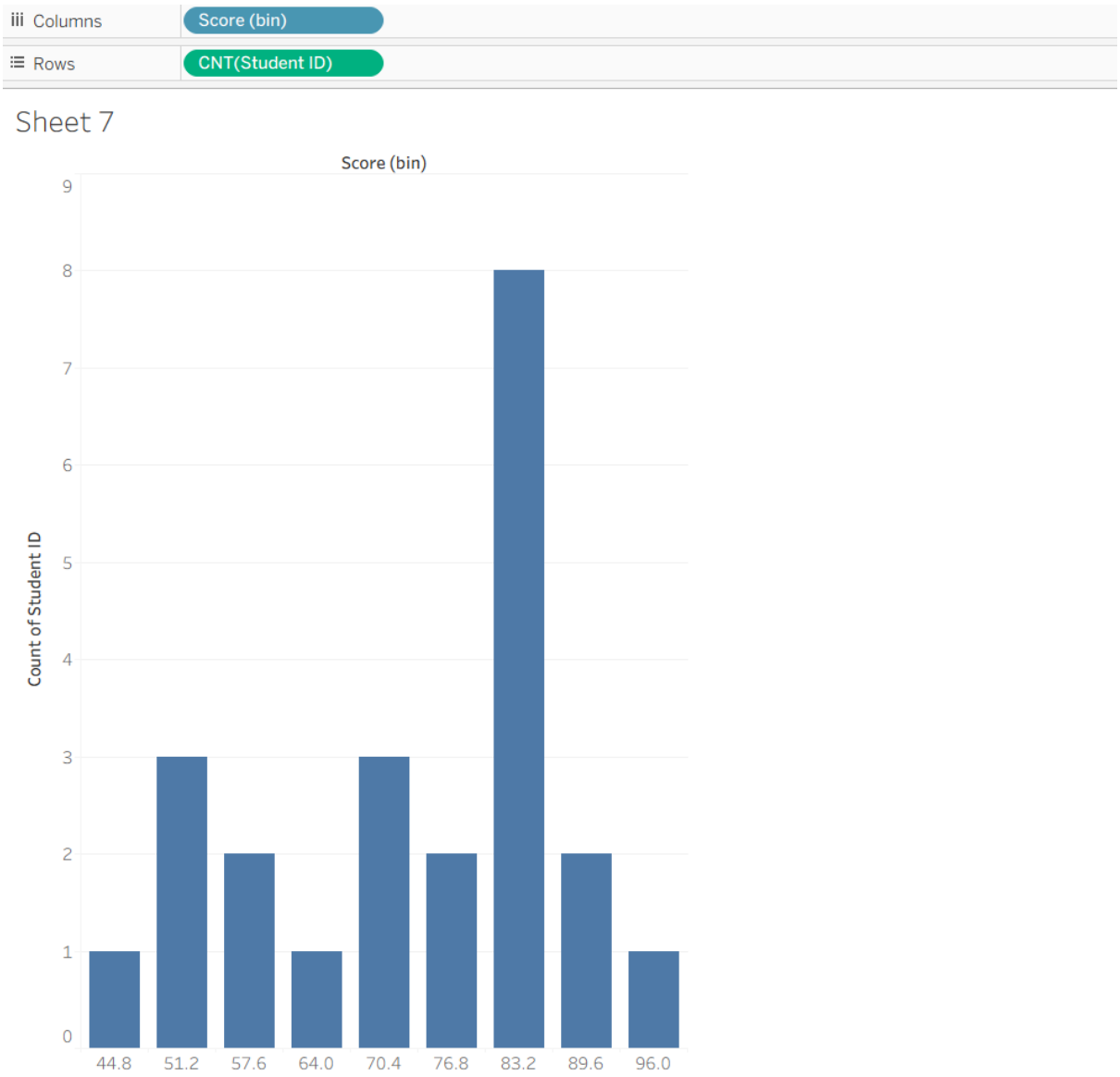
Step 3:-
Accept **default size → Ok**
A new field
Score(bins)
Appears

Step 4:-
Drag **Score(bins) → Column**
Different scores on the screen

Step 5:-
Drag **Number of record → Rows**
Histogram will appear on screen

Output:-

Histogram showing how many students fall in each score range



PRACTICAL NO. 3(F)

Aim:- Build box and whisker plots.

a.) What is box and whisper plot ?

A box and whisper plot, also called a Box Plot, is a chart that summarizes the distribution of numerical data using the minimum value, first quartile(Q1), median, third quartile(Q3), and maximum value. It can also highlight outlier.

Objective:-
Compare score distribution across department

Steps:-

Step 1:-
Open **new worksheet**

Step 2:-
Drag **Department → Column**
The Department field is added to the Columns shelf, creating separate categories.

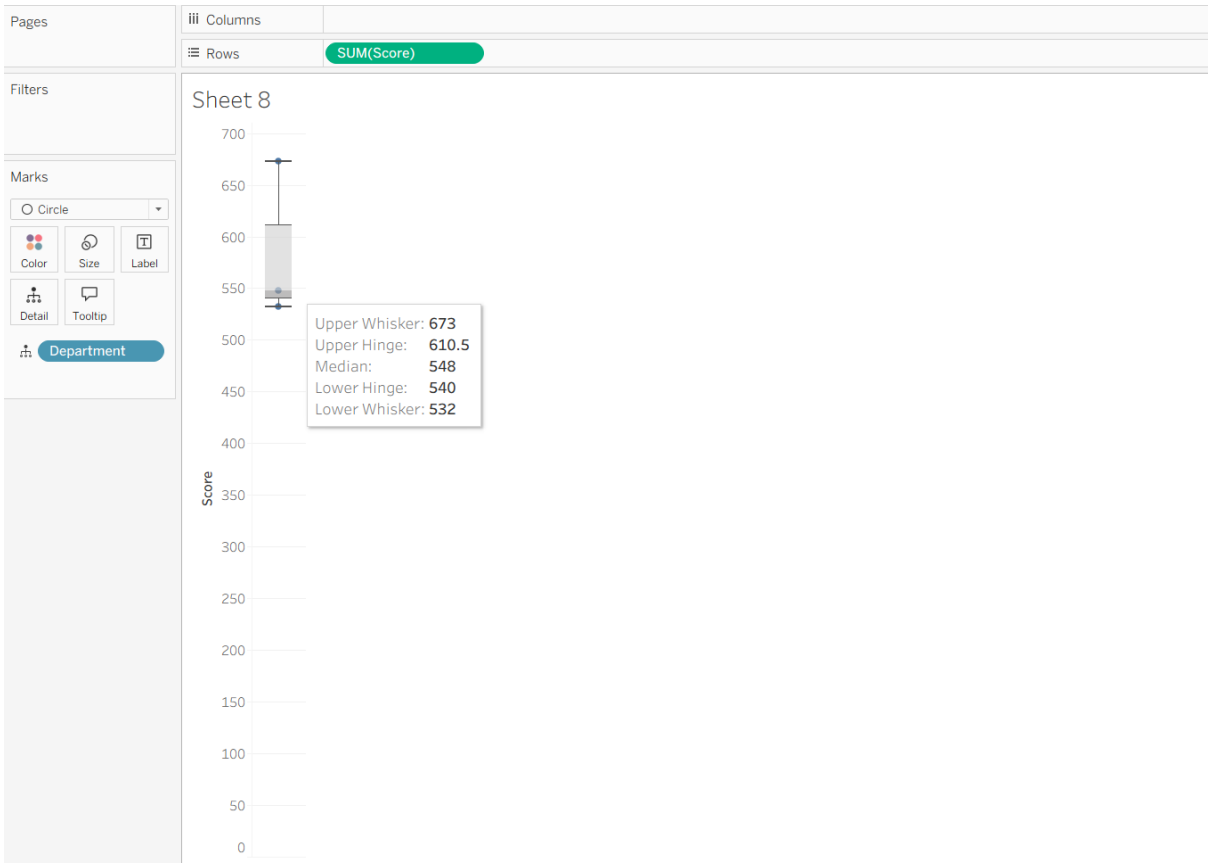
Step 3:-
Drag **Score → Rows**
The Score field is added to the Rows shelf, displaying score values for each department.

Step 4:-
Click **Show me → Box and whisper plot**
Tableau generates a Box-and-Whisker Plot showing the distribution, median, spread, and outliers of scores for each department.

Output:-

One box for each department showing:

- Minimum Score
- Maximum Score
- Median
- Quartiles
- Outliers(If any)



PRACTICAL NO. 3(G)

Aim:- Use gantt chart and advance date visualization.

a.) What is gantt chart ?

A gantt chart is a chart that displays tasks or activities over time using horizontal bars. The position of each bar shows when the activity starts, and the length of bar represents its duration.

Objective:-
Display activity duration using Hours.

Steps:-

Step 1:-
Open **new worksheet**

Step 2:-
Drag **StudentID → Rows**

Step 3:-
Drag **Activity Date → Column**
Choose Exact Date

Step 4:-
Marks → Gantt Chart

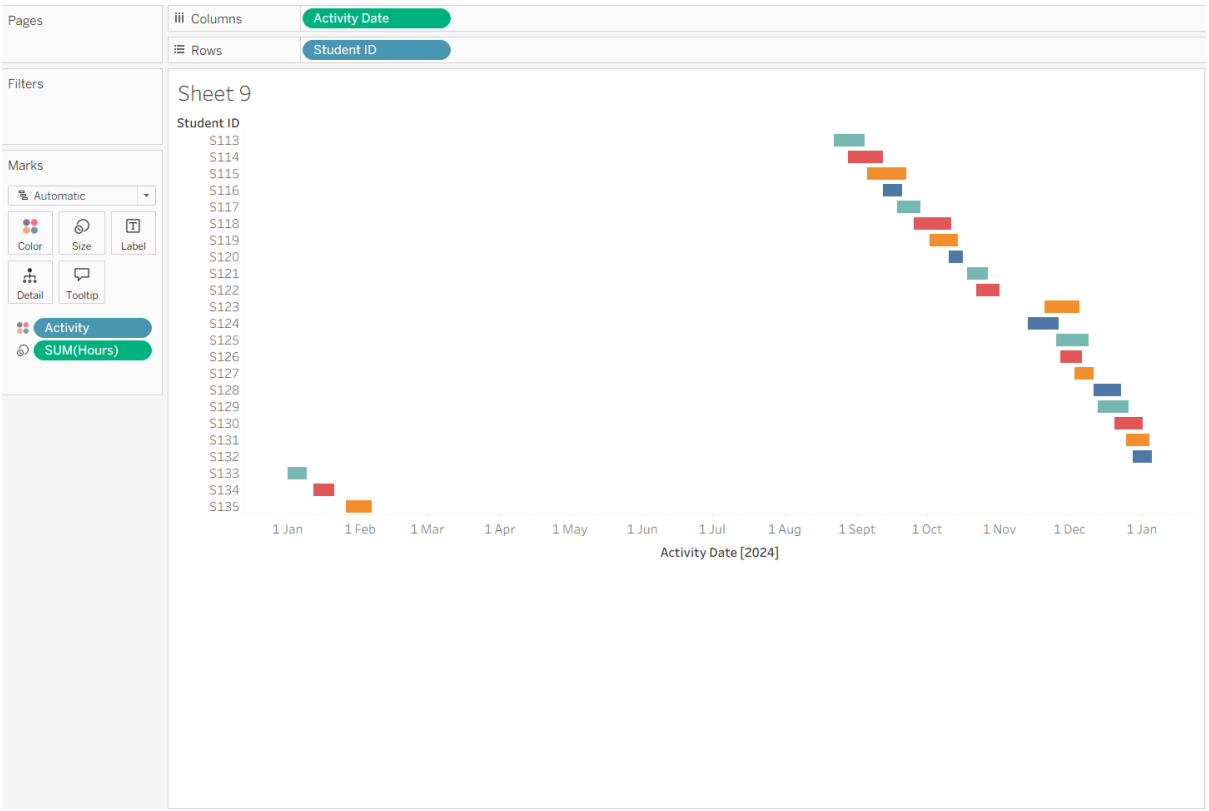
Step 5:-
Drag **Hours → Size**

Step 6:-
Drag **Activity → Colors**

Output:-

Each student has a Gantt bar.

Longer bars represents more activity hours.



Conclusion:

This practical demonstrated how different Tableau charts can be used to visualize and analyze data effectively for various purposes. It helped in comparing values, identifying relationships, understanding distributions, and presenting data in a clear and meaningful way using appropriate chart types. Overall, these visualizations improve data interpretation and support better decision-making.

Practical 6: Designing Effective and Visually Appealing Dashboards

Aim: To design professional and interactive dashboards in Tableau by applying formatting, colour schemes, customized tooltips, Viz in Tooltip, and visual storytelling techniques for effective business decision-making.

a) Apply worksheet-level and workbook-level formatting

b) Customize number, date, and null value formatting

c) Design effective colour schemes and layouts

d) Create informative tooltips

e) Implement "Viz in Tooltip"

f) Improve readability and user experience

g) Apply best practices for visual storytelling

A.) Aim:- Apply worksheet-level and workbook level formatting.

Step 1: Connect the Dataset

1. Open **Tableau Desktop**.
2. Under **Connect**, click **Microsoft Excel**.
3. Select **Retail_Store_Performance_500.xlsx**.
4. Drag the **RetailData** sheet onto the canvas.
5. Click **Sheet 1**.

Step 2: Create the Required Worksheets

Before starting sub-practicals (A) – (G), create the following worksheets.

Worksheet 1: Sales by Region

1. Rename **Sheet1** to **Sales by Region**.
2. Drag **Region** to the **Columns** shelf.
3. Drag **Sales** to the **Rows** shelf.
4. Click **Show Me** and select **Bar Chart**.
5. Click **Sort Descending**.
6. Drag **Sales** to **Label** on the Marks card.

Worksheet 2: Monthly Sales Trend

1. Create a new worksheet.
2. Rename it **Monthly Sales Trend**.
3. Drag **Order Date** to the **Columns** shelf.
4. Change the date level to **Month (Continuous)**.
5. Drag **Sales** to the **Rows** shelf.

Worksheet 3: Profit by Product Category

1. Create a new worksheet.
2. Rename it **Profit by Category**.
3. Drag **Product Category** to **Rows**.
4. Drag **Profit** to **Columns**.
5. Select **Horizontal Bar Chart**.
6. Sort in descending order.

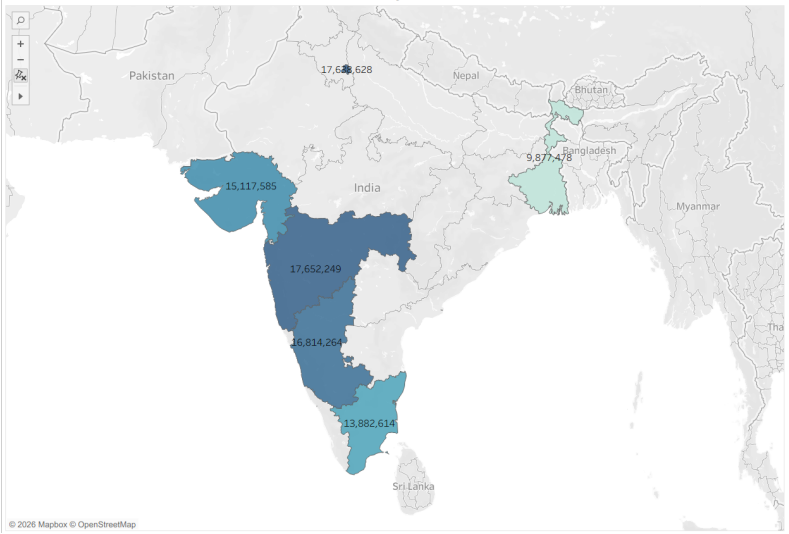
Worksheet 4: Sales by State

1. Create a new worksheet.
2. Rename it **Sales by State**.
3. Double-click **State**.
4. Tableau creates a map automatically.
5. Drag **Sales** to **Color**.
6. Drag **Sales** to **Size**.

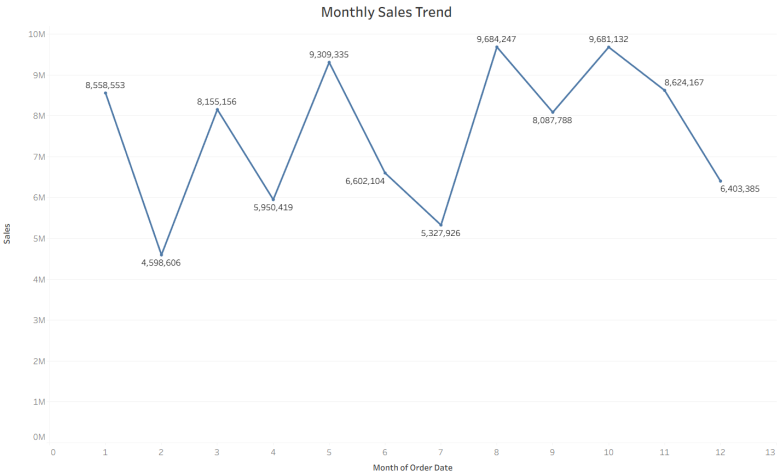
Worksheet 5: Customer Rating Distribution

1. Create a new worksheet.
2. Rename it **Customer Rating Distribution**.
3. Right-click **Customer Rating**.
4. Select **Create → Bins**.
5. Click **OK**.
6. Drag **Customer Rating (bin)** to **Columns**.
7. Drag **Number of Records** to **Rows**.
8. Select **Histogram**.

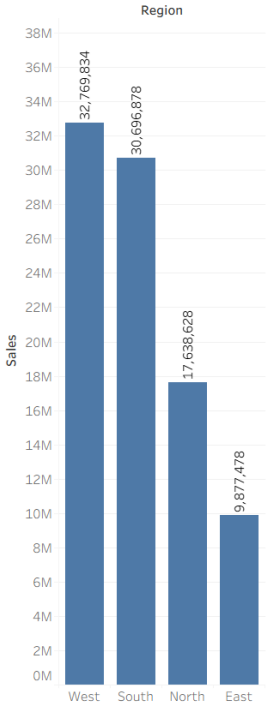
Sales By State



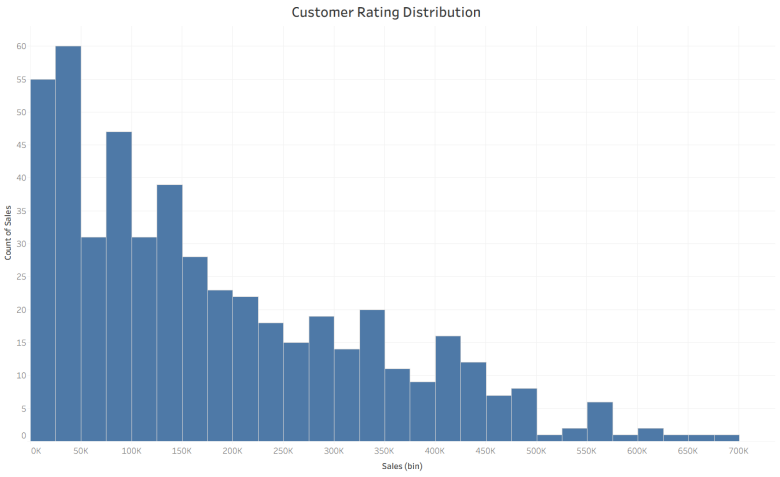
Monthly Sales Trend



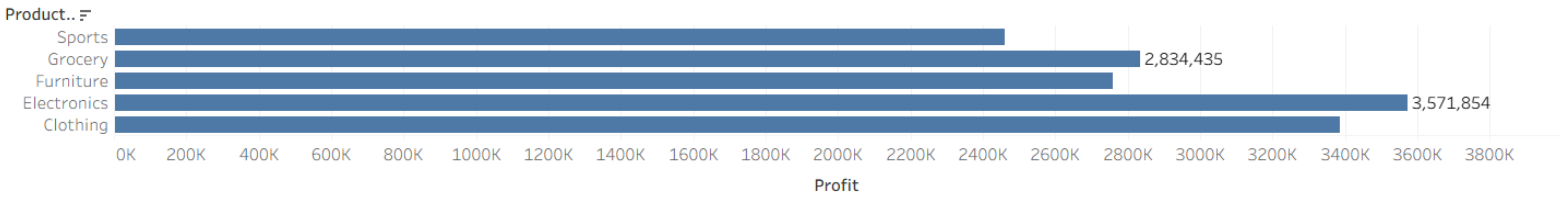
Region



Customer Rating Distribution



Profit By Product



Step 3: Create the Dashboard

1. Click **New Dashboard**.
2. Set **Dashboard Size → Automatic**.
3. Drag all five worksheets onto the dashboard.
4. Arrange them neatly.

After completing these steps, begin the following sub-practicals.

(a) Apply Worksheet-Level and Workbook-Level Formatting

Worksheet-Level Formatting

1. Open the **Sales by Region** worksheet.
2. Double-click the worksheet title and rename it to **Sales by Region**.
3. Click **Format → Font**.
4. Set:
    - Title: 16 pt, Bold
    - Header: 12 pt
    - Pane: 11 pt
5. Click **Format → Borders**.
6. Remove unnecessary row and column dividers.
7. Click **Format → Shading** and keep the worksheet background white.
8. Repeat the same formatting for all worksheets.

Workbook-Level Formatting

1. Open the dashboard.
2. Set **Dashboard Size → Automatic**.
3. Use the same font style throughout the dashboard.
4. Align all charts properly.
5. Maintain equal spacing between charts.
6. Add the dashboard title:
   Retail Store Performance Dashboard

Result:

A professionally formatted dashboard with a consistent appearance.

(b) Customize Number, Date, and Null Value Formatting

Steps:

1. Right-click **Sales**.
2. Select **Default Properties → Number Format**.
3. Choose **Currency (Custom)**.
4. Set decimal places to **0**.
5. Repeat the same steps for **Profit**.
6. Right-click **Discount (%)**.
7. Select **Number Format → Percentage**.
8. Right-click **Order Date**.
9. Select **Default Properties → Date Format**.
10. Choose **MMM YYYY**.
11. If null values are present, click **Format → Special Values**.
12. Replace **Null** with **No Data**.

Expected Output:

Sales and Profit are displayed as currency, Discount is shown as a percentage, dates are displayed in **MMM YYYY** format, and null values appear as **No Data**.

(c) Design Effective Color Schemes and Layouts

Steps:

1. Open the Sales by Region worksheet.
2. On the Marks card, click Color.
3. Choose a suitable color palette (for example, Blue, Green, or Orange).
4. Repeat the same process for all worksheets using consistent colors.
5. Avoid using too many different colors in the dashboard.
6. Open the Dashboard.
7. Arrange all worksheets neatly using Horizontal and Vertical Containers (optional).
8. Keep equal spacing between charts.

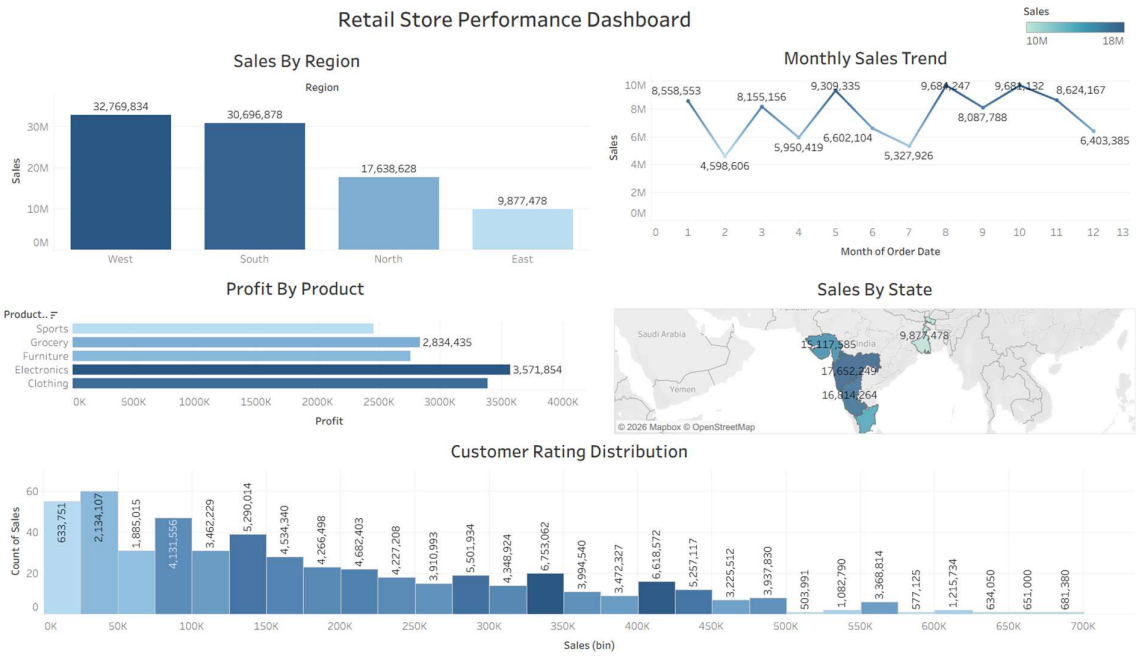
9. Resize charts so that each visualization is clearly visible.
10. Place legends and filters where they do not overlap with charts.
11. Use a white or light-colored dashboard background for better readability.

Expected Output:

The dashboard displays a consistent color theme with properly aligned charts, making the visualization attractive, organized, and easy to understand.

Use Layout Containers (Professional Method)

1. From the left panel, drag a **Horizontal Container** onto the dashboard.
2. Place **Sales by Region** and **Monthly Sales Trend** inside it.
3. Drag a second **Horizontal Container** below the first.
4. Place **Profit by Category** and **Sales by State** inside it.
5. Place **Customer Rating Distribution** below them (or in another container if desired).



(d) Create Informative Tooltips

Theory:

Tooltips provide additional information when the user places the mouse pointer over a mark in a visualization. Informative tooltips help users understand the data without displaying too much information directly on the chart.

Steps:

1. Open the **Sales by Region** worksheet.
2. On the **Marks** card, click **Tooltip**.
3. Delete the default tooltip text if required.
4. Insert the following fields:
    - Region
    - Sales
    - Profit
    - Customer Rating
    - Returned
5. Format the tooltip text with proper labels and spacing.
6. Click **OK**.
7. Repeat similar steps for other worksheets if required.
8. Hover the mouse over any bar to verify the tooltip information.

Example Tooltip

Region : <Region>

Sales : ₹<SUM(Sales)>

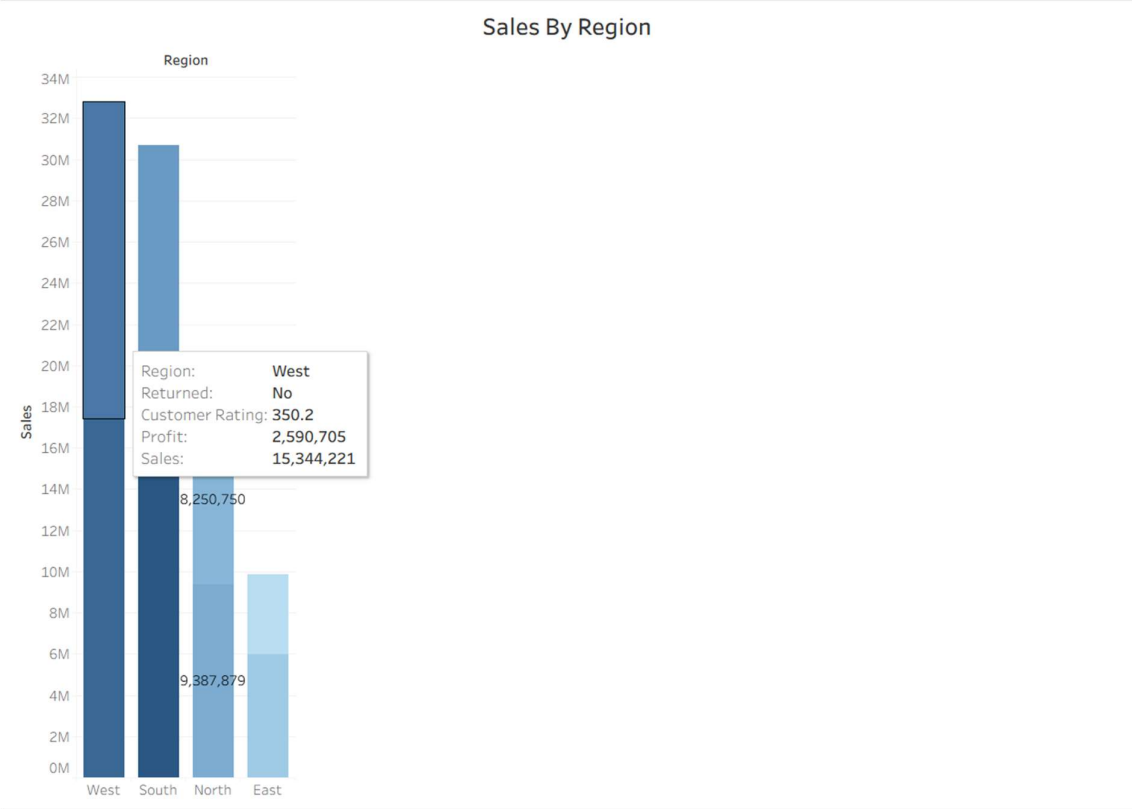
Profit : ₹<SUM(Profit)>

Customer Rating : <AVG(Customer Rating)>

Returned : <Returned>

Expected Output:

When the user hovers over any mark in the visualization, a well-formatted tooltip displays detailed business information related to that data point.



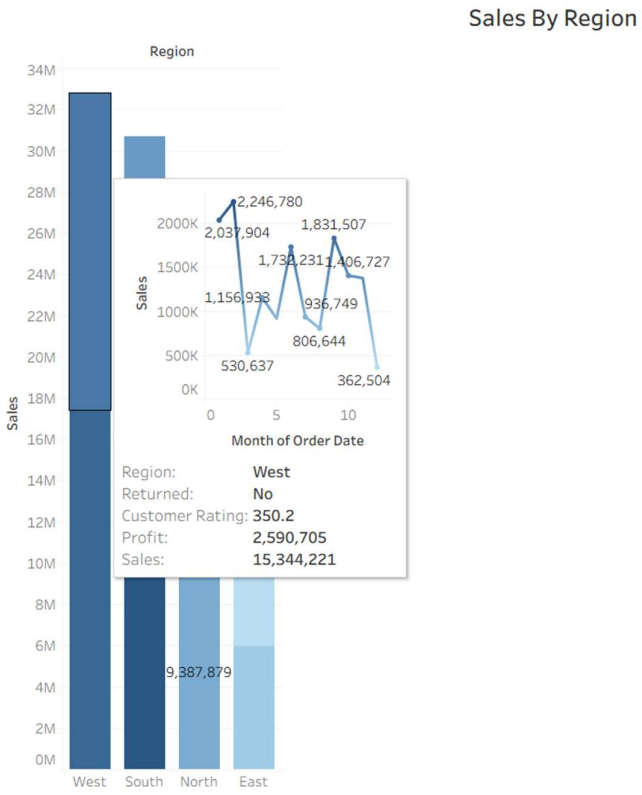
(e) Implement "Viz in Tooltip"

Steps:

1. Create a worksheet named **Monthly Sales Trend** (if not already created).
2. Open the **Sales by Region** worksheet.
3. Click **Tooltip** on the **Marks** card.
4. Click **Insert**.
5. Select **Sheets**.
6. Choose **Monthly Sales Trend**.
7. Click **OK**.
8. Hover the mouse over any region in the chart.

Expected Output:

When the user places the mouse pointer over a region, the tooltip displays a small monthly sales trend chart, providing additional insights without leaving the worksheet.



(f) Improve Readability and User Experience

Steps:

1. Use meaningful titles for all worksheets.
2. Increase font size for titles and labels if necessary.
3. Remove unnecessary grid lines and borders.
4. Keep equal spacing between charts.
5. Add filters such as **Region**, **Product Category**, or **Customer Segment**.
6. Place filters at the top or left side of the dashboard.
7. Display legends in an appropriate location.
8. Ensure charts do not overlap.
9. Maintain a consistent font style throughout the dashboard.

Expected Output:

The dashboard becomes clean, organized, and easy to understand. Users can quickly analyze information and interact with filters without confusion.

(g) Apply Best Practices for Visual Storytelling

Steps:

Step 1: Open the Completed Dashboard

1. Open the dashboard that contains all the worksheets:
    - Sales by Region
    - Monthly Sales Trend
    - Profit by Category
    - Sales by State
    - Customer Rating Distribution

Step 2: Add a Meaningful Dashboard Title

1. Click **Dashboard** from the menu bar.
2. Select **Show Title** (if it is not already enabled).
3. Double-click the dashboard title.
4. Type:

Retail Store Performance Dashboard

5. Change the font to:
    - **Arial**
    - **18 pt**
    - **Bold**
6. Click **OK**.

Purpose: A meaningful title helps users understand the dashboard immediately.

Step 3: Arrange Worksheets in a Logical Sequence

Arrange the worksheets in the following order:

Retail Store Performance Dashboard

Sales by Region Monthly Sales Trend

Profit by Category Sales by State

Customer Rating Distribution

Purpose: Place related charts together so users can understand the information easily.

Step 4: Place Important Charts at the Top

1. Drag the **Sales by Region** worksheet to the upper-left corner.
2. Drag **Monthly Sales Trend** beside it.
3. Place **Profit by Category** below Sales by Region.
4. Place **Sales by State** beside Profit by Category.
5. Place **Customer Rating Distribution** at the bottom.

Purpose: Users usually read from top to bottom, so important information should appear first.

Step 5: Use Consistent Colors

1. Open the **Sales by Region** worksheet.
2. Click **Color** on the **Marks** card.
3. Select a suitable color palette.
4. Repeat the same color theme for all worksheets.
5. Avoid using too many bright or different colors.

Example:

- Blue → Sales

- Green → Profit
- Orange → Ratings

Purpose: Consistent colors improve readability and make the dashboard look professional.

Step 6: Highlight Important Trends

Method 1: Show Labels

1. Open **Sales by Region**.
2. On the **Marks** card, click **Label**.
3. Check **Show Mark Labels**.

Now sales values appear on each bar.

Method 2: Add Annotation

1. Right-click the highest sales bar.
2. Select **Annotate → Mark**.
3. Enter the text:

Highest Sales Region

4. Click **OK**.

The annotation highlights the most important data point.

Step 7: Remove Unnecessary Elements

1. Remove unwanted legends (if not required).
2. Remove unnecessary grid lines:
    - Go to **Format → Lines**.
    - Set Grid Lines to **None**.
3. Remove extra borders if they make the dashboard cluttered.
4. Keep only the charts required for analysis.

Purpose: A clean dashboard is easier to understand.

Step 8: Review the Dashboard

Check the following:

✓ All charts are visible.

✓ Titles are meaningful.

✓ Fonts are readable.

✓ Charts are properly aligned.

✓ Colors are consistent.

✓ Filters are working correctly.

✓ No chart overlaps another.

Step 9: Identify Business Insights

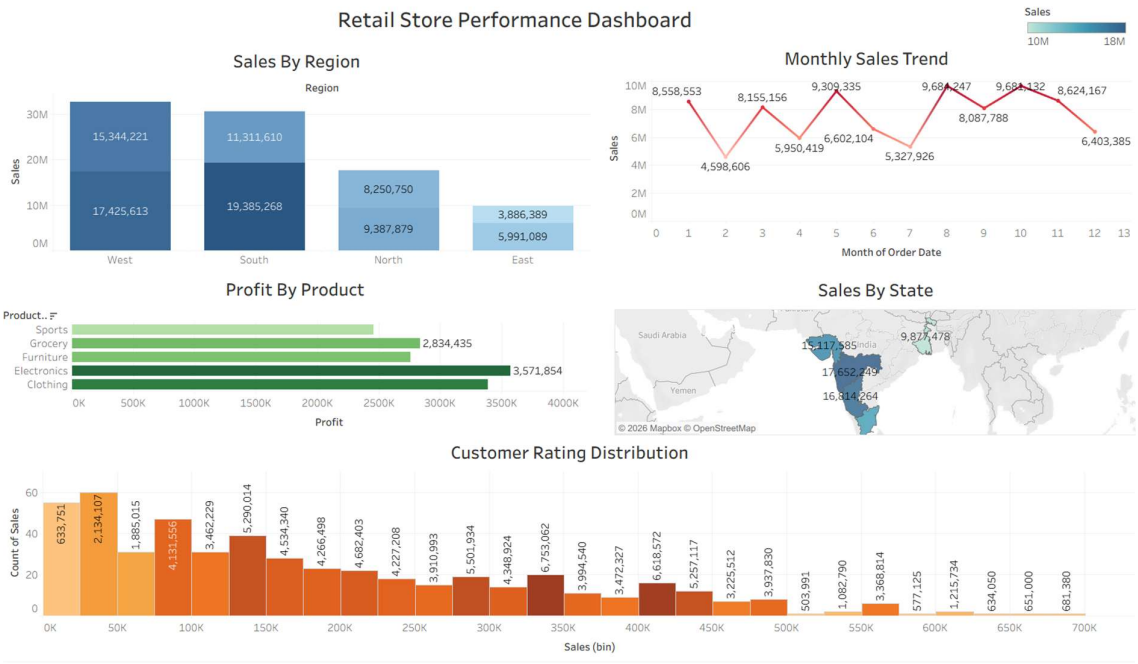
Observe the dashboard and write important findings.

Example Insights

- West Region recorded the highest sales.
- Electronics products generated the maximum profit.
- Customer ratings are generally above 4.5.
- Sales increased during the middle months of the year.
- Returned orders are comparatively low.

Expected Output

A professional dashboard with a clear title, logical chart arrangement, consistent colors, highlighted business insights, and an attractive layout that effectively communicates business performance.



Conclusion:-

This practical demonstrated how Tableau dashboards can be made more professional, interactive, and user-friendly through effective formatting and visualization techniques. Worksheet-level and workbook-level formatting helped maintain consistency, while customized number, date, and null-value formats improved clarity. Appropriate colour schemes and layouts made the dashboards visually appealing and easier to understand. Informative tooltips and Viz in Tooltip provided additional details without overcrowding the dashboard. Overall, applying readability principles and visual storytelling best practices helped transform raw data into meaningful insights that can support effective business decision-making.

PRACTICAL 8

Performing Advanced Analytics using Trends, Clustering, and Forecasting

Aim:- To perform advanced analytics in Tableau using trend lines, clustering, distribution analysis, forecasting, and statistical models.

Dataset: Weather & Energy Consumption:

The following dataset is used for performing all the activities in this practical.

Date	Temperature	Humidity	Wind Speed	Energy Consumption
01-01-2025	18	62	8	420
02-01-2025	19	60	7	415
03-01-2025	21	58	6	400
04-01-2025	22	55	5	390
05-01-2025	24	52	5	380
06-01-2025	25	50	4	370
07-01-2025	27	48	4	360
08-01-2025	26	51	5	365
09-01-2025	24	55	6	380
10-01-2025	22	58	7	395
11-01-2025	20	61	8	410
12-01-2025	19	64	9	425
13-01-2025	23	57	6	395
14-01-2025	25	53	5	380
15-01-2025	28	47	4	350

(a) Add and Interpret Trend Lines

Step 1: Connect the Dataset

1. Open Tableau Desktop.
2. Connect to the Excel file containing the Weather & Energy Consumption dataset.
3. Open a new worksheet.
4. Rename the worksheet as Trend Analysis.

Step 2: Create a Scatter Plot

5. Drag Temperature to the Columns shelf.
6. Drag Energy Consumption to the Rows shelf.
7. Tableau creates a scatter plot showing the relationship between temperature and energy consumption.

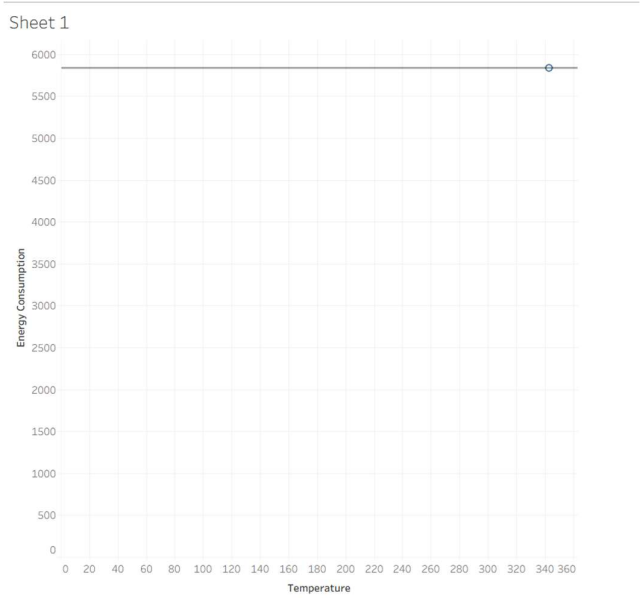
The scatter plot helps us understand whether changes in temperature are related to changes in energy consumption.

Step 3: Add a Linear Trend Line

8. Open the Analytics pane.
9. Locate Trend Line.
10. Drag Trend Line onto the visualization.
11. Select Linear.

A straight trend line is now displayed on the scatter plot.

Expected Output:-



Interpretation:-

The linear trend line represents the overall relationship between Temperature and Energy Consumption.

In this dataset, the trend is generally negative. This means that as temperature increases, energy consumption tends to decrease.

Linear Trend:-

A linear model can be represented as:

$Y = aX + b$

where:

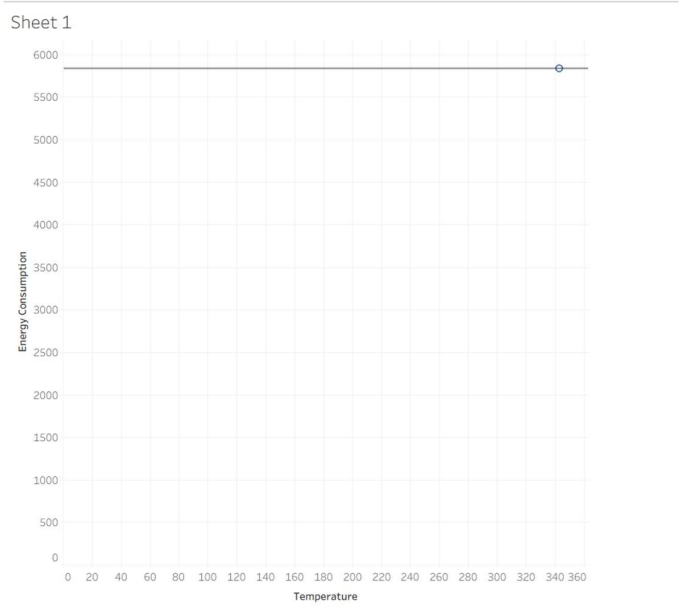
- Y = dependent variable
- X = independent variable
- a = slope
- b = intercept

Step 4: Add an Exponential Trend Line

12. Keep the same scatter plot.
13. Open the Analytics pane.
14. Drag Trend Line onto the visualization.
15. Select Exponential.

Tableau displays an exponential trend line.

Expected Output:-



Exponential Trend:-

An exponential model can be represented as:

$Y = ae^{bx}$

where:

- Y = predicted value
- X = independent variable
- a and b = model parameters

Interpretation of Trend Lines:-

A trend line helps identify the general direction of a relationship between two variables.

- Positive trend: As X increases, Y also increases.
- Negative trend: As X increases, Y decreases.
- Linear trend: Relationship is approximately represented by a straight line.
- Exponential trend: Relationship changes at an increasing or decreasing rate.

(b) Customize and Analyze Trend Models

Step 1: Open Trend Model Details

16. Right-click directly on the trend line.
17. Select Describe Trend Line.

Tableau displays statistical information about the trend model.

The information may include:

- Model equation
- R-squared value
- P-value
- Model type
- Statistical significance
- Model information

Step 2: Understand R-Squared

R-squared (R^2) indicates how well the trend model explains the variation in the data.

The value generally ranges from 0 to 1.

- R^2 close to 1: Very strong model fit.
- R^2 close to 0: Weak model fit.

For example, an R^2 value of 0.90 indicates that the model explains approximately 90% of the variation in the dependent variable.

Step 3: Understand P-Value

The p-value helps determine whether the relationship represented by the model is statistically significant.

A commonly used significance level is:

$p < 0.05$

If the p-value is less than 0.05, the relationship is generally considered statistically significant.

Step 4: Customize the Trend Model

Different trend models can be compared to determine which one provides a better representation of the data.

The main models available in Tableau include:

- Linear
- Logarithmic
- Exponential
- Polynomial
- Power

The appropriate model depends on the pattern present in the dataset.

Interpretation:-

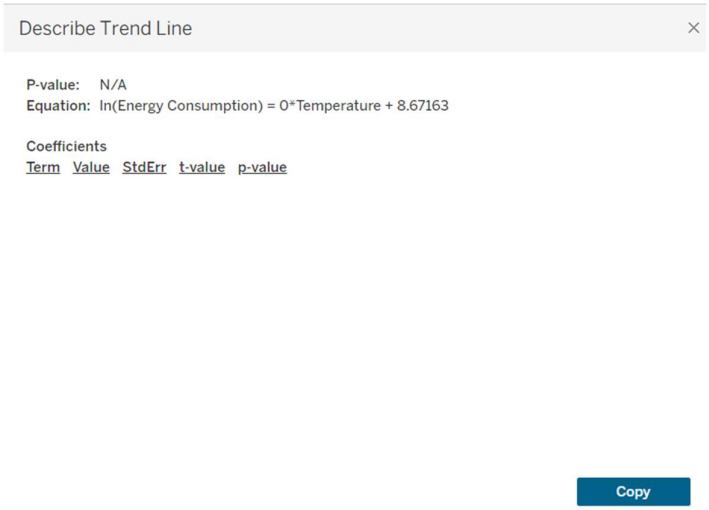
The trend model provides more information than simply looking at the trend line visually.

For example: Linear model

- Shows a straight-line relationship.
- Useful when the rate of change is approximately constant.

Exponential model

- Useful when the rate of change increases or decreases rapidly.
- Can represent relationships where changes occur proportionally.



(c) Perform Clustering on the Dataset

Clustering is used to divide data into groups containing observations with similar characteristics.

Step 1: Create a New Worksheet

18. Create a new worksheet.
19. Rename it as Weather Clustering.

Step 2: Create a Scatter Plot

20. Drag Temperature to Columns.
21. Drag Humidity to Rows.
22. Drag Date to Detail on the Marks card.

Each date is now represented as an individual observation.

Step 3: Apply Clustering

23. Open the Analytics pane.
24. Locate Cluster.
25. Drag Cluster onto the visualization.
26. The Clusters dialog box opens.

Step 4: Select the Number of Clusters

27. Set Number of Clusters to 3.
28. Tableau automatically divides the observations into three groups.

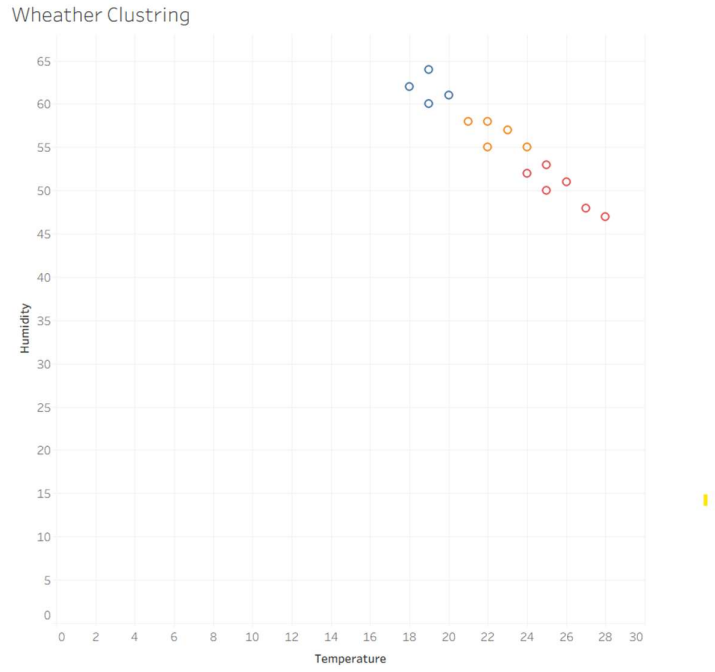
Step 5: Add Variables to the Cluster Model

The following variables can be used for clustering:

- Temperature
- Humidity
- Wind Speed
- Energy Consumption

Add the required fields to the Variables section of the Clusters dialog box.

Excepted Output:-



Interpretation:-

The clusters represent groups of observations having similar characteristics.

For example, the clusters may represent:

- Cluster 1: Higher temperature and lower humidity
- Cluster 2: Moderate temperature and humidity
- Cluster 3: Lower temperature and higher humidity

The exact cluster membership is determined automatically by Tableau.

Conclusion:-

Clustering helps identify natural groups within the dataset without manually assigning categories.

(d) Analyze Distributions Using Built-in Tools

Distribution analysis helps us understand how values are spread across a dataset.

Two useful Tableau tools are:

29. Histogram
30. Box Plot

Part 1: Create a Histogram

Step 1: Create Energy Consumption Bins

31. Right-click Energy Consumption in the Data pane.
32. Select Create → Bins.
33. Set the Size of bins to 20.
34. Click OK.

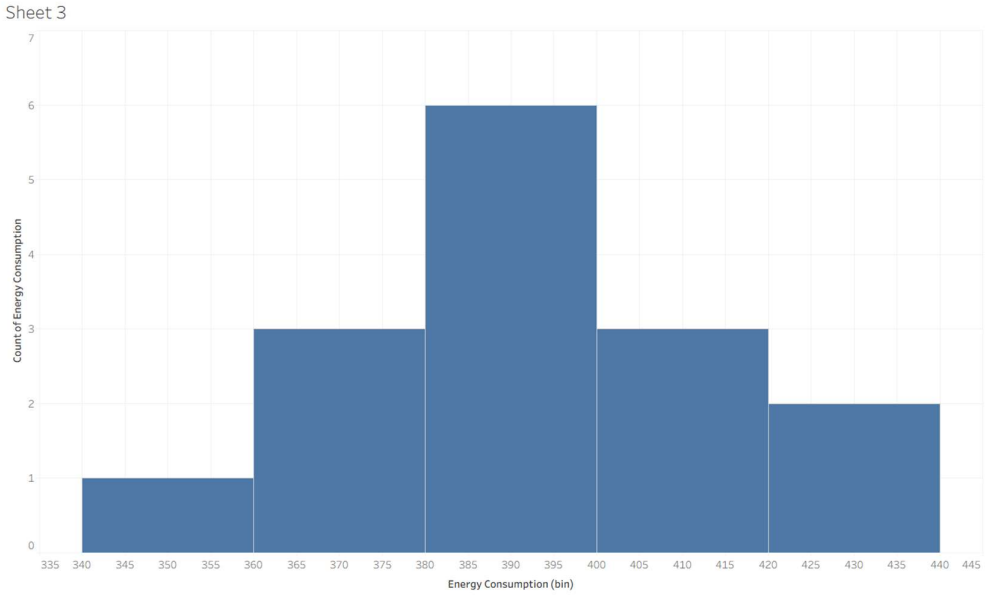
Tableau creates a new field called Energy Consumption (bin).

Step 2: Create the Histogram

35. Drag Energy Consumption (bin) to Columns.
36. Drag Energy Consumption to Rows.
37. If required, change the aggregation to COUNT.

The histogram displays the frequency of energy consumption values within different ranges.

Excepted Output:-



Interpretation of Histogram:-

The histogram helps identify:

- Frequency
- Distribution shape
- Concentration of values
- Possible gaps
- Possible unusual values

For example, if most observations fall between 380 and 420, that range represents the most common energy consumption values in the dataset.

Part 2: Create a Box Plot

Step 1: Create a New Worksheet

38. Create a new worksheet.
39. Rename it as Energy Box Plot.

Step 2: Add Fields

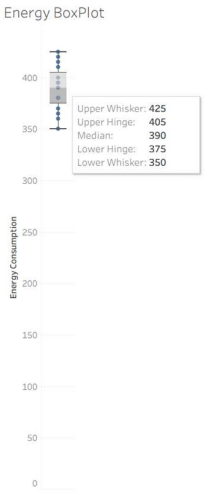
40. Drag Energy Consumption to Rows.
41. Drag Date to Columns.

Step 3: Create the Box Plot

42. Open the Analytics pane.
43. Locate Box Plot.
44. Drag Box Plot onto the visualization.

Alternatively, use Show Me if the Box Plot option is available.

Expected Output:-



Interpretation:-

A box plot displays:

- Median
- Lower quartile
- Upper quartile
- Spread of the data
- Possible outliers

Histogram vs Box Plot:

Histogram	Box Plot
Shows frequency of values	Shows statistical distribution
Uses bins	Uses quartiles
Shows distribution shape	Shows median and spread
Useful for frequency analysis	Useful for identifying outliers

(e) Apply Forecasting Techniques

Forecasting is used to estimate future values based on historical data.

In this practical, Tableau is used to forecast future Energy Consumption.

Step 1: Create a New Worksheet

45. Create a new worksheet.
46. Rename it as Energy Consumption Forecast.

Step 2: Create a Time-Series View

47. Drag Date to Columns.
48. Drag Energy Consumption to Rows.
49. Make sure Date is a true date field.
50. Use a continuous date so that the date appears as a green pill.

Tableau creates a time-series line chart.

Step 3: Apply Forecast

51. Open the Analytics pane.
52. Locate Forecast.
53. Drag Forecast onto the visualization.

Tableau automatically generates a forecast based on the historical energy consumption values.

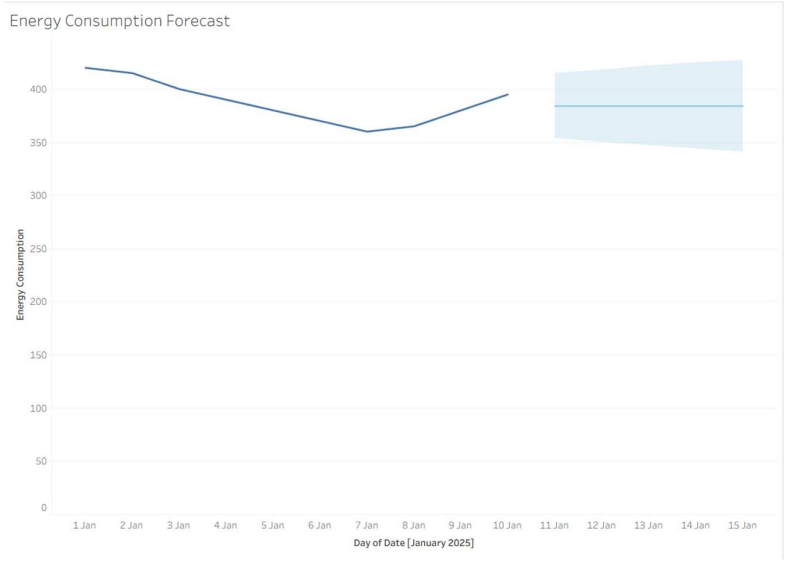
Step 4: Set Forecast Length

To specify the number of future periods:

54. Go to Analysis.
55. Select Forecast.
56. Select Forecast Options.
57. Set the forecast length to the required number of periods.

For demonstration, the forecast length can be set to 5 periods.

Excepted Output:-



Step 5: Analyze the Forecast

The forecast displays:

- Historical values
- Predicted future values
- Forecast range
- Confidence or prediction interval

The forecast line represents the estimated future energy consumption.

What Does Forecasting Do?

Forecasting uses historical patterns to estimate future values.

For example, if the historical data shows that energy consumption is generally decreasing as temperature increases, Tableau uses the available historical pattern to estimate future energy consumption.

The forecast is not an exact prediction. It is an estimate based on the patterns present in the historical data.

Describe the Forecast

To examine the forecasting model:

58. Go to Analysis.
59. Select Forecast.
60. Select Describe Forecast.

Tableau displays information about the forecasting model and its results.

(f) Compare Different Statistical Models

Different statistical models can be compared to determine which model provides a better fit to the data.

In this practical, Linear and Exponential model are compared.

Step 1: Create a Scatter Plot

61. Drag Temperature to Columns.
62. Drag Energy Consumption to Rows.
63. Ensure that the visualization is a scatter plot.

Step 2: Add Linear Trend Line

64. Open the Analytics pane.
65. Drag Trend Line onto the visualization.
66. Select Linear.

Step 3: Analyze the Linear Model

67. Right-click the linear trend line.
68. Select Describe Trend Line.
69. Note the following:

- Equation
- R-squared value

- P-value

Step 4: Add Exponential Trend Line

70. Add another trend line.
71. Select Exponential.
72. Right-click the trend line.
73. Select Describe Trend Line.

Record:

- Equation
- R-squared value
- P-value

Step 5: Compare the Models

Model	Equation	R^2	Interpretation
Linear	$Y = aX + b$	Note Tableau value	Measures straight-line relationship
Exponential	$Y = ae^{bx}$	Note Tableau value	Measures exponential relationship

Interpretation:-

The model with the higher R^2 value generally provides a better fit to the observed data.

However, R^2 should not be considered alone. The model's equation, statistical significance, and the actual pattern of the data should also be considered.

Important Note:

If both Linear and Exponential models show an R^2 value very close to 1, it means that both models fit the current dataset extremely well.

This can happen when the dataset contains a very strong and predictable relationship between the variables.

Therefore, for meaningful model comparison, the dataset should contain some natural variation.

(g) Export and Interpret Statistical Results

Statistical results obtained from Tableau can be recorded and used for further analysis.

Step 1: Open Trend Line Statistics

74. Open the worksheet containing the trend line.
75. Right-click the trend line.
76. Select Describe Trend Line.

Step 2: Examine the Results

The statistical information may include:

- Model equation
- R-squared value
- P-value
- Model type
- Number of observations
- Statistical significance

Step 3: Record the Results

The displayed statistical information can be copied manually into Excel or Word for documentation.

Create a table such as:

Model	**Equation**	**R^2**	**P-value**	**Interpretation**
Linear	Tableau result	Tableau result	Tableau result	Evaluate model fit
Exponential	Tableau result	Tableau result	Tableau result	Evaluate model fit

Step 4: Interpret the Statistical Results

The results can be interpreted using the following guidelines:

R^2

A higher R^2 generally indicates that the model explains more of the variation in the dependent variable.

P-value

A p-value below 0.05 is commonly interpreted as evidence that the relationship is statistically significant.

Equation

The equation describes the mathematical relationship between the independent and dependent variables.

Overall Interpretation:-

Tableau provides several advanced analytical features for understanding data.

Trend lines help identify relationships between variables. Clustering divides observations into groups with similar characteristics. Histograms and box plots help understand the distribution of data. Forecasting estimates future values using historical patterns. Statistical models allow different relationships to be compared using measures such as R^2 and p-value.

These features make Tableau useful not only for creating visualizations but also for performing advanced data analysis and supporting data-driven decision making.

Conclusion:-

In this practical, advanced analytical techniques were performed using Tableau. Linear and exponential trend lines were added and interpreted, statistical models were analyzed using R^2 and p-values, observations were grouped using clustering, distributions were examined using histograms and box plots, and future energy consumption was estimated using forecasting.

Thus, Tableau can be effectively used to perform advanced analytics and interpret relationships, patterns, distributions, and future trends in data.

Main Motive of Practical 8

The main motive of this practical is to learn how to use Tableau not just for making charts, but for performing advanced data analysis.

Students should mainly learn these 5 things:

77. Find relationships → Use Trend Lines to understand whether two variables are related and whether the relationship is positive or negative.
78. Find groups/patterns → Use Clustering to automatically group similar data points.
79. Understand data distribution → Use Histogram and Box Plot to see how data is spread and identify unusual values.
80. Predict future values → Use Forecasting to estimate what may happen in the future based on historical data.
81. Compare statistical models → Use R^2, p-value, and model equations to decide which statistical model fits the data better.

Term	Simple Meaning
Trend Lines	A line that shows the general direction of data, such as whether values are increasing, decreasing, or staying stable.
Clustering	Grouping similar data items together automatically. For example, grouping customers with similar buying habits.
Distribution Analysis	Understanding how data is spread. For example, finding whether most students scored between 60–80 marks.
Forecasting	Using old or past data to predict future values. For example, predicting next month's temperature or future sales.
Statistical Models	Mathematical methods used to analyze data, find patterns, and make predictions.

PRACTICAL 10

DATA PREPARATION AND TRANSFORMATION USING TABLEAU PREP

MAIN PURPOSE OF THIS PRACTICAL:-

We are learning how to prepare raw or messy data before using it for analysis and visualization in Tableau.

Real-world data is usually not ready to use directly. It may contain:

- Missing values
- Duplicate or inconsistent data
- Data in the wrong format
- Data spread across multiple files
- Data arranged in a difficult format
- Data from different databases

So, in this practical, we learn how to clean, combine, organize, and restructure the data so that Tableau can create correct and meaningful charts, dashboards, and analysis.

AIM:- To prepare, clean, transform, and organize data using Tableau Prep for analysis and visualization.

DATASET TO USE

We will use a simple Employee Department Dataset.

You can create an Excel file with the following data:

Employee_ID	Employee_Name	Department	City	Salary	Joining_Year
E101	Amit	IT	Mumbai	45000	2022
E102	Neha	it	Mumbai	50000	2021
E103	Rahul	IT Dept	Thane	48000	2023
E104	Priya	HR	Mumbai	40000	2022
E105	Sneha	Human Resource	Thane	42000	2021
E106	Karan	Sales	Palghar	38000	2023
E107	Riya	sales	Palghar	41000	2022
E108	Ajay	IT	Nashik	52000	2020
E109	Pooja	HR	Nashik	43000	2023
E110	Vijay	IT Dept	Mumbai	55000	2021

This dataset is deliberately made with inconsistent values so that students can practice cleaning.

For example:

- IT
- it
- IT Dept

actually refer to the same department.

Similarly:

- HR
- Human Resource

refer to the same department.

A) CONNECT AND LOAD DATA INTO TABLEAU PREP

STEP 1: Open Tableau Prep

Open:

Tableau Prep Builder

You will see the start screen.

STEP 2: Connect the Excel file

On the left side, under Connections:

Click:

Microsoft Excel

Select your Excel file.

For example:

Employee_Data.xlsx

STEP 3: Select the sheet

Tableau Prep will show the sheets in your Excel file.

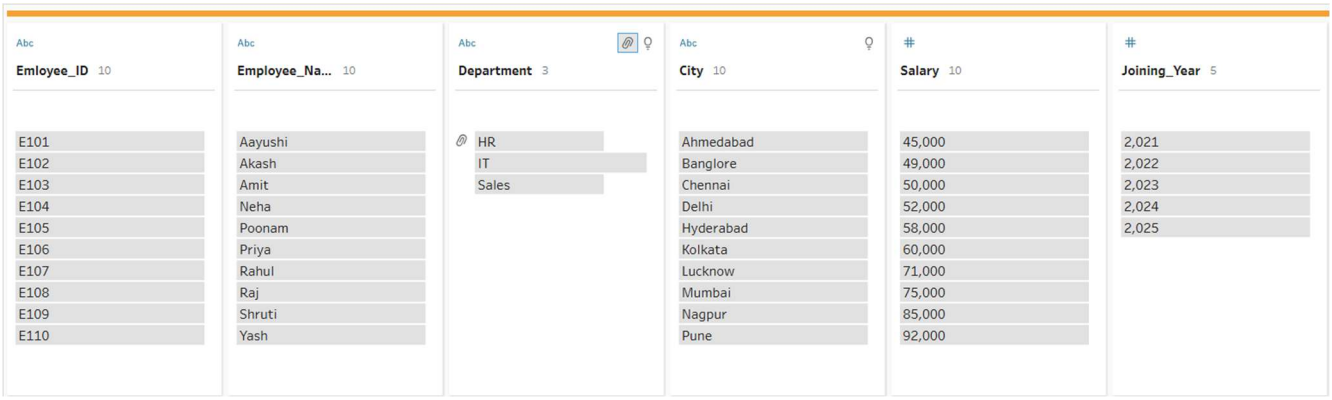
Select:

Employee Data

Drag it to the flow area.

You will see:

Input



Emloyee_ID 10	Employee_Na... 10	Department 3	City 10	Salary 10	Joining_Year 5
E101	Aayushi	HR	Ahmedabad	45,000	2,021
E102	Akash	IT	Banglore	49,000	2,022
E103	Amit	Sales	Chennai	50,000	2,023
E104	Neha		Delhi	52,000	2,024
E105	Poonam		Hyderabad	58,000	2,025
E106	Priya		Kolkata	60,000	
E107	Rahul		Lucknow	71,000	
E108	Raj		Mumbai	75,000	
E109	Shruti		Nagpur	85,000	
E110	Yash		Pune	92,000	

STEP 4: Add a Cleaning Step

Click the + symbol after Input.

Select:

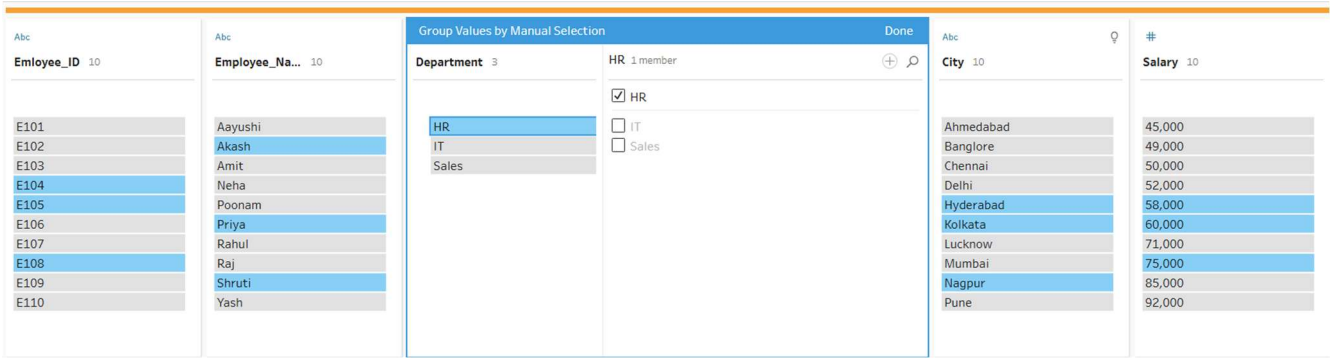
Clean Step

Your flow will look like:

Input → Clean

This is where we will clean the data.

B) CLEAN AND PROFILE DATA



Emloyee_ID 10	Employee_Na... 10	Department 3	City 10	Salary 10
E101	Aayushi	HR	Ahmedabad	45,000
E102	Akash	IT	Banglore	49,000
E103	Amit	Sales	Chennai	50,000
E104	Neha		Delhi	52,000
E105	Poonam		Hyderabad	58,000
E106	Priya		Kolkata	60,000
E107	Rahul		Lucknow	71,000
E108	Raj		Mumbai	75,000
E109	Shruti		Nagpur	85,000
E110	Yash		Pune	92,000

What does cleaning mean?

Cleaning means:

Finding and correcting problems in the data.

Examples:

- Missing values
- Spelling mistakes
- Extra spaces
- Different formats
- Duplicate records
- Incorrect values

STEP 1: Examine the data

Click the Clean Step.

Tableau Prep displays the fields.

You can see:

- Employee_ID
- Employee_Name
- Department
- City
- Salary
- Joining_Year

Tableau Prep also shows the distribution of values.

This is called data profiling.

Simple meaning:

Data profiling = checking what is inside our data.

STEP 2: Check Department

Look at the Department field.

You will notice:

IT

it

IT Dept

HR

Human Resource

Sales

sales

There are inconsistencies.

We will fix them in the next step.

C) GROUP AND STANDARDIZE VALUES

This is one of the most important parts of Tableau Prep.

What does standardization mean?

It means:

Changing different versions of the same value into one common value.

For example:

IT

it

IT Dept

becomes:

IT

Do this:

1. In your Clean Step, find Department.
2. Click the drop-down/menu on the Department field.
3. Select Group Values.
4. Tableau Prep will show the different values:

- IT
- it
- IT Dept
- HR
- Human Resource
- Sales
- sales

5. Select IT, it, and IT Dept.
6. Click Group.

They will become one group.

7. Rename the group to IT.

Then do the same for:

HR + Human Resource → HR

and

Sales + sales → Sales

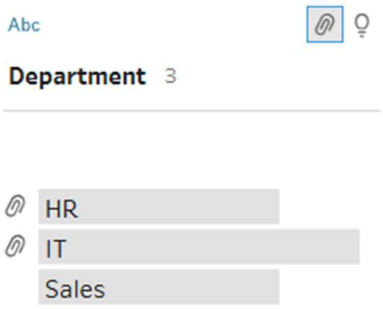
Your final Department values should be:

Before	After
IT	IT
it	IT
IT Dept	IT
HR	HR
Human Resource	HR
Sales	Sales
sales	Sales

So, just click “Group Values” — you are on the right option. ✅

If you want, we can continue from this exact step to sub-practical (d) Merge Mismatched Fields, one clicks at a time.

Expected Output:-



D) MERGE MISMATCHED FIELDS

For this part, we will use two files and combine them using a common field.

Our first file:

Employee_Data.xlsx

contains:

Employee_ID	Employee_Name	Department	City	Salary
E101	Amit	IT	Mumbai	45000
E102	Neha	IT	Mumbai	50000

E103	Rahul	IT	Thane	48000

Second file:

Create another Excel file called Employee_Location.xlsx:

Employee_ID	Location
E101	Mumbai Office
E102	Mumbai Office
E103	Mumbai Office
E104	Mumbai Office
E105	Thane Office

The common field is:

Employee_ID

STEP 1: Add the second file

In Tableau Prep:

8. Go to the Connections area on the left.
9. Click +.
10. Select Microsoft Excel.
11. Select Employee_Location.xlsx.
12. Select the sheet.
13. Drag the sheet into the flow.

You should now have two inputs:

Employee Data

and

Employee Location

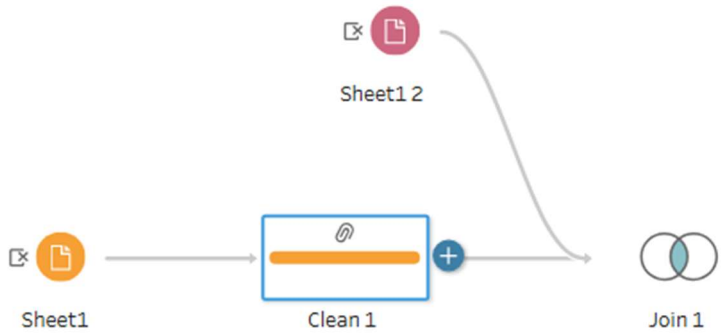
STEP 2: Create a Join

We want to combine these two tables.

14. Click the + after your Employee Data step.
15. Select Join.
16. Drag/connect the second table to the Join step.

You should see something like:

Employee Data → Join ← Employee Location



STEP 3: Select the matching field

Tableau Prep needs to know:

Which column is common in both files?

Select:

Employee_ID = Employee_ID

This is called the Join Clause.

STEP 4: Check the joined data

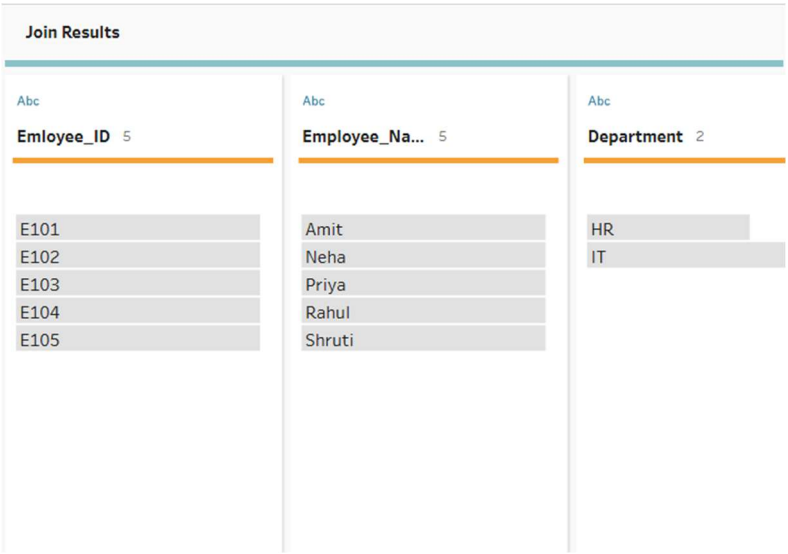
Click the Join step.

You should now see fields from both files.

For example:

Employee_ID	Employee_Name	Department	Salary	Location
E101	Amit	IT	45000	Mumbai Office
E102	Neha	IT	50000	Mumbai Office
E103	Rahul	IT	48000	Thane Office

Now the two files have been combined.



SIMPLE MEANING

Join = combining two tables using a common field.

Here:

Employee_ID is the common field.

(After joining first count filter and clean data and after that count aggregate function.)

E) PERFORM AGGREGATION

Now let's continue.

STEP 1: Add Aggregate

17. Click the + after the Join step.
18. Select Aggregate.



STEP 2: Group by Department

In the Aggregate step:

Drag:

Department → Grouped Fields

This means:

"Give me a separate result for each department."

STEP 3: Calculate Average Salary

Find:

Salary

Drag it into Aggregated Fields.

Change the aggregation to:

Average

STEP 4: Count Employees

Drag:

Employee_ID

to Aggregated Fields.

Set it to:

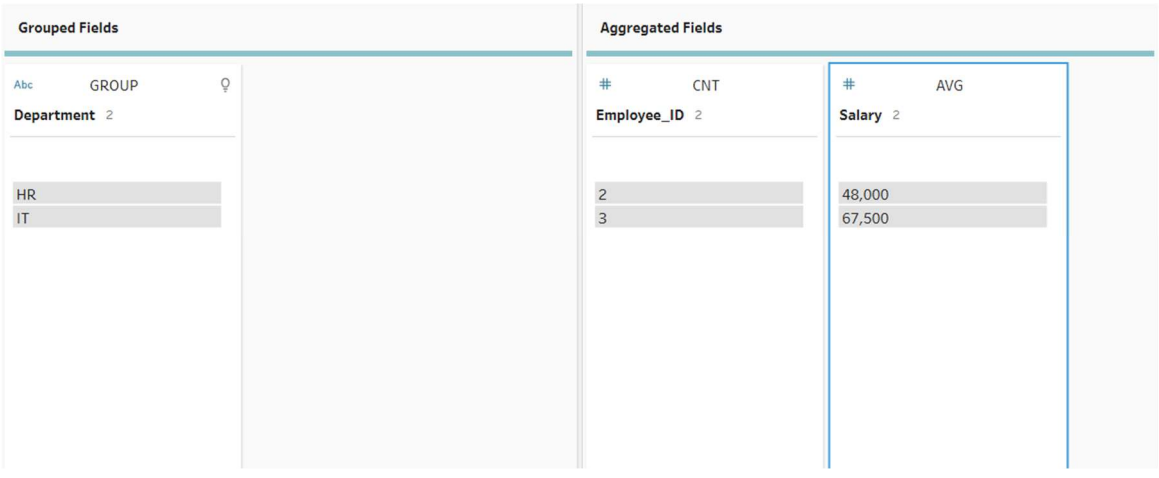
Count

Your result will look approximately like:

Department	Count of Employee_ID	Average Salary
IT	4	50,000
HR	2	41,500
Sales	2	39,500

The exact numbers depend on the complete dataset.

Expected Output:-



F) APPLY FILTERS AND TRANSFORMATIONS

Now let's move to the next sub-practical.

STEP 1: Add a Clean Step

Click the + after the Aggregate step.

Select:

Clean Step

You should now have:

Join → Aggregate → Clean

STEP 2: Filter the data

Suppose we want only employees whose salary is more than ₹40,000.

In the Clean step:

19. Find the Salary field.
20. Click its drop-down/menu.
21. Look for Filter.
22. Set the condition:

Salary > 40000

23. Apply the filter.

Now Tableau Prep will keep only records where salary is greater than ₹40,000.

Open the field menu for Joining_Year. how to open it

In Tableau Prep, the field menu is the small menu attached to the field card.

For Joining_Year:

24. Open your Clean Step.
25. Find the field card named Joining_Year.
26. Move your mouse over the Joining_Year field.
27. Look for a small down-arrow (▼) or three-dot menu (⋯) on the field card.
28. Click it.
29. A menu will open.
30. Choose Rename Field.
31. Type:

Year

32. Press Enter.

If you don't see Joining_Year

It may have disappeared because you are currently looking at the Aggregate step. Click the Clean Step before the Aggregate where all the original fields are visible.

If your screen looks different, send me a screenshot of your Clean Step and I can point out exactly where the menu is.

there is no card name like joining year

For Practical 10, do this instead:

Since you have already completed:

Input → Clean → Group Values → Join → Aggregate

we can skip the Joining_Year rename step.

For the Filter and Transformation sub-practical, use a field that you can actually see.

Let's filter Salary

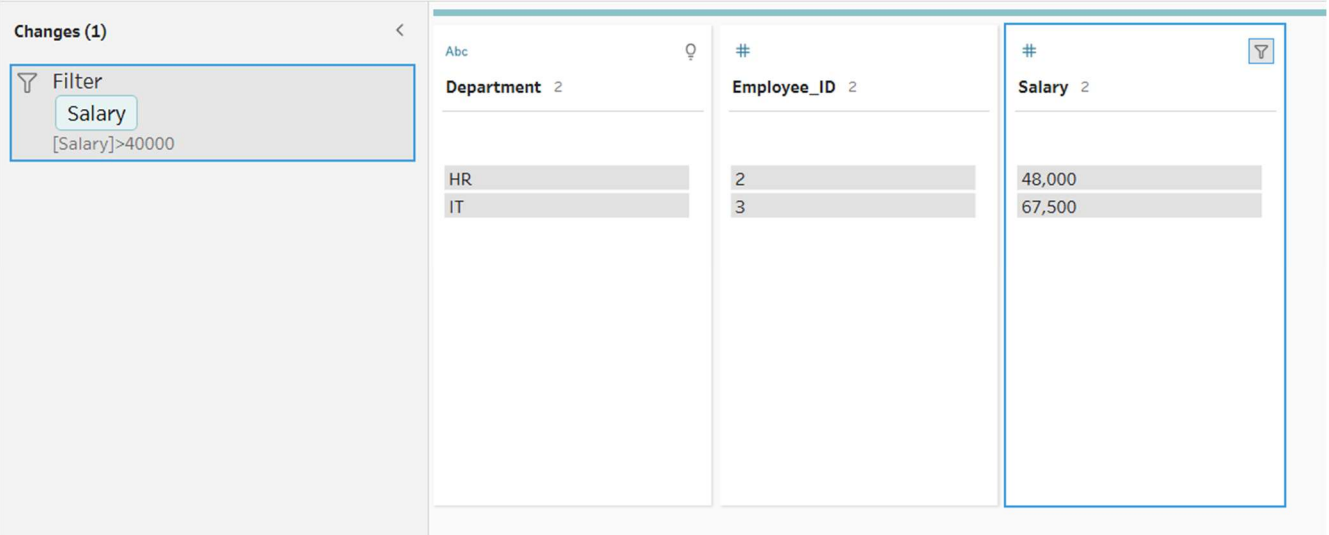
33. Click your Clean Step after the Join/Aggregate step.
34. Look at the fields displayed.
35. Find Salary.
36. Click the small drop-down arrow / menu on the Salary field.
37. Choose the filtering option.
38. Set:

Salary > 40000

39. Apply the filter.

That's enough to demonstrate filtering.

Expected Output:-



Conclusion:-

In this practical, we learned how to prepare, clean, transform, and organize raw data using Tableau Prep. We identified and handled missing, duplicate, and inconsistent values to improve data quality. We also learned how to change data formats, combine data from different sources, and restructure data into a suitable format. Tableau Prep makes the data preparation process easier and more efficient before analysis. The cleaned and transformed data can be used to create accurate charts, dashboards, and meaningful visualizations in Tableau. Overall, this practical helped us understand the importance of data preparation in obtaining reliable and meaningful results from data analysis.

PRACTICAL 11

IMPLEMENTING ADVANCED VISUALIZATION TECHNIQUES AND CUSTOMIZATIONS

Dataset used:

You can use a dataset with fields such as\:You can use your existing dataset too—the field names may be different, but the procedure is similar.

Employee	Department	State	Salary	Year
Amit	IT	Maharashtra	50000	2024
Ravi	IT	Maharashtra	55000	2025
Priya	HR	Gujarat	45000	2024
Neha	HR	Gujarat	52000	2025
Kiran	Sales	Goa	35000	2024
Sneha	Sales	Goa	40000	2025
Rahul	Finance	Rajasthan	48000	2024
Pooja	Finance	Rajasthan	55000	2025
Arjun	IT	Punjab	42000	2024
Meena	IT	Punjab	47000	2025
Sameer	Sales	Karnataka	60000	2024
Anjali	HR	Kerala	50000	2024
Nitin	Sales	Karnataka	68000	2025
Riya	HR	Kerala	58000	2025
Varun	Finance	Tamil Nadu	62000	2024
Kavya	Finance	Tamil Nadu	70000	2025

A) CREATE A SLOPE CHART

What is a slope chart?

A slope chart shows how a value changes between two points, usually two years.

For example:

2024 Salary → 2025 Salary

Step 1: Open Tableau

1. Open Tableau.
2. Open your workbook.
3. Go to a new Worksheet.
4. Rename it: **Slope Chart**

Step 2: Create the basic chart

Drag:

- Department → Rows
- Salary → Columns
- Year → Columns

You may initially get a bar-type visualization.

Step 3: Change Marks type

On the Marks card:

5. Click the drop-down currently showing Automatic.
6. Select Line.

Step 4: Connect the two years

Drag:

Year → Path

on the Marks card.

Now Tableau connects the 2024 and 2025 values.

Step 5: Add labels

Drag:

Salary → Label

on the Marks card.

You now have a basic slope chart.

Expected output

You will see something like:

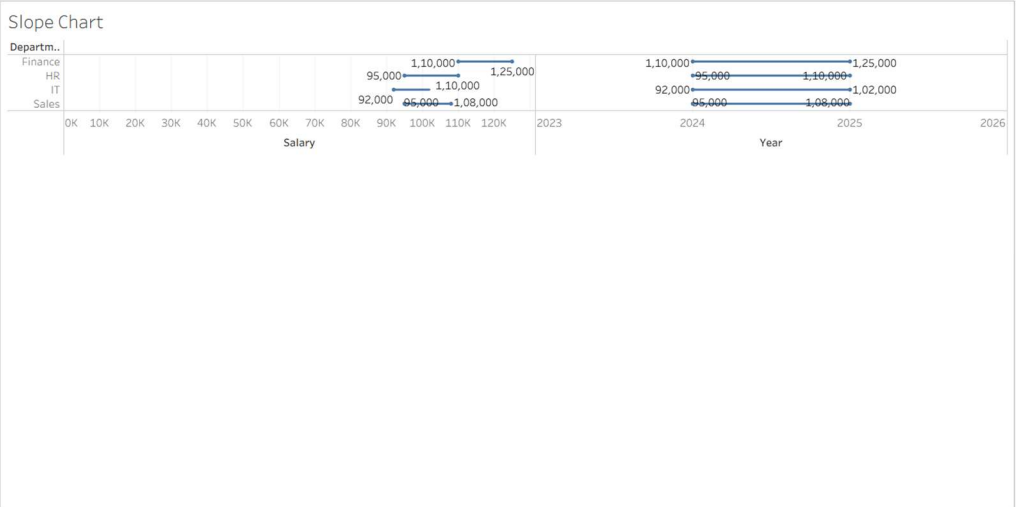
Department

IT ●————————●

HR ●————————●

Sales ●————————————●

The left point represents 2024 and the right point represents 2025.



B) CREATE A LOLLIPOP CHART

What is a lollipop chart?

A lollipop chart is basically a bar chart with a circle at the end.

It makes comparisons visually attractive.

Step 1: Create a new worksheet

Click:

New Worksheet

Rename it:

Lollipop Chart

Step 2: Create the bar

Drag:

Department → Rows

Drag:

Sales → Columns

You will get a horizontal bar chart.

Step 3: Create a dual axis

Drag Sales to the Columns shelf again.

Now you have two Sales axes.

Right-click the second Sales pill.

Select:

Dual Axis

Step 4: Change the first Marks card

You will now see two Marks cards:

- SUM(Sales)
- SUM(Sales) (2)

Select the first one.

Choose:

Bar

Step 5: Change the second Marks card

Select the second Sales Marks card.

Change:

Automatic → Circle

Increase the Size slightly.

Step 6: Synchronize axes

Right-click one of the axes.

Select:

Synchronize Axis

Step 7: Hide the second axis

Right-click the axis.

Select:

Show Header

to remove the duplicate axis if necessary.

Expected result

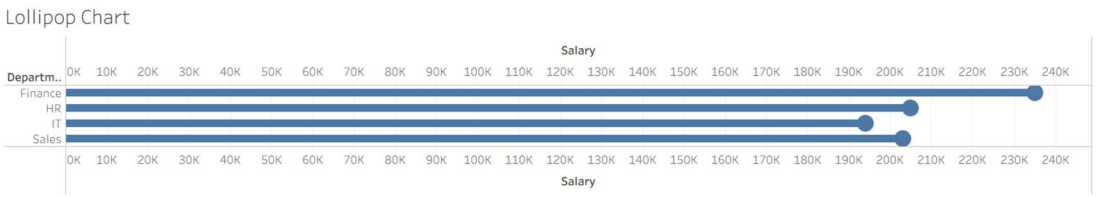
You get:

IT ──────●

HR ─────●

Sales ──────────────●

The bar is the stick and the circle is the lollipop head.



C) DESIGN A WATERFALL CHART

What is a waterfall chart?

A waterfall chart shows how individual values increase or decrease a total.

Example:

Starting Sales → Increase → Decrease → Final Sales

Step 1: Create a new worksheet

Rename:

Waterfall Chart

Step 2: Use a category

Drag:

Department → Columns

Drag:

Sales → Rows

Step 3: Change Marks type

On the Marks card:

Automatic → Gantt Bar

Step 4: Create a calculated field

Go to:

Analysis → Create Calculated Field

Name it:

Sales Base

Formula:

SUM([Sales])

Click OK.

Step 5: Add Sales Base

Drag:

Sales Base → Size

on the Marks card.

You should now see the waterfall effect.

Step 6: Add labels

Drag:

Sales → Label

Step 7: Format

Use:

Color to distinguish increases and decreases if needed.

Expected output

Conceptually:

Starting

[redacted]

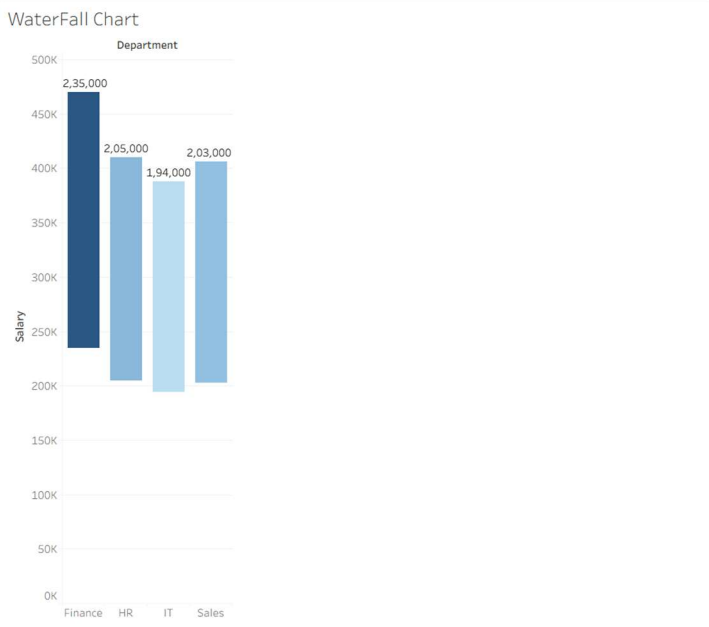
[redacted] Increase

[redacted]

[redacted] Decrease

[redacted] Final

The chart visually explains how different components contribute to the final total.



D) BUILD SPARKLINES FOR TREND ANALYSIS

What is a sparkline?

A sparkline is a small line chart that shows a trend without using a large chart area.

For example:

Department	Trend
IT	
HR	
Sales	

Step 1: Create a new worksheet

Rename:

Sparklines

Step 2: Add Department

Drag:

Department → Rows

Step 3: Add Date/Year

Drag:

Year → Columns

Step 4: Add Sales

Drag:

Sales → Rows

Step 5: Change Marks type

Marks:

Automatic → Line

Now each department has its own small trend line.

Step 6: Remove unnecessary elements

Go to:

Analysis → Table Layout

You can adjust row/column sizing.

You can also hide unnecessary headers/gridlines from:

Format → Lines

Step 7: Make it compact

Reduce the row height so that the chart looks like a small trend indicator.

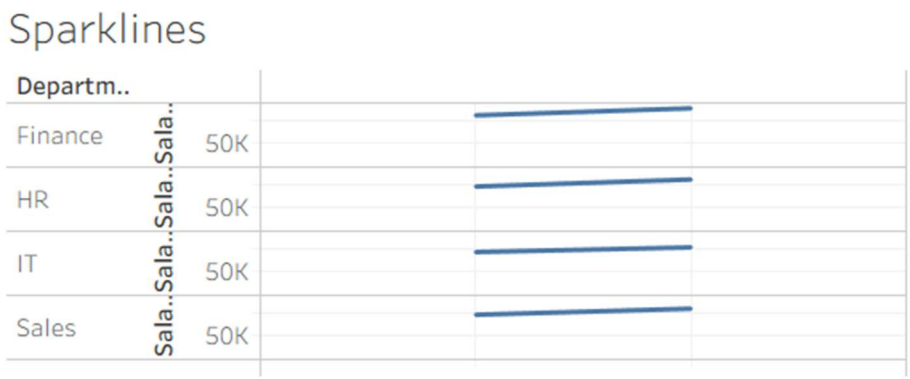
Expected result

IT ／＼＼＿＼＿＼＿／

HR ＼＼＿＼＿／／

Sales ／／／／

The purpose is to quickly identify whether a value is increasing, decreasing, or fluctuating.



E) TRUE SHEET SWAPPING IN TABLEAU

We will swap:

Sales Chart ↔ Salary Chart

using one parameter.

STEP 1: Create the two worksheets

Worksheet 1 — Department Chart

Create:

- Department → Rows
- Employee → Columns

Rename it:

Department Chart

Worksheet 2 — Salary Chart

Create:

- Department → Rows
- Salary → Columns

Rename it:

Salary Chart

STEP 2: Create the parameter

Go to:

Analysis → Create Parameter

Set:

Name: Select Chart

Data type: String

Under List, select:

List of values

Enter:

Department

Salary

Click OK.

STEP 3: Create a calculated field for Department Chart

Go to:

Analysis → Create Calculated Field

Name it:

Show Department

Enter:

[Select Chart] = "Department"

Click OK.

This returns:

- TRUE when Department is selected.
- FALSE when Salary is selected.

STEP 4: Create a calculated field for Salary Chart

Again:

Analysis → Create Calculated Field

Name:

Show Salary

Formula:

[Select Chart] = "Salary"

Click OK.

STEP 5: Put the calculated field on Department Chart

Go to your Department Chart worksheet.

Find:

Show Department

in the Data pane.

Drag:

Show Department → Filters

A Filter window will appear.

Select:

True

Click OK.

Now:

- Select Chart = Department → Department Chart is TRUE → it appears.
- Select Chart = Salary → Department Chart is FALSE → it disappears.

STEP 6: Put the calculated field on Salary Chart

Go to the Salary Chart worksheet.

Drag:

Show Salary → Filters

Select:

True

Click OK.

Now:

- Select Chart = Salary → Salary Chart appears.
- Select Chart = Department → Salary Chart disappears.

STEP 7: Create the Dashboard

Click:

New Dashboard

Drag:

Department Chart

onto the dashboard.

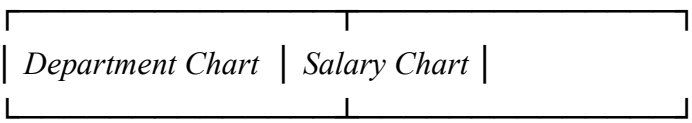
Then drag:

Salary Chart

onto the same location/container.

Important:-

If you simply place them side-by-side:



you will still see two areas.

Instead, put them into the same dashboard area, preferably using a floating layout.

STEP 8: Make the worksheets floating

On the Dashboard:

7. Select Department Chart.
8. Change it to Floating.
9. Select Salary Chart.
10. Change it to Floating.
11. Move them so they occupy exactly the same position and size.

They will overlap.

STEP 9: Show the parameter

On the dashboard, select:

Select Chart

from the worksheet's parameter options and choose:

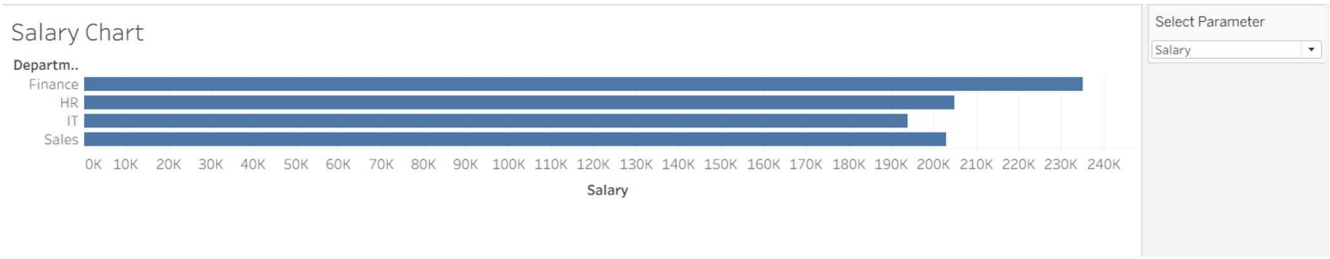
Show Parameter

You should now get a control such as:

Select Chart

Department ▼

Expected Output:-



STEP 10: Test the sheet swap

Choose:

Sales

The dashboard shows:

Department Chart

IT [redacted]

HR [redacted]

Sales [redacted]

Then change the parameter to:

Salary

The Department worksheet becomes FALSE and disappears, while the Salary worksheet becomes TRUE and appears.

Salary Chart

IT [redacted]

HR [redacted]

Sales [redacted]

Then the important part:-

In the Dashboard, click the Sales worksheet:

Department sheet → Filters → Department calculation → select TRUE

Then click the Salary worksheet:

Salary sheet → Filters → Salary calculation → select TRUE

So the result is:

Department selected → Sales Sheet visible

Salary selected → Salary Sheet visible

That is a true sheet swap.

Expected Output:-





F) CREATE CUSTOM GEOGRAPHIC TERRITORIES

What are territories?

Territories allow you to group geographical locations into your own business regions.

For example:

Mumbai

Pune

Nashik

Nagpur

could be grouped as:

Maharashtra Territory

Step 1: Open your geographic worksheet

Create a new worksheet.

Drag:

State → Detail

on the Marks card.

Step 2: Create a group

Right-click:

State

Select:

Create → Group

Step 3: Select states

For example, select:

- Maharashtra
- Gujarat
- Goa

Click:

Group

Rename the group:

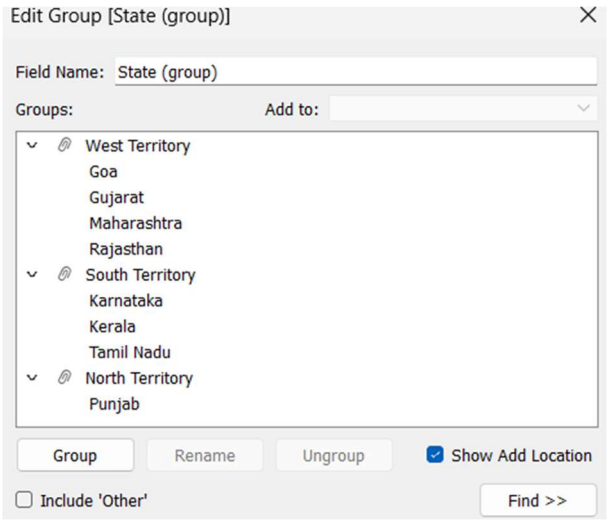
West Territory

Similarly, create:

North Territory

South Territory

etc.



Step 4: Create the map

Double-click:

State

Tableau should automatically create a map if the geographic role is correctly recognized.

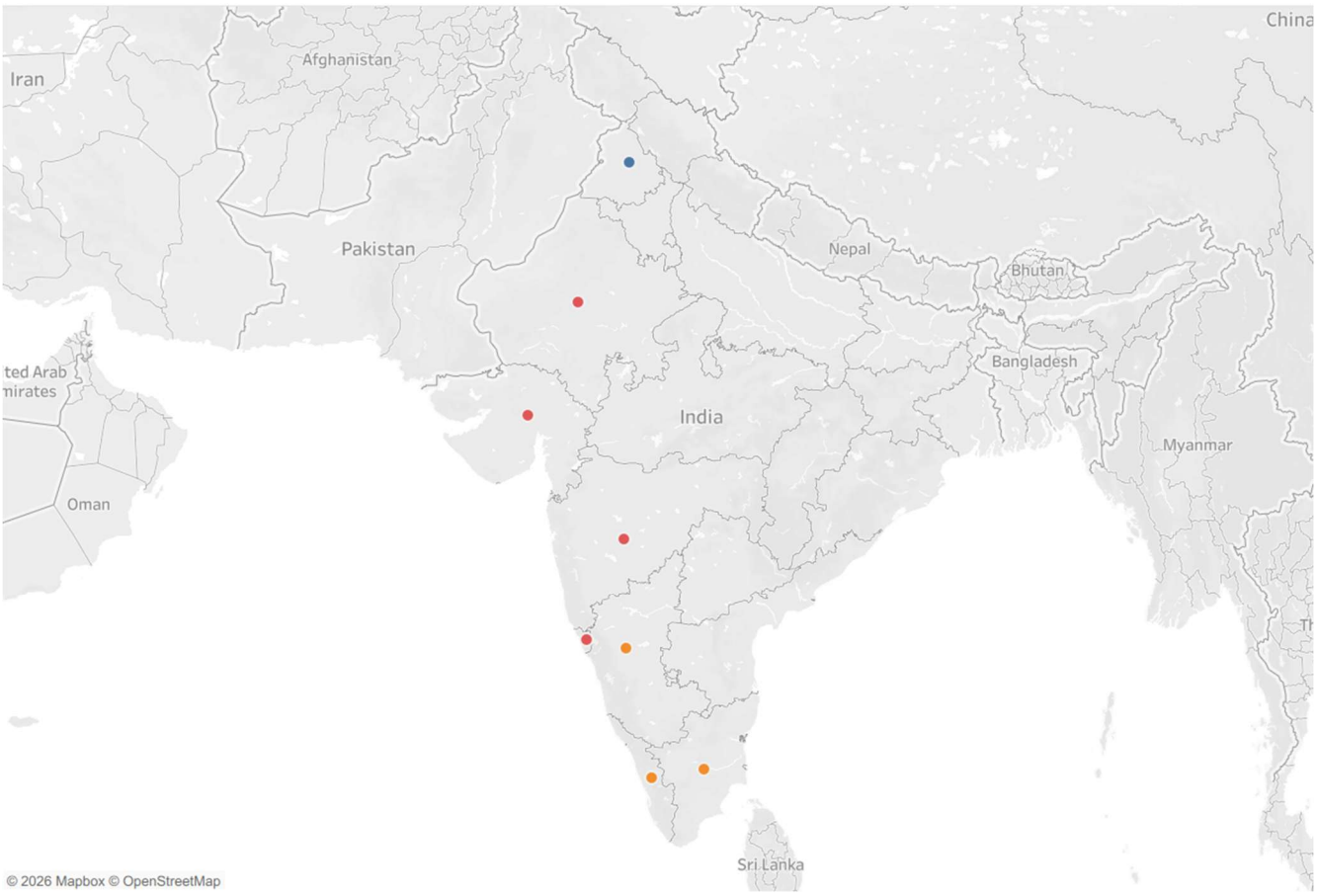
Step 5: Add territory

Drag your newly created:

State Group → Color

Expected output:-

Your map will now show your custom territories, rather than treating every state as a completely separate business region.



If State does not create a map:

Right-click State:

Geographic Role → State/Province

Then double-click it again.

G) USE BACKGROUND IMAGES IN DASHBOARDS

This allows you to place a dashboard visualization over an image.

For example:

Office layout + charts

or

India map image + data visualization

Step 1: Prepare an image

Keep a JPG or PNG image ready.

For example:

office_layout.png

Step 2: Open a dashboard

Click:

New Dashboard

Step 3: Add the image

From the Objects section:

Select:

Image

Step 4: Select your image

Browse to your image.

Click:

Open

Step 5: Adjust the image

Choose:

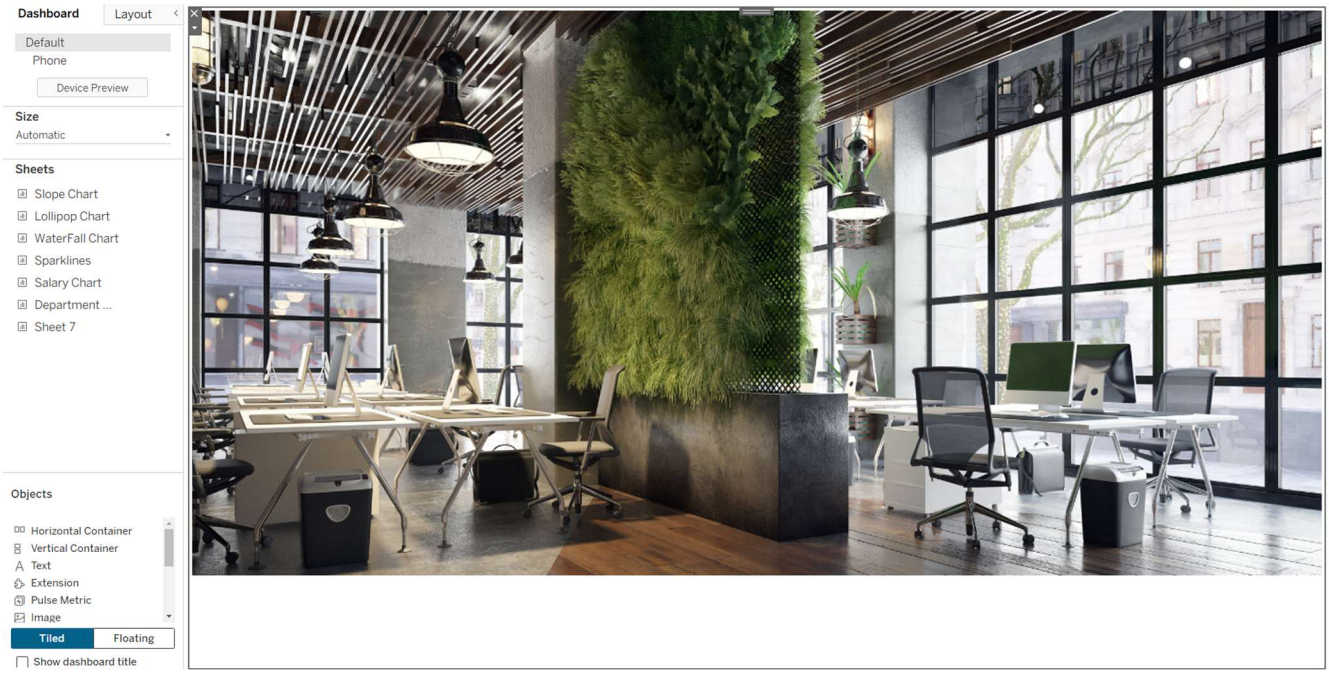
Fit Image

or adjust its size manually.

Important distinction:-

If your practical specifically means a background image behind a worksheet, Tableau also supports adding images through the worksheet/dashboard formatting options depending on the Tableau version.

For a simple practical demonstration, adding an Image object to the dashboard is easiest.



H) APPLY ANIMATION AND TRANSPARENCY

1. Animation

Tableau can animate changes in views when data changes.

Step 1:-

Open your worksheet/dashboard.

Go to:

Format → Animations

In newer Tableau versions, animation controls may appear under:

Format → Animations

Step 2:-

Turn animation:

On

You can adjust:

Duration

For example:

0.5 seconds

Drag Department →Filter

Step 3: Test

Change a filter or parameter.

The chart should transition smoothly instead of changing instantly.

2. Transparency

Transparency controls how see-through a mark is.

Step 1:-

Select the Marks card.

Step 2:-

Click:

Color

Step 3:-

Find:

Opacity

Move the slider.

For example:

100% → completely visible

50% → partially transparent

20% → highly transparent

Why use transparency?

Transparency is useful when many marks overlap.

For example, in a scatter plot:

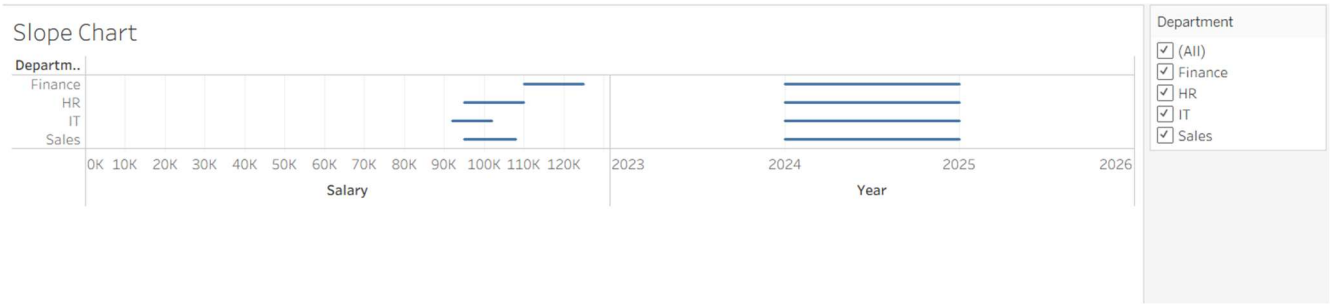
●

●

●●

●●●

Using transparency makes overlapping points easier to identify.



COMPLETE PRACTICAL FLOW

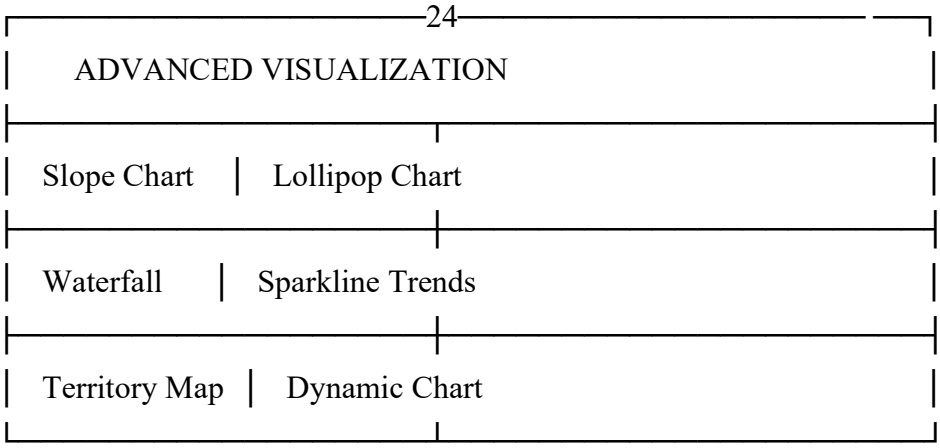
Your Practical 11 can therefore be demonstrated as seven separate worksheets/features:

Part	Visualization/Feature	Main Tableau technique
A	Slope Chart	Line + Path
B	Lollipop Chart	Dual Axis + Bar + Circle
C	Waterfall Chart	Gantt Bar + Calculation
D	Sparkline	Line chart + compact formatting
E	Sheet Swapping	Parameter + Dashboard
F	Custom Territories	Geographic Group
G	Background Image	Dashboard Image
H	Animation & Transparency	Format + Marks → Color

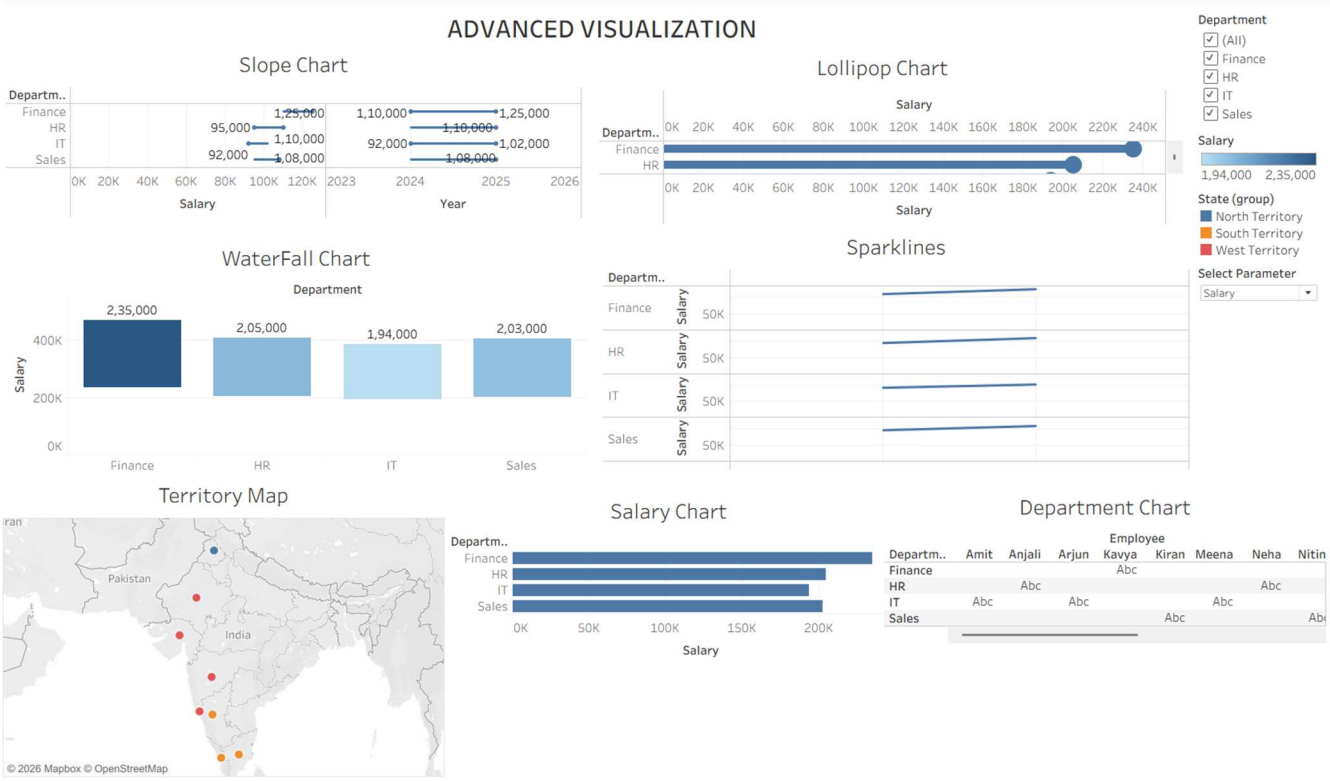
Suggested final dashboard:-

You can create one final dashboard containing:

ADVANCED VISUALIZATION DASHBOARD



This gives you a clear practical demonstration of all the advanced visualization techniques in one Tableau workbook.



Conclusion:-

In this practical, we learned how to create and customize different advanced visualizations and features in Tableau. We explored the creation of slope charts, lollipop charts, waterfall charts, and sparklines to represent data effectively and make comparisons easier. We also learned how to use dual-axis charts, Gantt bars, calculated fields, parameters, and dashboards to enhance data analysis and interactivity. Additionally, we understood how to create custom territories using geographic groups, add background images to dashboards, and apply animation and transparency to improve the visual presentation of data. Through these techniques, we gained practical knowledge of Tableau's advanced charting, formatting, and dashboard customization capabilities. Overall, this practical helped us develop the skills required to transform raw data into meaningful, interactive, and visually appealing representations that support better data interpretation and decision-making.

Practical 4: Performing Data Analysis Using Calculations in Tableau.

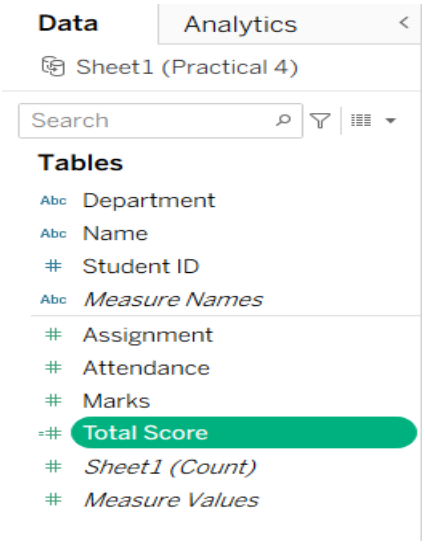
Aim :
To create calculated fields, perform row-level and aggregate calculations, implement Level of Detail (LOD) expressions, create parameters, and use calculations for data analysis.
a) Create basic calculated fields using formulas
b) Perform row-level calculations
c) Perform aggregate-level calculations
d) Implement FIXED Level of Detail (LOD) calculations
e) Use INCLUDE and EXCLUDE LOD expressions
f) Create and use parameters for dynamic analysis
g) Apply calculations to fix data issues and enhance insights

(A)Aim:- Create Basic Calculated Fields Using Formulas.

Step 1:-
Open Tableau and connect to "**Practical_2.csv".**

Step 2:-
Go to **Analysis → Create Calculated Field**.

<u>Example 1: Total Score</u>
Name:
Total Score
Formula:
[Marks] + [Assignment]
Click **OK**.



Step 2: Display the Result

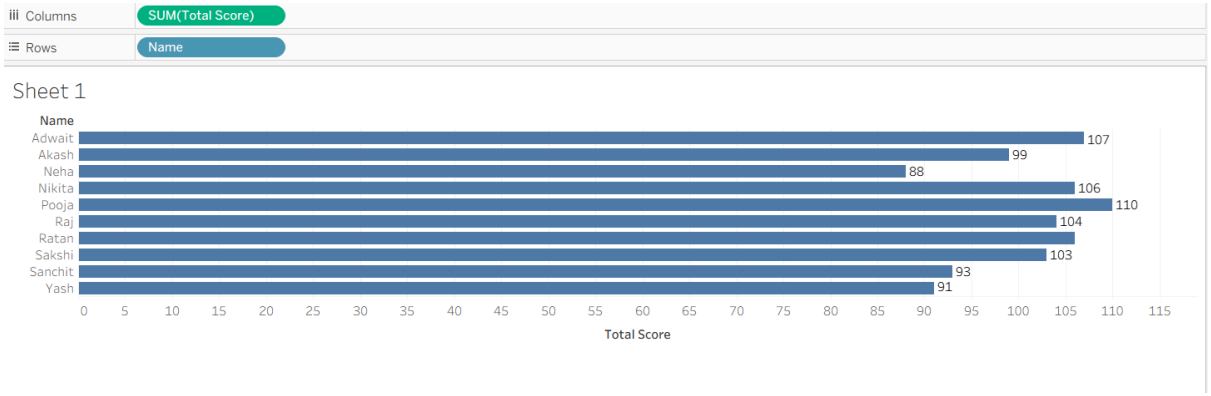
Open **Sheet 1**

Drag:

- **Name** → Rows
- **Total Score** → Text (Marks Card)

Drag

- **Name** → Rows
- **Total Score** → Columns



Example 2: Pass/Fail

Name: Result

Formula:

IF [Marks] >= 40 THEN

"Pass"

ELSE

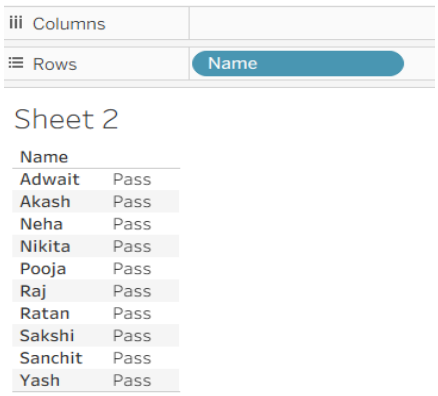
"Fail"

END

Display the Result:

Drag

- **Name → Rows**
- **Result → Text**



(B)Aim:- Perform Row-Level Calculations.

A row-level calculation is performed **for each record**.

Example:
Percentage after adding assignment marks:
Percentage=120 (Marks + Assignment)×100
([Marks] + [Assignment]) / 120 * 100
Each student gets their own percentage.

Result :
Drag:

- **Name** → Rows
- **Percentage** → Text (Marks Card)

Student 1: Amit
Marks = **75**
Assignment = **15**
75 + 15 = 90
90 / 120 = 0.75
0.75 × 100 = 75%
Result: 75%



Sheet 3

Name	
Adwait	89.17
Akash	82.50
Neha	73.33
Nikita	88.33
Pooja	91.67
Raj	86.67
Ratan	88.33
Sakshi	85.83
Sanchit	77.50
Yash	75.83

(C)Aim:- Perform Aggregate-Level Calculations.

Aggregate calculations summarize data.

Example 1

Average Marks

- Method 1: Show Overall Average

It means it shows total no.of student's average marks.

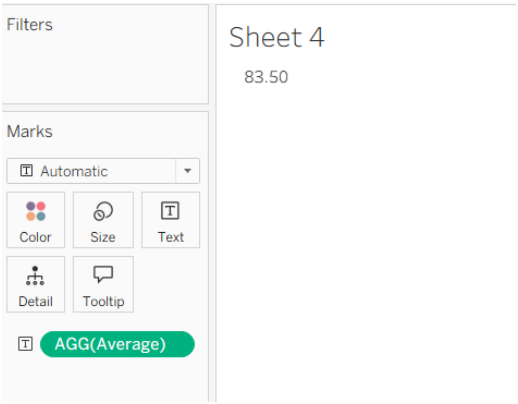
85+72+91+65+78+55+35+96+67+81/10

AVG([Marks])

Result :

Drag **Average Marks** → **Text** on the Marks card.

(Displayed total no .of student's average marks.)

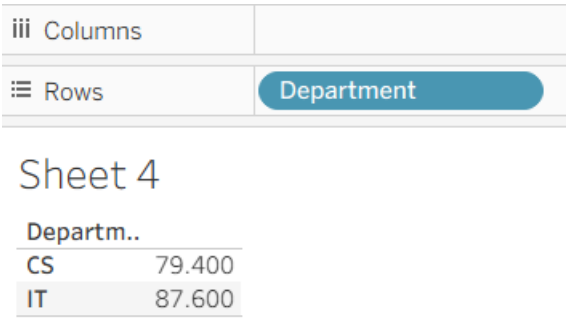


- Method 2: Department-wise Average

Drag:

Department → Rows

Average Marks → Text



Example 2:

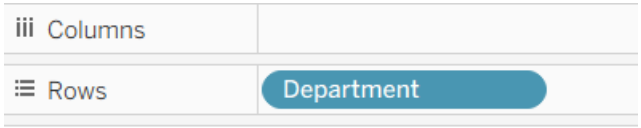
Total Marks
SUM([Marks])
Result :
Drag

- **Total Marks** → Text

(Student wise total marks)
Drag

- **Department** → Rows
- **Total Marks** → Text

(Department wise total marks)



Sheet 4

Departm..	
CS	397
IT	438

Example 3:

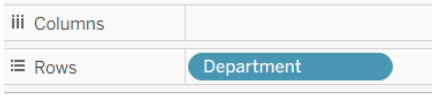
Highest Marks
MAX([Marks])
Drag

- **Highest Marks** → Text

(Department-wise Highest Marks)

Drag

- **Department** → Rows
- **Highest Marks** → Text



Sheet 5

Departm..	
CS	87
IT	92

(D)Aim:- Implement FIXED Level of Detail (LOD).

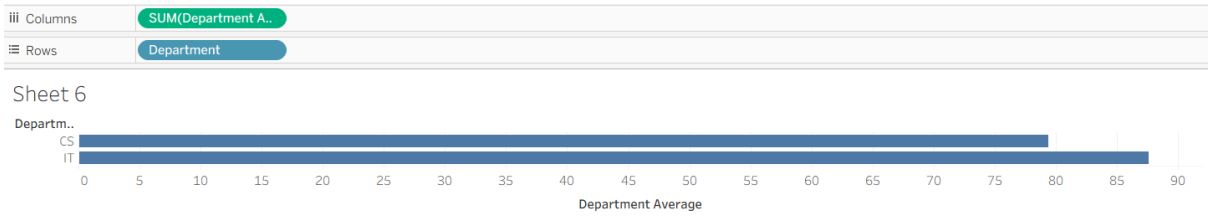
- **Definition:-**
  A **FIXED Level of Detail (LOD)** expression calculates a value at a specified level of detail **regardless of what is displayed in the worksheet**.

  Formula
  { FIXED [Department] : AVG([Marks]) }
  Name:
  Department Average

- **What does this mean?**
  It means
  "Calculate the average marks for every Department."
  For IT
  85+91+78+67/4
  =80.25
  For CS
  72+65+55+96+81/5
  =73.8

  Result :
  Drag → Department to rows.
  Drag → Department average to text.

  (another way)
  Drag:
  Department → Rows
  Department Average → Columns



(E)Aim:- Use INCLUDE and EXCLUDE LOD Expressions.

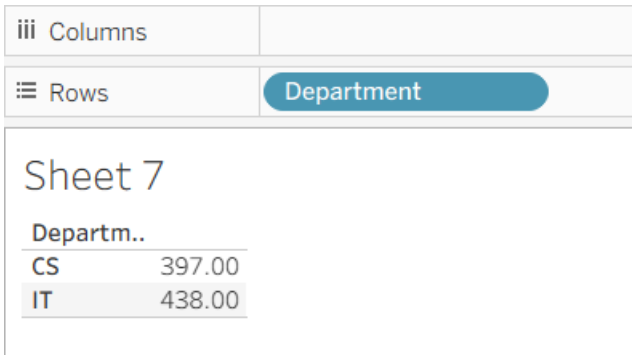
INCLUDE means
"Include an extra dimension while calculating."

INCLUDE:
Include Student Name while calculating department average.
{ INCLUDE [Name] : AVG([Marks]) }
The view only has **Department**, so Tableau aggregates these student averages again.
IT Department
Student averages:

- Amit = 80
- Neha = 70
- Ravi = 90

Department Average
80+70+903/3=451
Drag:

- **Department** → Rows
- **Include Average** → Text



EXCLUDE:
EXCLUDE removes one or more dimensions from the calculation.
Ignore Student Name and calculate only department average.

What exactly did EXCLUDE do?
Suppose your data is:

Department	Name	Marks
IT	Amit	80
IT	Neha	70
IT	Ravi	90

Your worksheet contains:

- Department
- Name

So normally Tableau sees the data like this:
IT → Amit → 80
IT → Neha → 70
IT → Ravi → 90
If you use:
AVG([Marks])
Tableau calculates for **each student** because both **Department** and **Name** are in the view.
Result:

Department	Name	AVG(Marks)
IT	Amit	80
IT	Neha	70
IT	Ravi	90

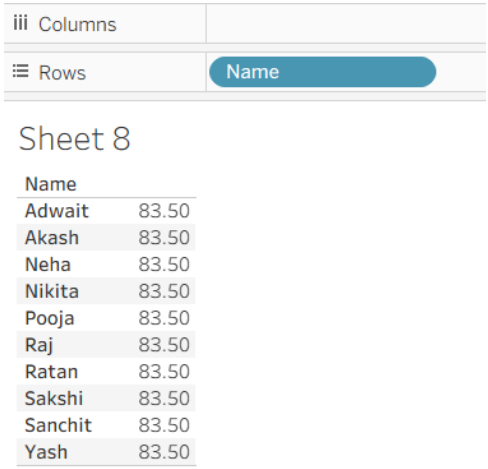
Now use EXCLUDE
{ EXCLUDE [Name] : AVG([Marks]) }
Here Tableau says:
"I know **Name** is present in the worksheet, but **ignore Name while calculating**."
So, Tableau temporarily changes this:
Department + Name
to
Department only

Now it calculates:
IT Department Average
(80+70+90)/3=80(80 + 70 + 90) / 3 = 80(80+70+90)/3=80
This value (80) is then shown for every student in the IT department.
Result:

Department	Name	EXCLUDE Result
IT	Amit	80
IT	Neha	80
IT	Ravi	80

Notice that the **student names are still visible** in the worksheet. **EXCLUDE does not remove the Name column from the view.** It only removes **Name from the calculation**.

{ EXCLUDE [Name] : AVG([Marks]) }



(F)Aim:- Create and Use Parameters for Dynamic Analysis.

A **Parameter** is **not connected to the data**. It is a user-controlled value. When the user changes the parameter, Tableau automatically updates any calculated fields that use it.
A **Parameter** is a dynamic value that allows users to interact with a Tableau visualization by changing values such as pass marks, discount percentage, year, or threshold.

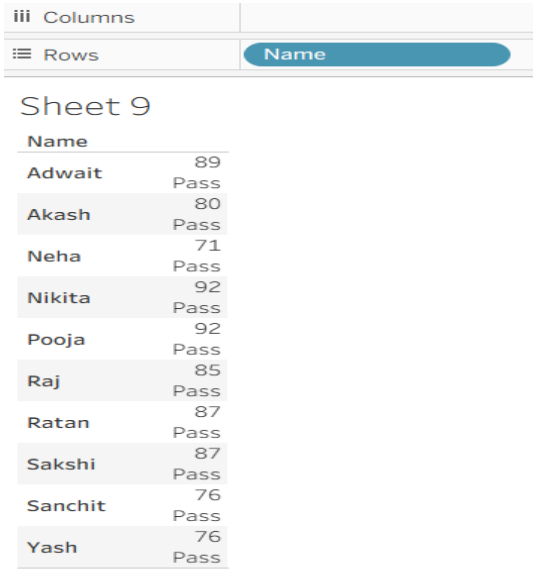
Step 1:
Right-click in the Data pane.
Choose: Create Parameter

Step 2:
Create:
Name: Pass Marks
Data Type: Integer
Current Value:40
Minimum:0
Maximum:100
Step Size:1
Click **OK**.

Step 3:
Create another Calculated Field.
Formula:
IF [Marks] >= [Pass Marks]
THEN "Pass"
ELSE "Fail"

END
Now changing the parameter automatically changes the pass/fail result.

Result :
Drag:
Name → Rows
Marks → Text
Result → Text.

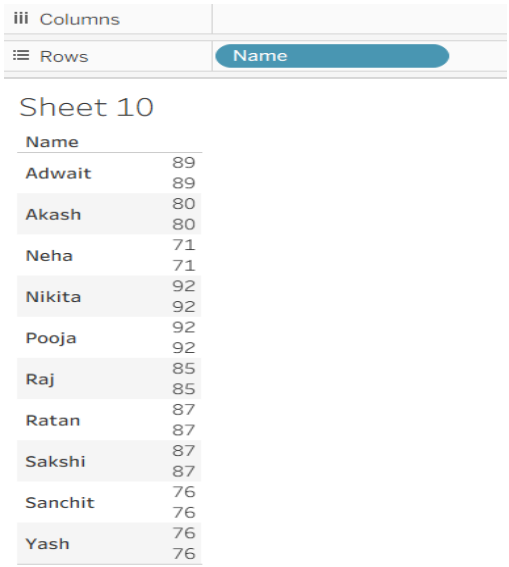


(G)Aim:- Apply Calculations to Fix Data Issues and Enhance Insights.

Example 1: Replace Null Values

IFNULL([Marks],0)
Drag:

- **Name** → Rows
- **Marks** → Text
- **Marks After IFNULL** → Text



Example 2: Attendance Category

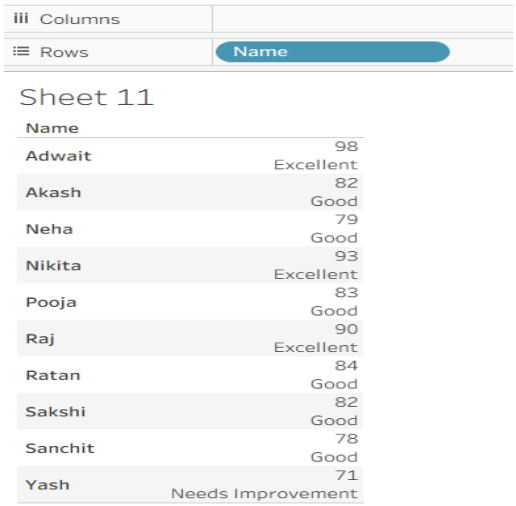
IF [Attendance] >= 90 THEN
"Excellent"

ELSEIF [Attendance] >= 75 THEN
"Good"

ELSE
"Needs Improvement"
END

Result :

Name → Rows
Attendance → Text
Attendance category → Text



Example 3: Grade

IF [Marks] >= 90 THEN "A"
ELSEIF [Marks] >= 75 THEN "B"
ELSEIF [Marks] >= 60 THEN "C"
ELSE
"Fail"
END

Result :
Name → Rows
Marks →Text
Grade → Text

